

SMRTTECH 4SC3

Smart Cities and Communities Lab

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**Lab Instructor:
Dr. Ahmad Mohammadi**

**Week 2 | Session 1:
Intelligent Transportation Systems
(Autonomous Vehicles)**

Fall 2026

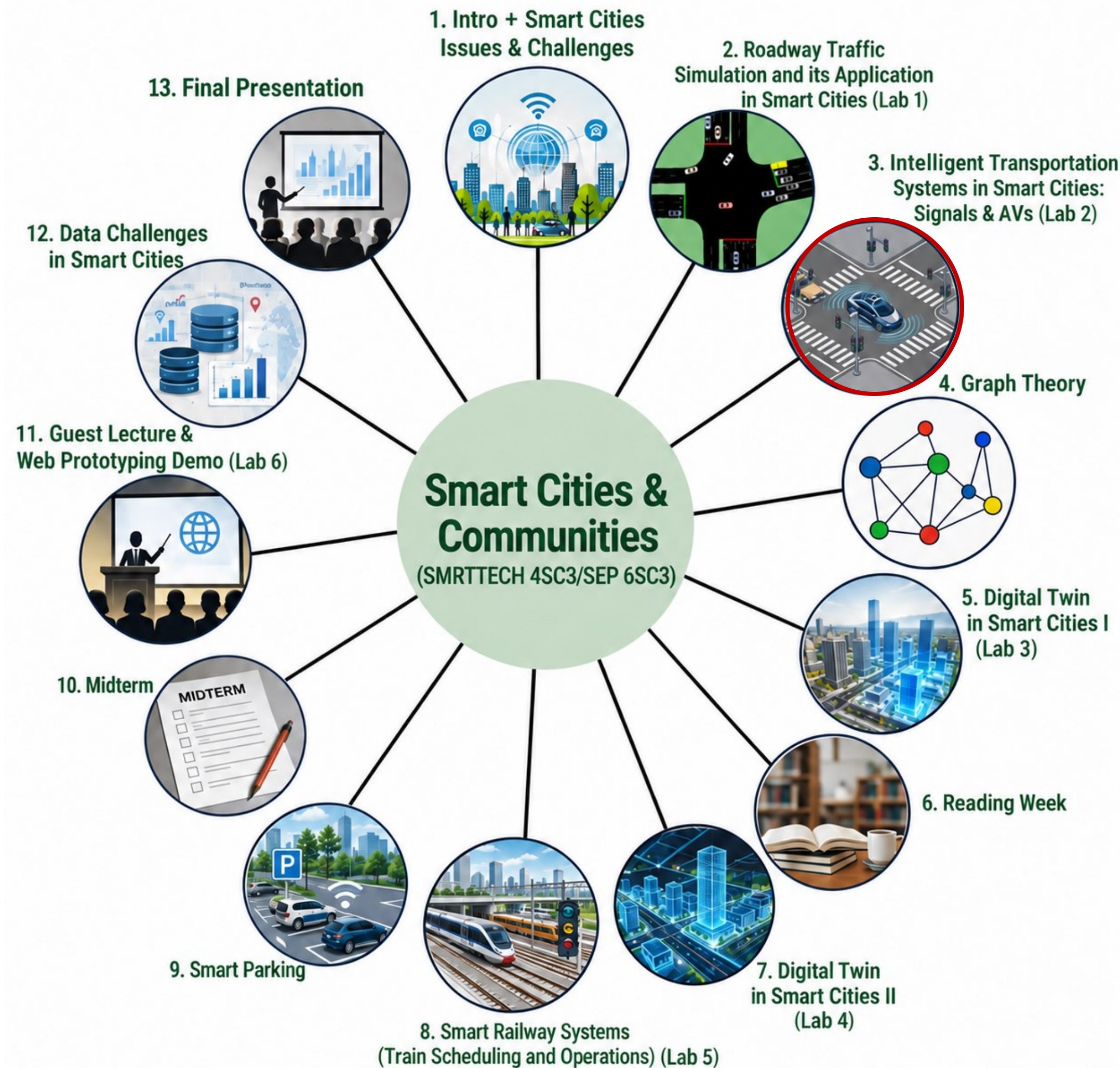


Engineering
W Booth School of Engineering
Practice and Technology

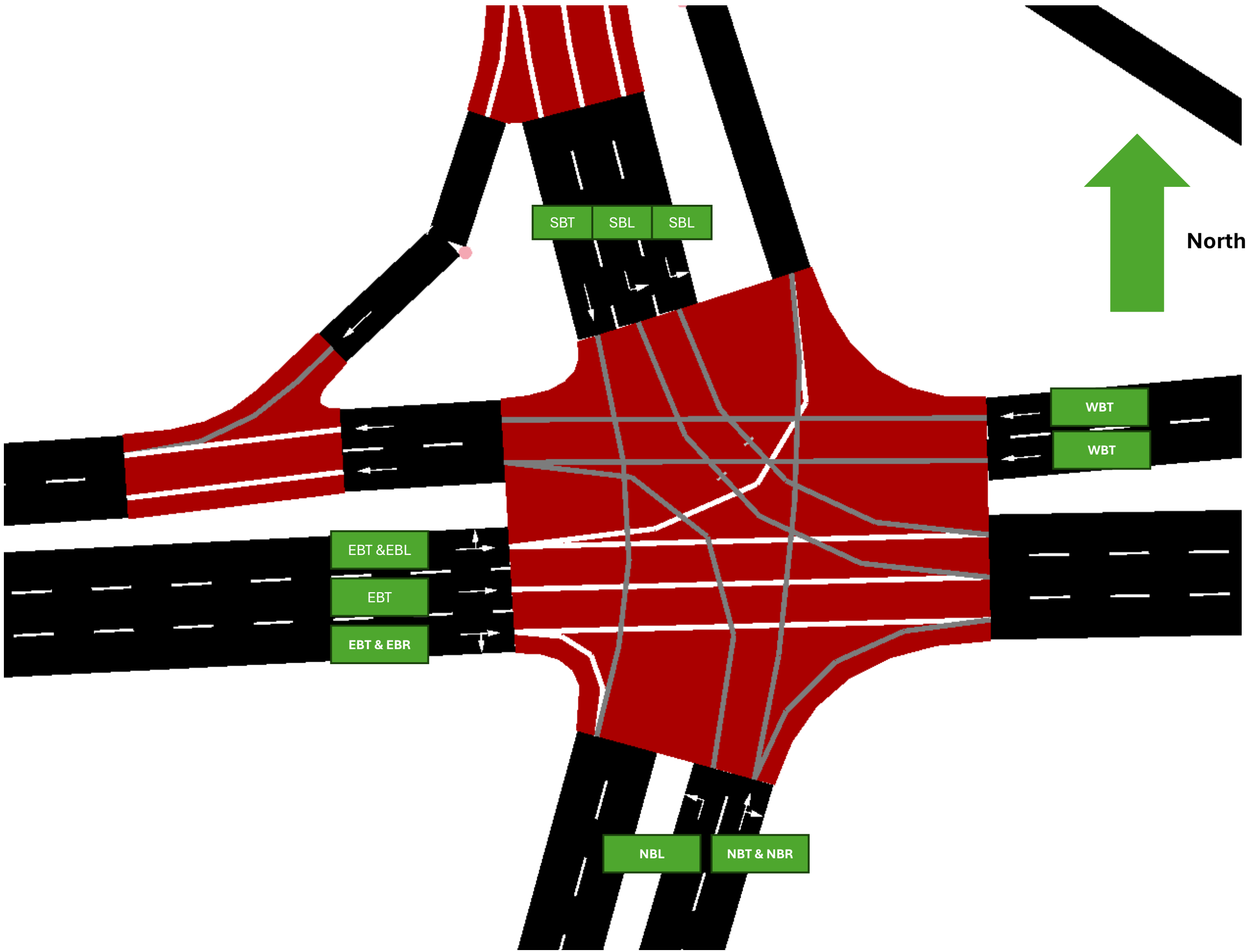


Agenda

- ☐ Intro
- ☐ Milestone 1: Traffic Flow
- ☐ Milestone 2: Traffic Signal
- ☐ Deliverable 1
- ☐ Milestone 3: Adjust Right of Way
- ☐ Milestone 4: Simulate Human Driven Vehicles (HDVs) & Connected Autonomous Vehicles (CAVs)
- ☐ Deliverable 2
- ☐ Deliverable 3
- ☐ Milestone 5: Transportation Intervention
- ☐ Explanation about completing Mini Project A



Final Reference Lane Configuration and Turning Movements – Lab01



Abbreviation	Meaning
NBT	Northbound Through
NBL	Northbound Left Turn
NBR	Northbound Right Turn
SBT	Southbound Through
SBL	Southbound Left Turn
SBR	Southbound Right Turn
EBT	Eastbound Through
EBL	Eastbound Left Turn
EBR	Eastbound Right Turn
WBT	Westbound Through
WBL	Westbound Left Turn
WBR	Westbound Right Turn



Seven Steps in Microsimulation Modeling

Step2 Data Collection requires:

- ☐ Vehicle Characteristics Data – Vehicle Dynamics
- ☐ Driver Characteristics Data – Car Following and Lane Changing Models

Reference link:

<https://ops.fhwa.dot.gov/publications/fhwahop18036/fhwahop18036.pdf>

Traffic Analysis Toolbox Volume III: Guidelines for Applying Traffic Microsimulation Modeling Software

2019 Update to the 2004 Version



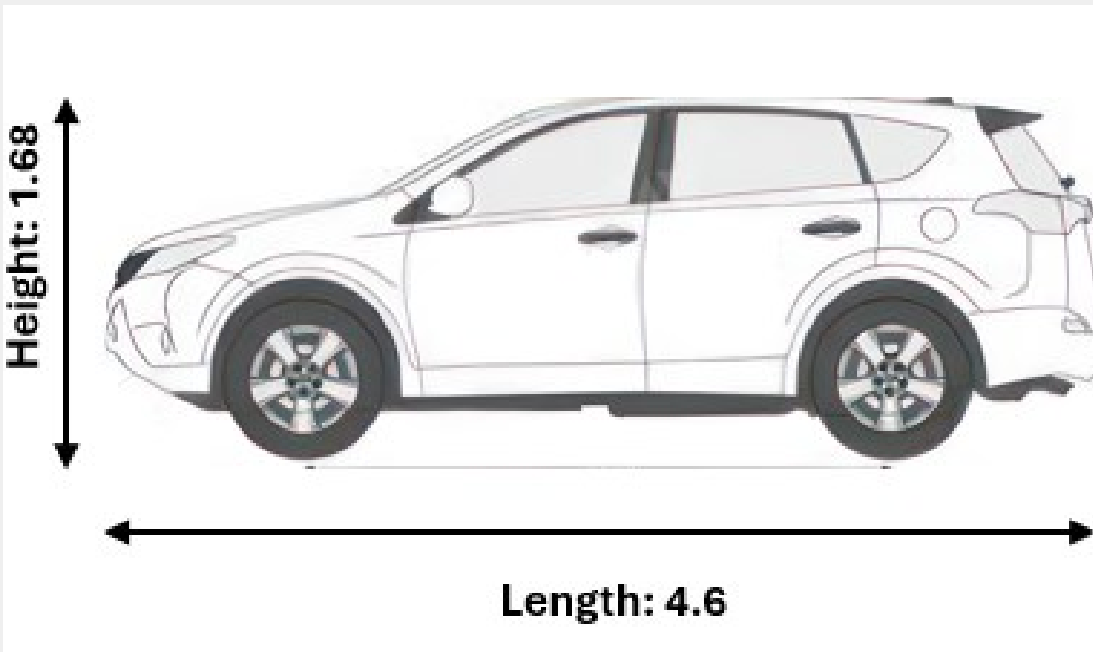
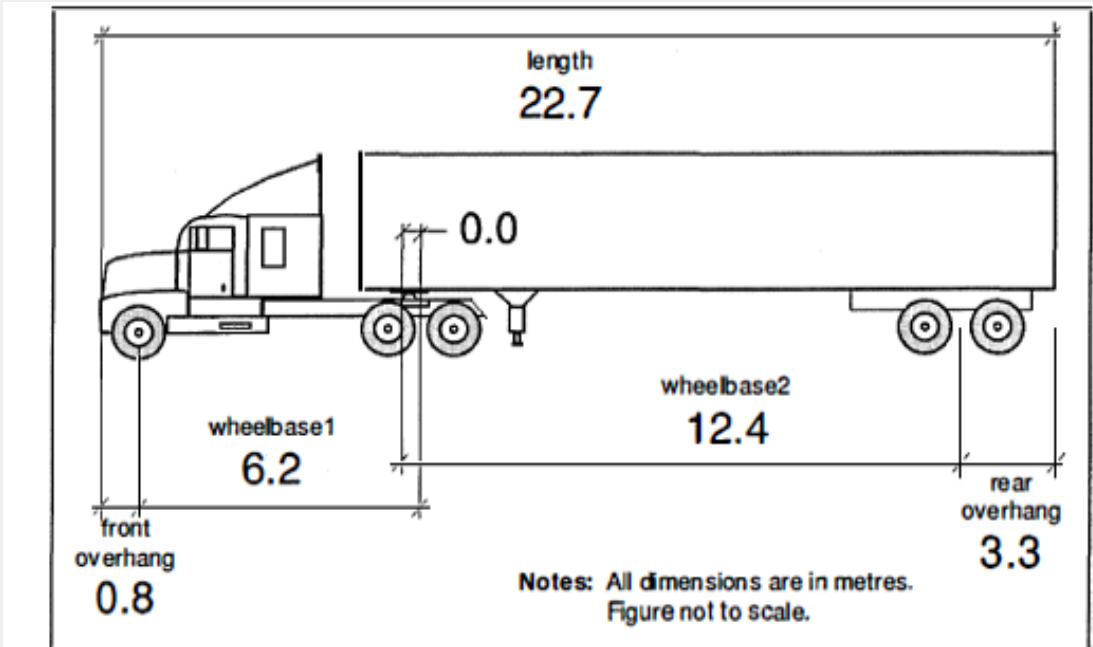
April 2019



U.S. Department of Transportation
Federal Highway Administration

Vehicles Characteristics

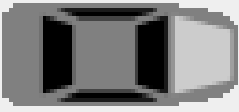
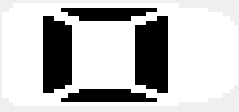
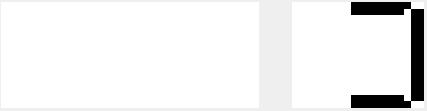
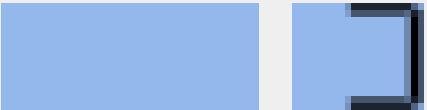
Real-World Dimension



3D Computer Graphic (VR)



2D Traffic Simulation



Vehicles Dynamics

Modelling the Behavior

Steering



Braking/Acceleration



Real-World

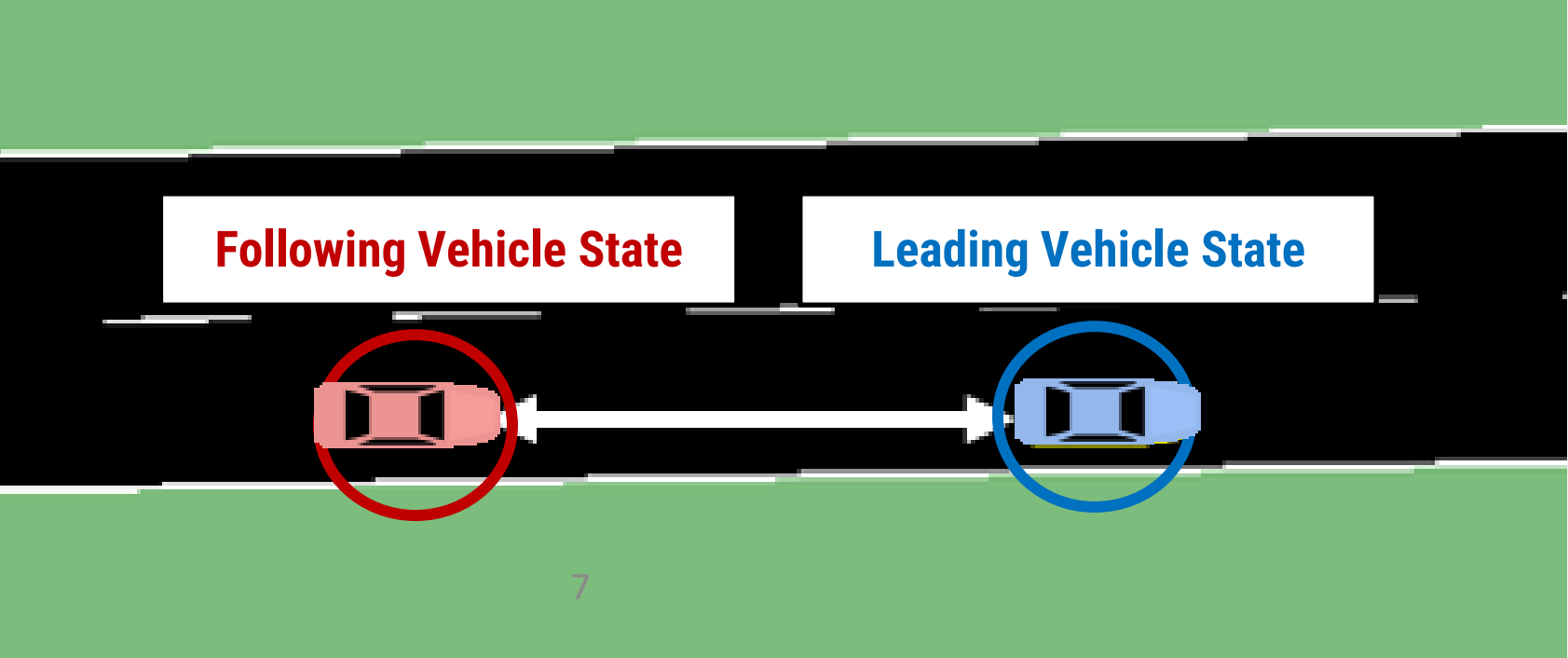
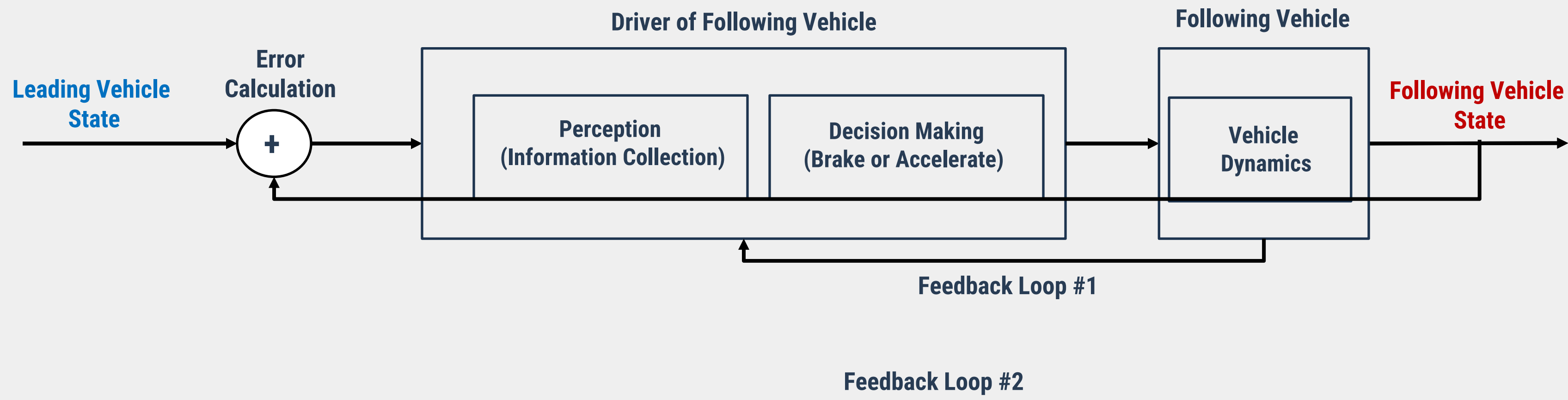


Simulation





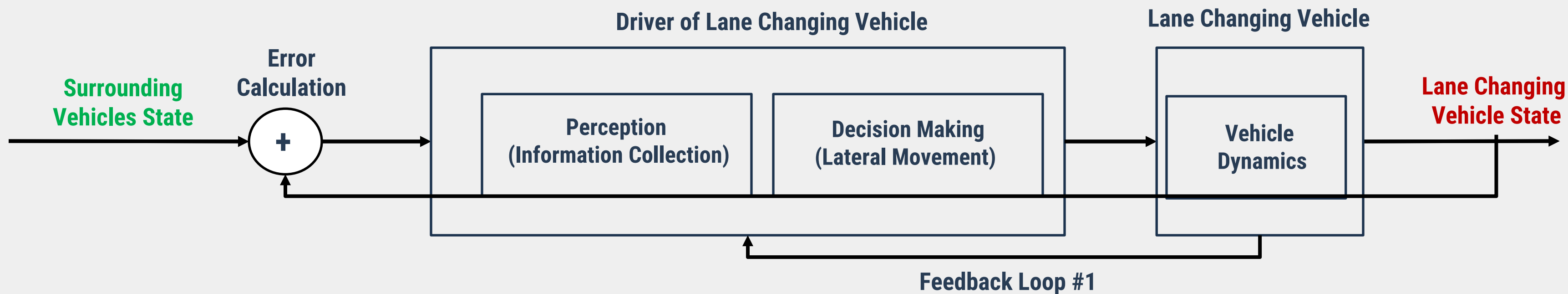
Car Following Model Flow Chart



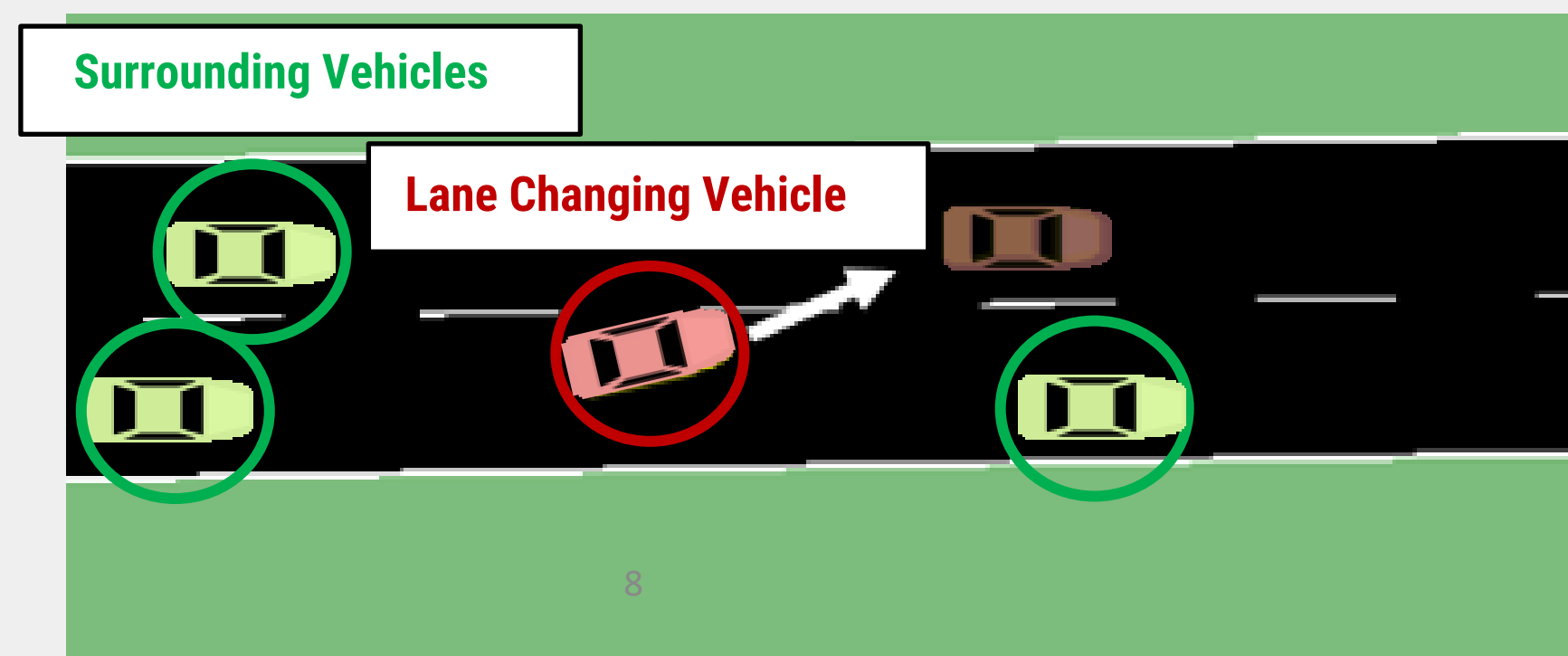
Reference:
Decision Making Flow in Car Following and Lane Changing Model, Rothery (2001)



Lane Changing Model Flow Chart

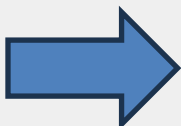


Feedback Loop #2





Vehicle Characteristics & Dynamics



Vehicle Type attributes			
vClass		guiShape	
 passenger		passenger	
id	DEFAULT_VEHTYPE	probability	1.00
 color	240,236,236,0	personCapacity	4
length	4.60	containerCapacity	0
minGap	2.50	boardingDuration	0.50
maxSpeed	55.56	loadingDuration	90.00
desiredMaxSpeed	2777.78	latAlignment	center
speedFactor	normc(1.00,0.10,0.20,2.00)	minGapLat	0.12
emissionClass	HBEFA3/PC_G_EU4	maxSpeedLat	1.00
width	1.80	actionStepLength	0.00
height	1.60	carriageLength	-1.00
imgFile		locomotiveLength	-1.00
osgFile	car-normal-citrus.obj	carriageGap	1
laneChangeModel	default	Edit parameters	



Vehicle Characteristics & Dynamics Parameter

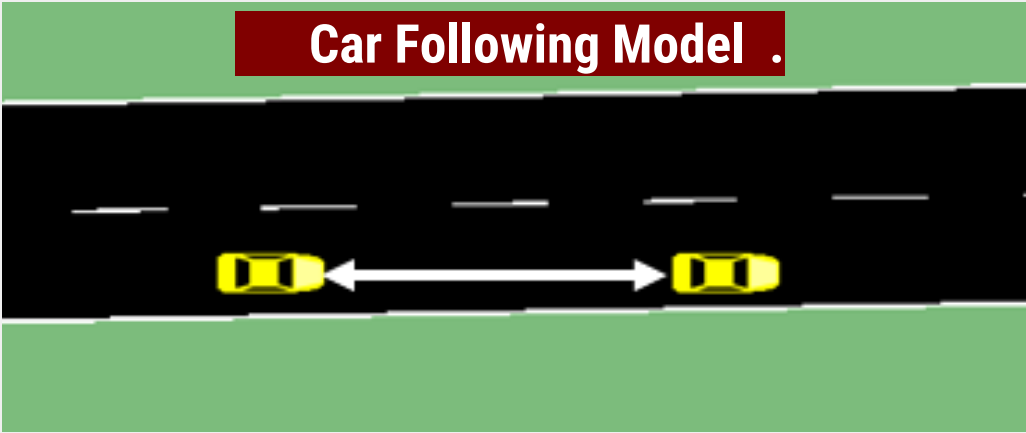
Passenger Cars and Light Delivery													
vClass	example	guiShape	length width height	mass	minGap	accel	decel	emer gency Decel	maxSpeed	desired MaxSpeed	person Capacity	emission Class HBEFA3	speed Dev
passenger		passenger 	5 ⁽²⁹⁾ 1.8m ⁽²⁹⁾ 1.5m ⁽²⁹⁾	1500kg	2.5m	2.6m/s ²⁽²⁹⁾	4.5m/s ²⁽²⁷⁾	9m/s ²	200km/h ⁽²⁹⁾	-	4	PC_G_EU4	0.1
taxi		taxi 	5 ⁽²⁹⁾ 1.8m ⁽²⁹⁾ 1.5m ⁽²⁹⁾	1500kg	2.5m	2.6m/s ²⁽²⁹⁾	4.5m/s ²⁽²⁷⁾	9m/s ²	200km/h ⁽²⁹⁾	-	4	PC_G_EU4	0.05
evehicle		evehicle 	5 ⁽²⁹⁾ 1.8m ⁽²⁹⁾ 1.5m ⁽²⁹⁾	1500kg	2.5m	2.6m/s ²⁽²⁹⁾	4.5m/s ²⁽²⁷⁾	9m/s ²	200km/h ⁽²⁹⁾	-	4	zero	0.1
emergency		emergency 	6.5m ⁽³⁷⁾ 2.16m ⁽³⁷⁾ 2.86m ⁽³⁷⁾	5000kg	2.5m	2.6m/s ²⁽²⁹⁾	4.5m/s ²⁽²⁷⁾	9m/s ²	200km/h ⁽²⁹⁾	-	3	LDV	0
delivery		delivery 	6.5m ⁽³⁷⁾ 2.16m ⁽³⁷⁾ 2.86m ⁽³⁷⁾	5000kg	2.5m	2.6m/s ²⁽²⁹⁾	4.5m/s ²⁽²⁷⁾	9m/s ²	200km/h ⁽²⁹⁾	-	3	LDV	0.05



Car Following Model Parameters

Car Following Model attributes	
Algorithm	Krauss
accel	2.60
decel	4.50
apparentDecel	4.50
emergencyDecel	9.00
sigma	0.50
tau	1.00

- Algorithm Type
- The maximum acceleration a vehicle can apply (in m/s^2)
- The maximum deceleration a vehicle can apply (in m/s^2).
- The braking deceleration (in m/s^2) of leading vehicle perceived by following vehicles
- The maximum deceleration ability of vehicles of this type in case of emergency (in m/s^2)
- The driver imperfection (0 denotes perfect driving, 0.5 human like reaction, 0.7-1 aggressive driving))
- The driver's desired (minimum) time headway.





Lane Changing Parameters

Lane Change Model attributes

strategic	1.0
cooperative	1.0
speedGain	1.0
keepRight	1.0
sublane	1.0
opposite	1.0
pushy	0.00
pushyGap	0.00
assertive	1.0
impatience	0.00
timeToImpatience	infinity
accelLat	1.0
lookaheadLeft	2.0
speedGainRight	0.1
maxSpeedLatStanding	0.00
maxSpeedLatFactor	1.00
turnAlignDistance	0.00
overtakeRight	0.00
keepRightAcceptanceTime	-1
overtakeDeltaSpeedFactor	0.00

- The eagerness for performing strategic lane changing. Higher values result in earlier lane-changing
- The willingness for performing cooperative lane changing. Lower values result in reduced cooperation
- The eagerness for performing lane changing to gain speed. Higher values result in more lane-changing.
- The eagerness for following the obligation to keep right. Higher values result in earlier lane-changing

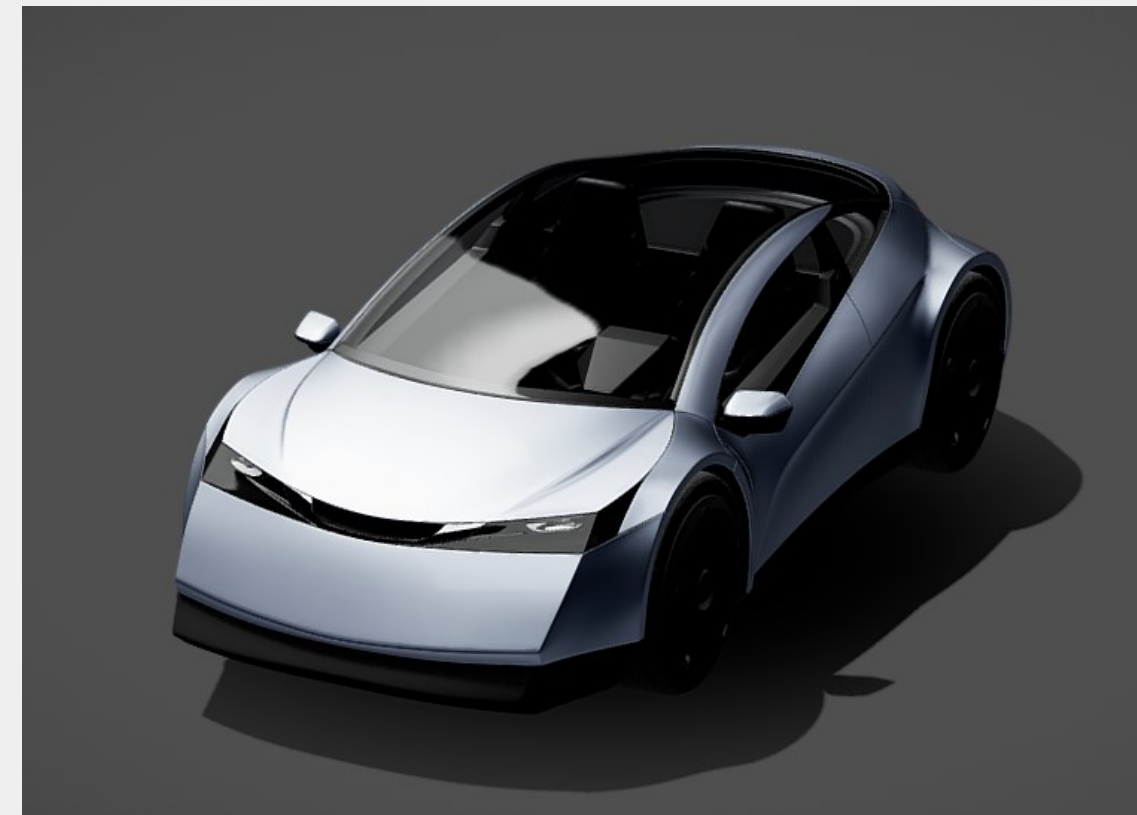


Human-Driven Vehicle vs Autonomous Vehicle

Human-Driven Vehicle (HDV)



Autonomous Vehicle (AV)



Lecture Reference Slides

What Makes a Vehicle Real in Simulation?

🕒 1:00

Think about a real vehicle you see every day - what physical and behavioral characteristics would you need to capture to represent it accurately in a traffic simulation?



List your characteristics here.

✍️ Walk through the next slides to learn about this concept

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What Must a Digital Traffic Signal Include?

If you were building a digital replica of a real traffic signal for a traffic simulation - what parts of the real system would you need to represent?

🕒 3:00



List the components here.

1. _____ 4. _____

2. _____ 5. _____

3. _____

✍️ Walk through the next slides to learn about this concept

SMRTTECH 4SC3 - Smart Cities and Communities.



CAV - Technology

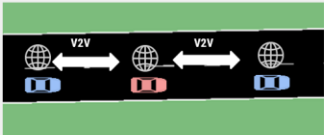
❑ **Cooperative Adaptive Cruise Control (CACC):** Each car has ACC and use vehicle-to-vehicle communication (V2V) to share speed and positioning data.

Benefits Over ACC:

❑ **Extended Awareness:** Sensors typically have a limited range, so a vehicle might only detect the one directly ahead. V2V communication allows vehicles to receive information about speed and acceleration from multiple vehicles further up the platoon, offering a broader view of traffic conditions.

❑ **Reduced Latency:** Communication between vehicles can deliver real-time data more quickly than sensors that need to process reflected signals, helping vehicles react faster to changes.

❑ **Enhanced Predictability:** By knowing the exact speed and intended maneuvers of preceding vehicles, a car can anticipate changes (like deceleration or lane changes) earlier, leading to smoother and safer adjustments.



4

What Changes When AVS Join the Road?

🕒 3:00

When autonomous vehicles are mixed with human-driven vehicles on the same road, what do you expect to change about how traffic behaves?



I expect these things to change:

1. _____

2. _____

3. _____

✍️ Walk through the next slides to learn about this concept



SMRTTECH 4SC3 Smart Cities and Communities

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Download Materials

1. Download SMRTTECH4SC3-Lab2.zip from (<https://github.com/RoadwayVR/SmartCities>) and extract it
2. Open SMRTTECH4SC3-Lab2a folder
3. Open Exercise2.netecfg

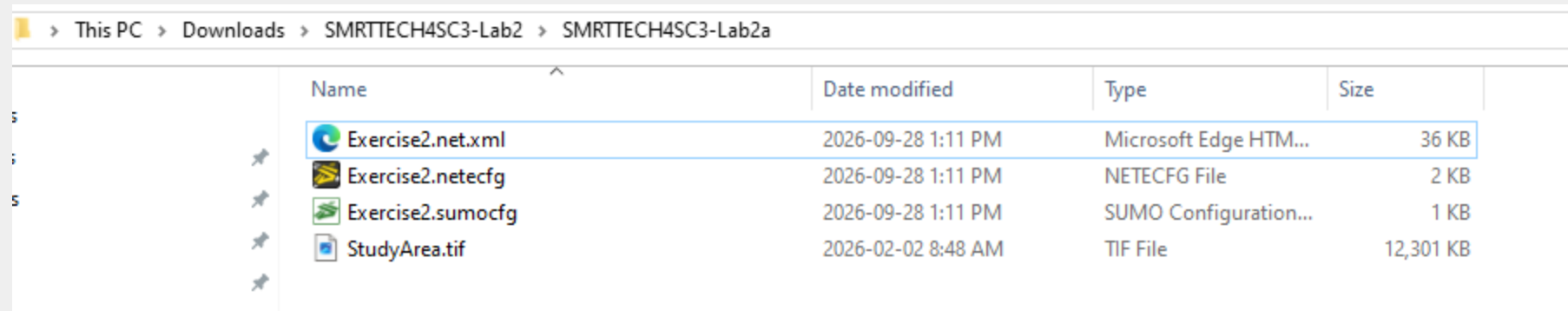
Download Materials

4. Assign the Background Map → See next slide for a tip about this



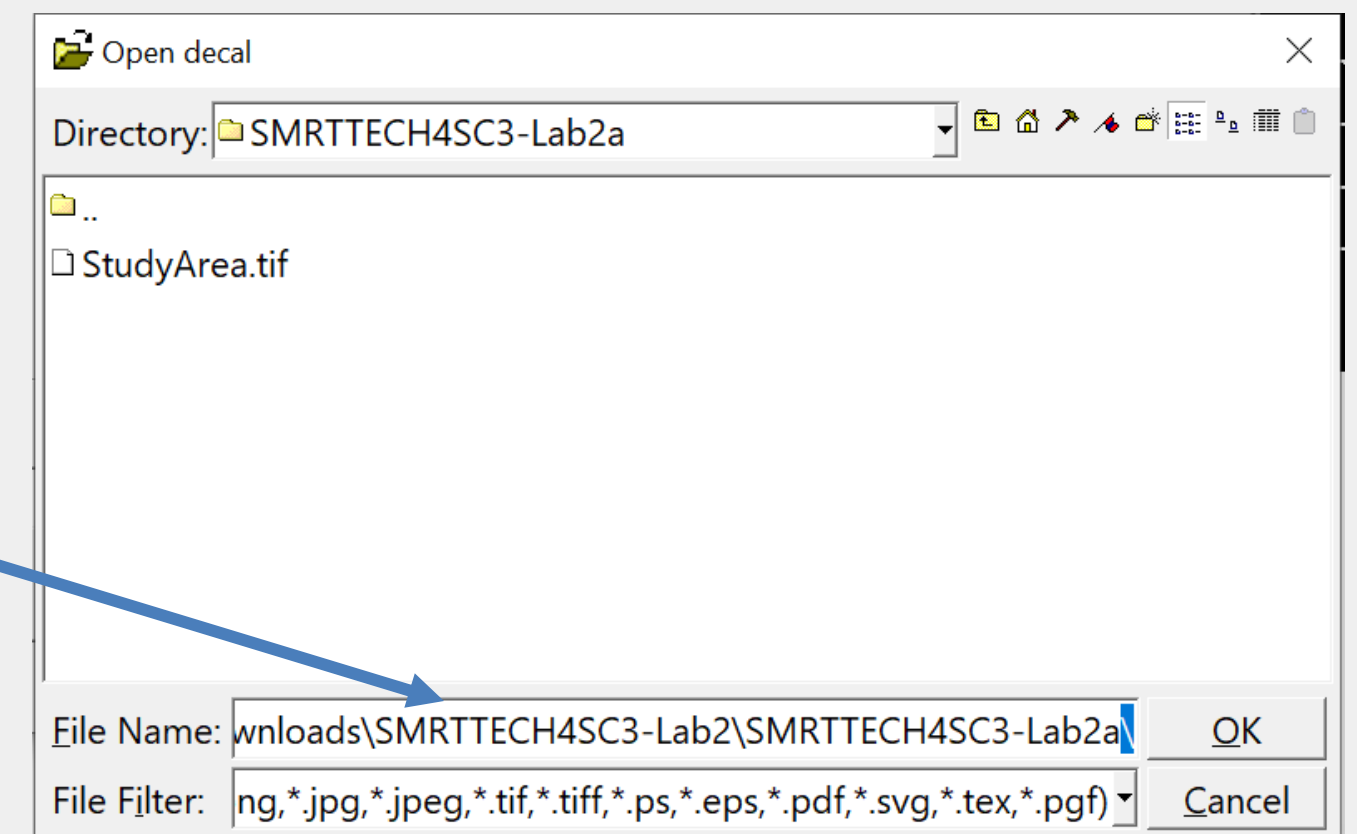
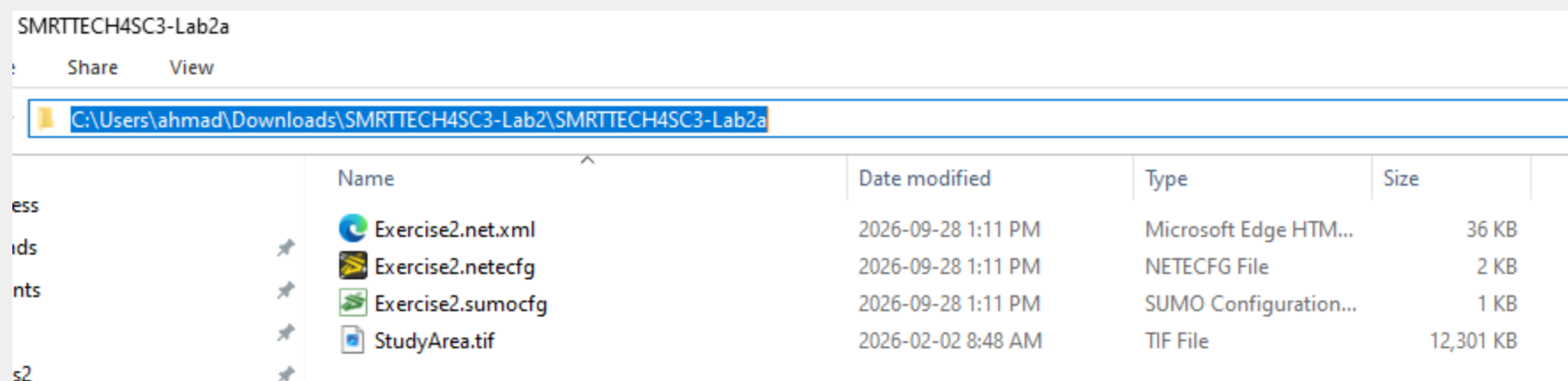
Tip about Finding Your Directory

- ☐ You usually need to save files or find image or find your directory in Netedit
- ☐ Finding your directory is a bit difficult in Netedit
- ☐ For example, I want to add the background GIS image



- ☐ Put the address in File name
- ☐ Add “\” at the end
- ☐ Select “Ok”
- ☐ It will go in your directory of interest → select image now and select ok

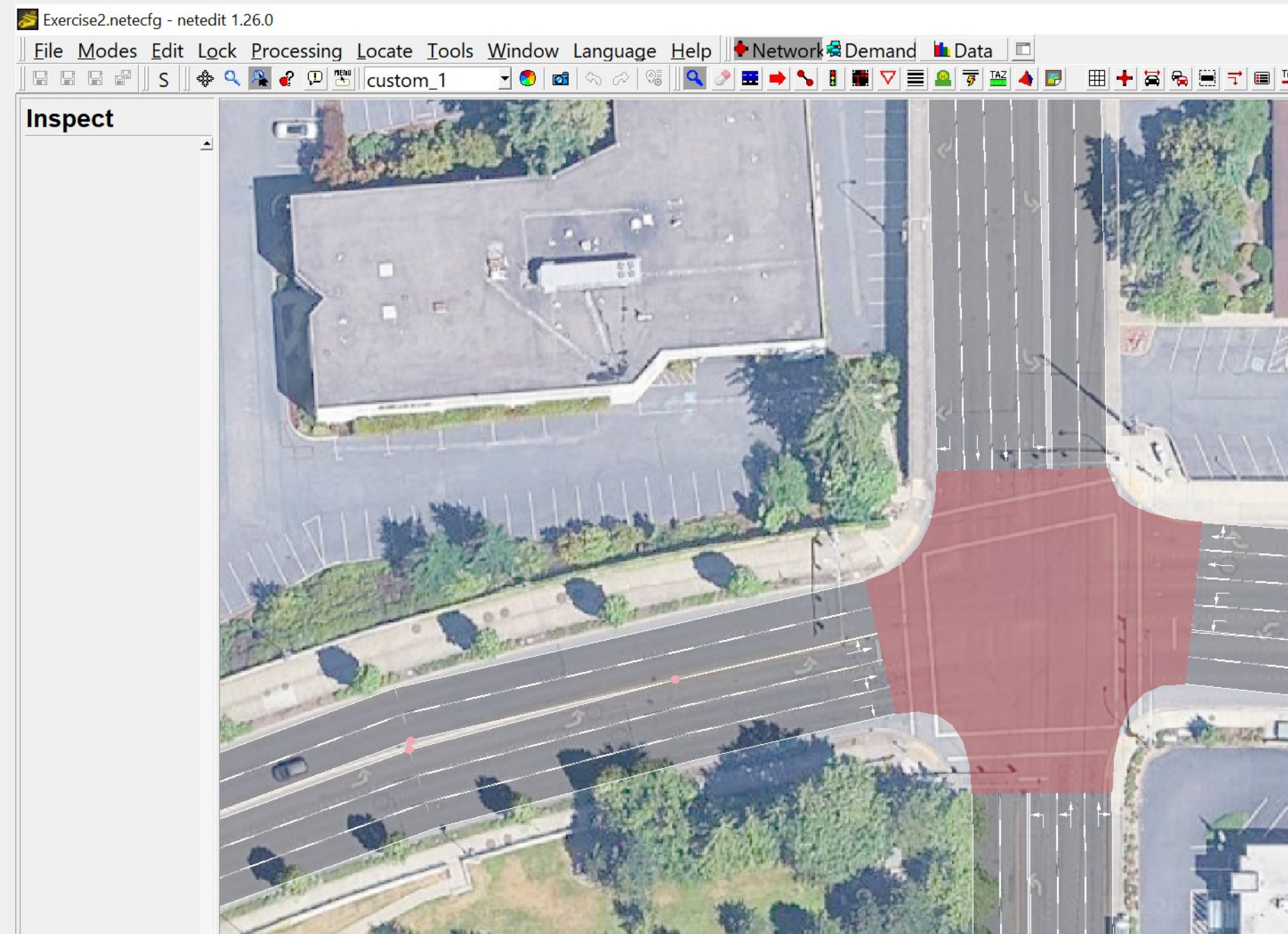
Select & Copy and Paste the directory address



Enlarge the Entire UI

❑ **Adjust transparency to roads and junctions (Important)**

❑ Netedit → Edit Visualization → Tab “OpenGL” → UI Scaling = 1.25 → Select “Ok” → Restart NetEdit



Milestone 1: Traffic Flow

Traffic Movement

(Read Carefully: You need to do this in your project)



Lane designations ordered from left to right by approach:

- ❑ Northbound Approach: NBL, NBT, NBR & NBT
- ❑ Southbound Approach (top approach, heading down):
- ❑ Southbound Approach: SBR & SBT, SBT, SBL & SBT
- ❑ Eastbound Approach: EBL, EBT, EBR & EBT
- ❑ Westbound Approach: WBR, WBT, WBL

Assumption:

The traffic flow of each traffic movement is 500 veh/h

Baseline Traffic Movement & Equal Distribution

(Read Carefully: You need to do this in your project)



Lane Mapping

Explicit lane-by-lane designations mapped across all 4 intersection approaches.

Uniform Flow Assumption

Initial baseline sets volume uniformly at **500 veh/h** per movement.

The Aggregation Challenge

Certain movements span multiple lanes (e.g., EBT split across two), requiring consolidation.

Simplifying Traffic Flow via Aggregation

(Read Carefully: You need to do this in your project)

1. Merge Multi-Lane Movements

Sum Volumes

Combine parallel lanes serving identical directional paths into a single unified volume node.

2. Consolidate Eastbound Through (EBT)

1000 veh/h

Merge dual EBT streams (500 + 500) into a single aggregate flow of **1,000 veh/h**.

3. Repeat for all Traffic Flow

Eastbound Aggregation Example

BEFORE AGGREGATION

EBT (Lane 1): 500 veh/h | EBT (Lane 2): 500 veh/h



AFTER AGGREGATION

EBT Total Volume = 1,000 veh/h

Simplifying Traffic Flow via Aggregation

(Read Carefully: You need to do this in your project)

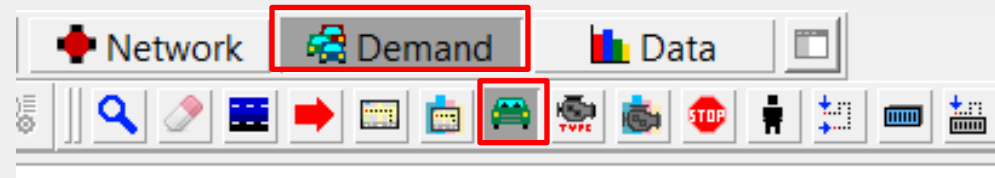


- ❑ Aggregate the values of traffic movement for each movement
- ❑ EBT has two Traffic Movement
→ make it one and aggregate traffic flow

Assign Traffic Flows (Demand) in Netedit

(Do Carefully)

❑ Add f_5 flow (demand)



Vehicles

▲ Vehicles

flow (from-to edges)

Parent vType

DEFAULT_VEHTYPE

Internal attributes

id f_0

color yellow

departLane first

departPos base

departSpeed 0

arrivalLane current

arrivalPos max

arrivalSpeed current

line

personNumber 0

containerNumber 0

departPosLat center

arrivalPosLat center

insertionChecks all

begin 0.00

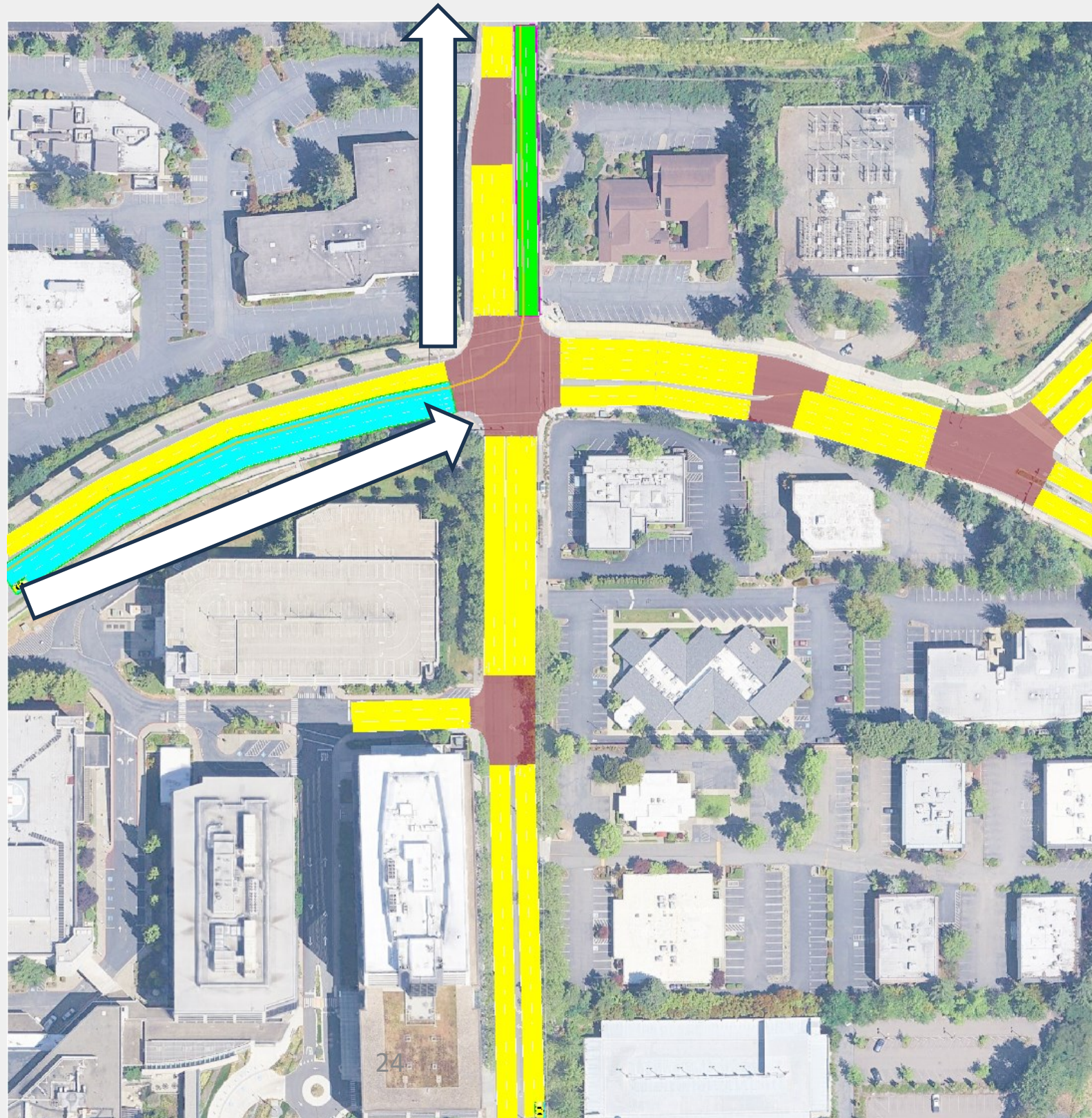
Flow attributes

terminate end

spacing 1sPerHour

end 3600.00

vehsPerHour 500



Vehicles

▲ Vehicles

flow (from-to edges)

Parent vType

DEFAULT_VEHTYPE

Internal attributes

id f_0

color yellow

departLane first

departPos base

departSpeed 0

arrivalLane current

arrivalPos max

arrivalSpeed current

line

personNumber 0

containerNumber 0

departPosLat center

arrivalPosLat center

insertionChecks all

begin 0.00

Flow attributes

terminate end

spacing 1sPerHour

end 3600.00

vehsPerHour 500

Parameters

Edit parameters

Netedit attributes

route file :ise2.rou.xml

Help

Route creator

- Selected edges: 2
- Path edges: 2
- Length: 243.04
- Average speed: 13.89

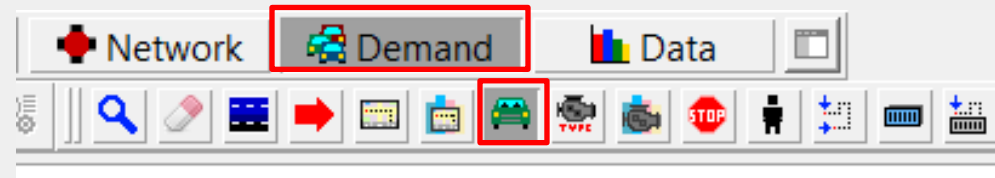
Finish route creation

Abort route creation

Remove last edge

Assign Traffic Flows (Demand) in Netedit

❑ Add f_4 flow (demand)



Vehicles

Vehicles

flow (from-to edges)

Parent vType

DEFAULT_VEHTYPE

Internal attributes

id f_1

color yellow

departLane first

departPos base

departSpeed 0

arrivalLane current

arrivalPos max

arrivalSpeed current

line

personNumber 0

containerNumber 0

departPosLat center

arrivalPosLat center

insertionChecks all

begin 0.00

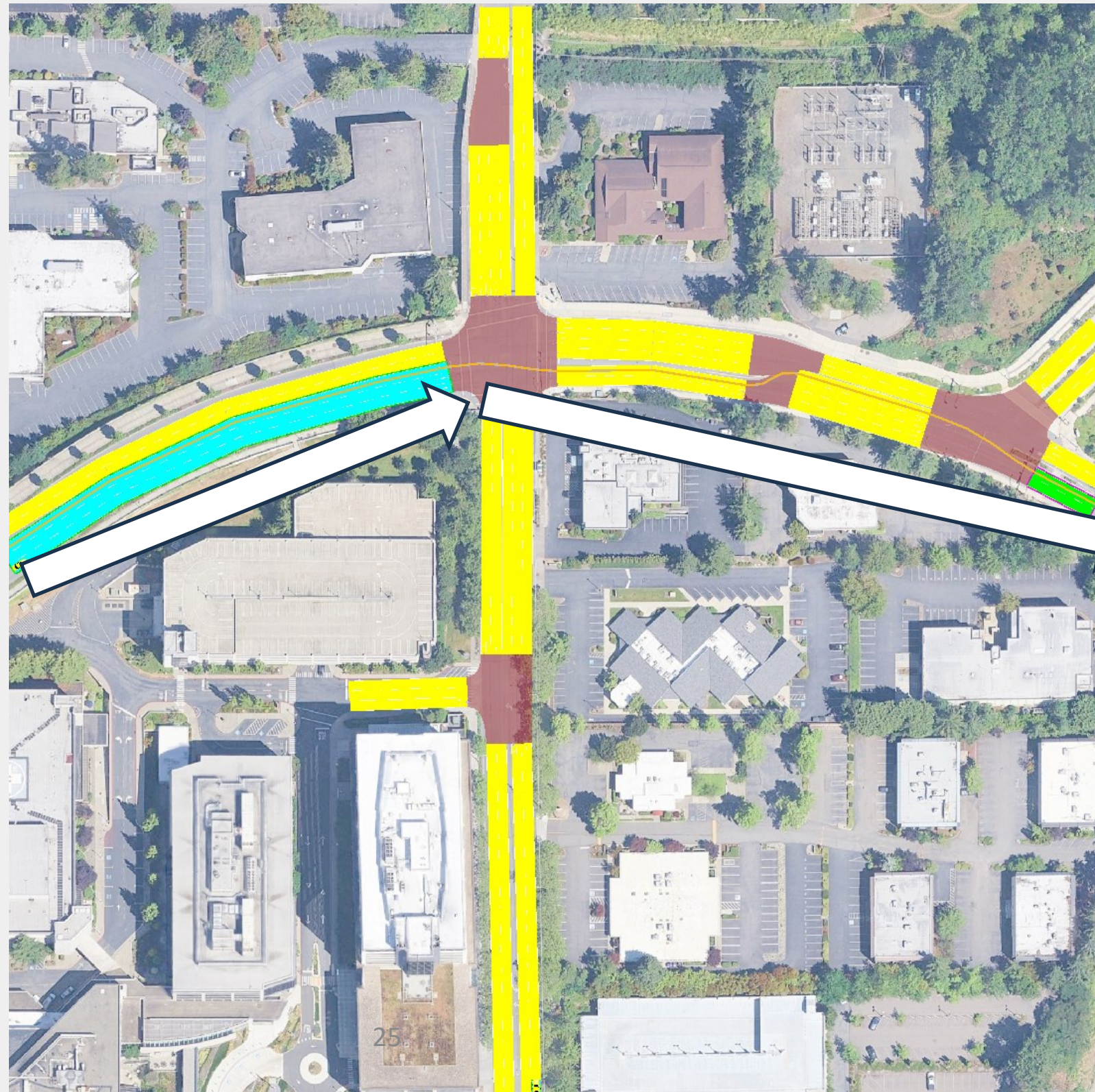
Flow attributes

terminate end

spacing sPerHour

end 3600.00

vehsPerHour 1000



Vehicles

Vehicles

flow (from-to edges)

Parent vType

DEFAULT_VEHTYPE

Internal attributes

id f_1

color yellow

departLane first

departPos base

departSpeed 0

arrivalLane current

arrivalPos max

arrivalSpeed current

line

personNumber 0

containerNumber 0

departPosLat center

arrivalPosLat center

insertionChecks all

begin 0.00

Flow attributes

terminate end

spacing sPerHour

end 3600.00

vehsPerHour 1000

Parameters

Edit parameters

Netedit attributes

route file rise2.rou.xml

Help

Route creator

- Selected edges: 2
- Path edges: 4
- Length: 276.82
- Average speed: 13.89

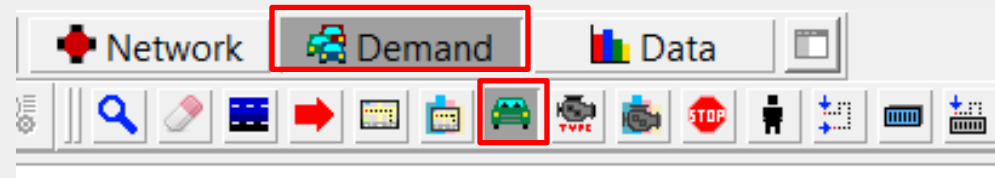
Finish route creation

Abort route creation

Remove last edge

Assign Traffic Flows (Demand) in Netedit

❑ Add f_3 flow (demand)



Vehicles

Vehicles
flow (from-to edges)

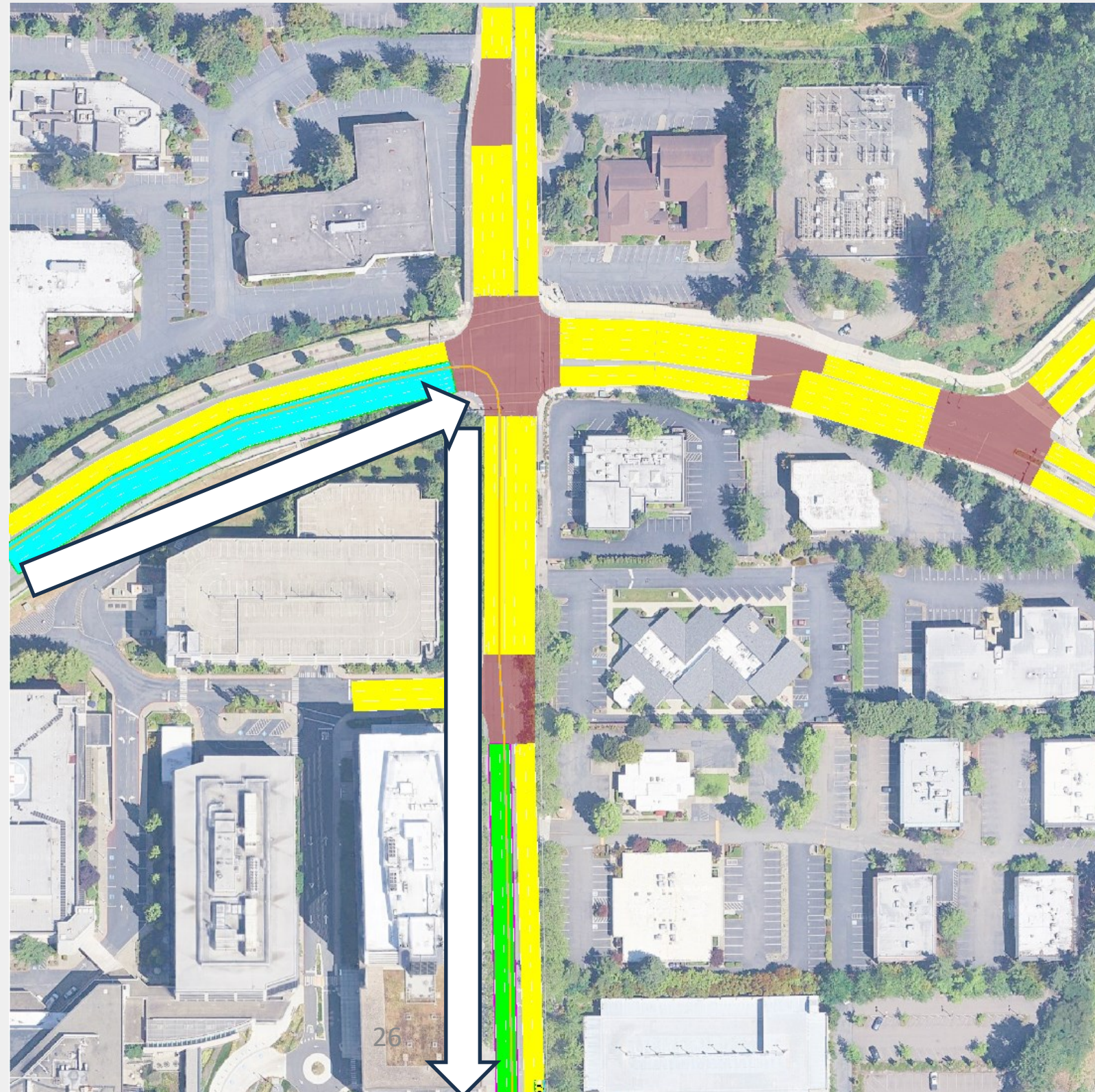
Parent vType
DEFAULT_VEHTYPE

Internal attributes

id	f_2
color	yellow
departLane	first
departPos	base
departSpeed	0
arrivalLane	current
arrivalPos	max
arrivalSpeed	current
line	
personNumber	0
containerNumber	0
departPosLat	center
arrivalPosLat	center
insertionChecks	all
begin	0.00

Flow attributes

terminate	end
spacing	rsPerHour
end	3600.00
vehsPerHour	500



Vehicles

Vehicles
flow (from-to edges)

Parent vType
DEFAULT_VEHTYPE

Internal attributes

id	f_2
color	yellow
departLane	first
departPos	base
departSpeed	0
arrivalLane	current
arrivalPos	max
arrivalSpeed	current
line	
personNumber	0
containerNumber	0
departPosLat	center
arrivalPosLat	center
insertionChecks	all
begin	0.00

Flow attributes

terminate	end
spacing	rsPerHour
end	3600.00
vehsPerHour	500

Parameters

Edit parameters

Netedit attributes

route file: ise2.rou.xml

Help

Route creator

- Selected edges: 2
- Path edges: 3
- Length: 338.55
- Average speed: 13.89

Finish route creation

Abort route creation

Remove last edge

Assign Traffic Flows (Demand) in Netedit

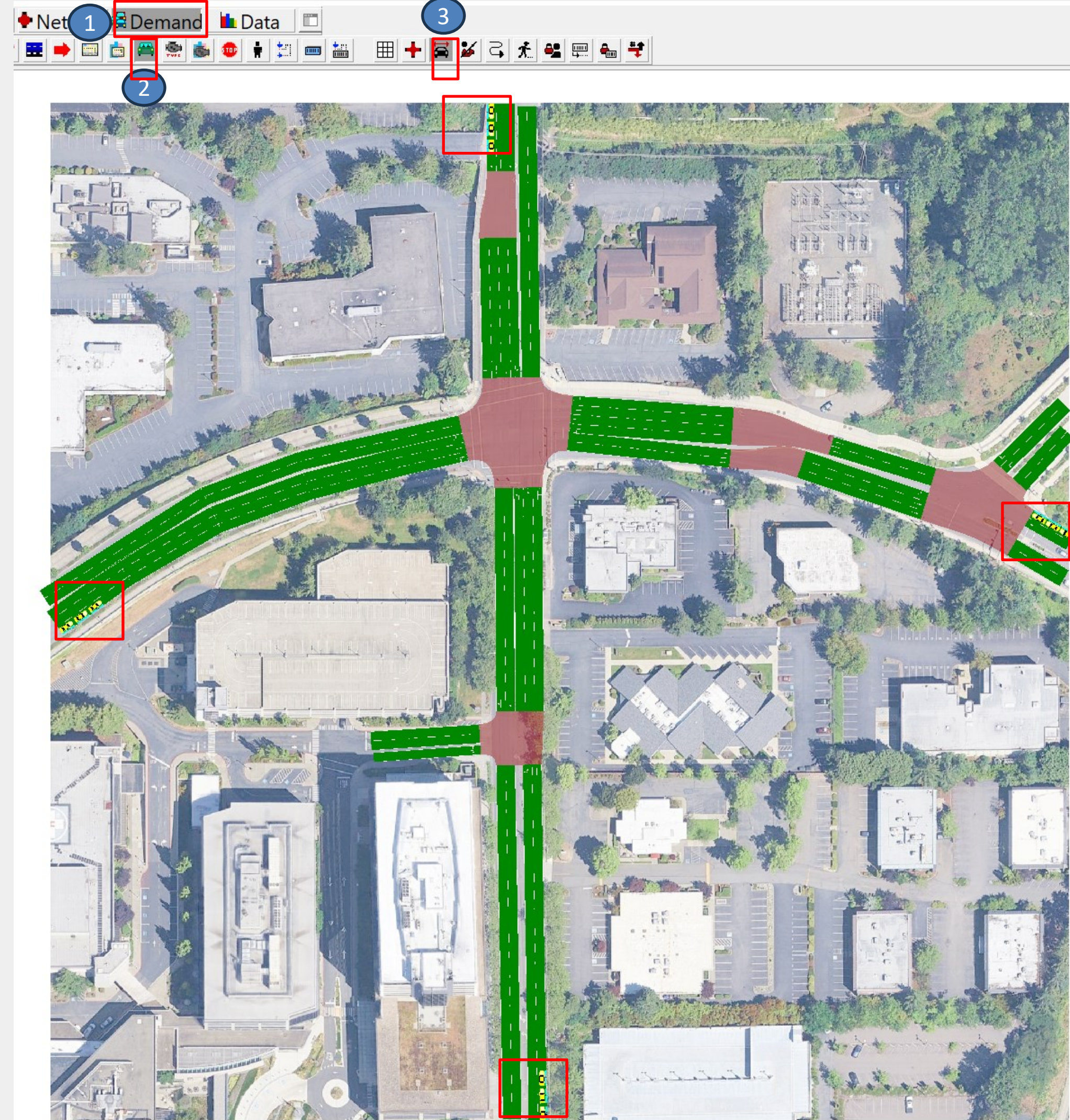
- ❑ Add f_3, f_4, and f_5 for the northbound approach (f_3 : NBL – f_4: NBT – f_5 : NBR): See previous slides for determining the traffic volume .
- ❑ Add f_6, f_7, and f_8 for the westbound approach (f_6 : WBL – f_7: WBT – f_8 : WBR)
- ❑ Add f_9, f_10, and f_11 for the southbound approach (f_9: WBL – f_10: WBT – f_11: WBR)

- ❑ Save all files



End of Milestone 1: Add Traffic Flow

- ☐ Select  and you should see three traffic flows added in each Direction
- ☐ Raise Your Hands for instructor Confirmation
- ☐ In the Meantime, Save and Run SUMO GUI to Observe Traffic Movements



Milestone 2: Traffic Signal

Traffic Signal

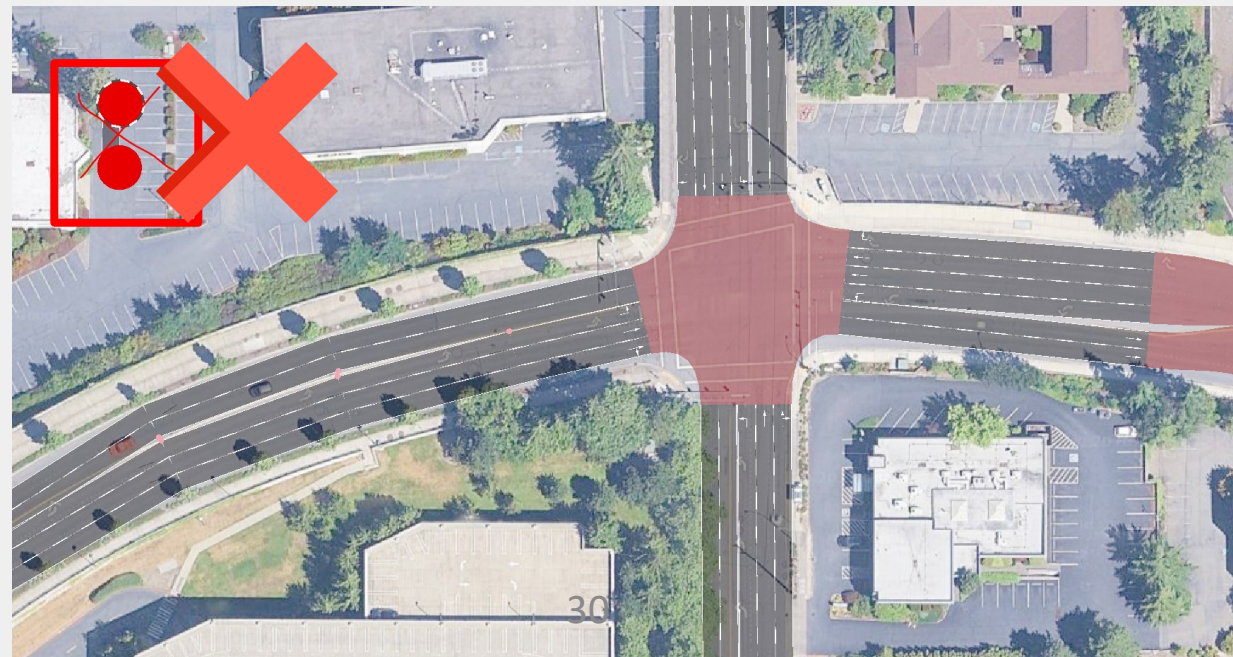
- ☐ Select Junction
- ☐ Make sure the name is “J1”



Net: junction

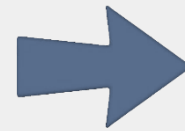
Internal attributes	
id	J1
pos	-16.02,60.58
type	affic_light
shape	4 -1.38,73.63
radius	default
keepClear	☒ true
rightOfWay	default
fringe	default
name	
tlType	static
tlLayout	default
tl	J33_#4more
isRoundabout	☐ false
Parameters	

- ☐ If you get an error because J1 already exists, remove the dummy junctions/roads using the Eraser tool.



Traffic Signal

❑ Select Traffic Signal



Edit Traffic Light

Traffic Light	
Junction ID	ion selected
TLS ID	
type	static
Join	Disjoin
Traffic Light Programs	
programID	
Create	Reset single
Delete	Reset all
Save	Cancel
Traffic Light Attributes	
offset	
parameters	
Assign E1 detectors	
Expand	Phases
Clean States	Add States
Group Sig.	Ungroup Sig.
TLS Program File	
Load	Save

Traffic Signal

Edit Traffic Light

▲ Traffic Light

Junction ID J33_#4more

TLS ID J33_#4more

type static ▼

Join Disjoin

▲ Traffic Light Programs

programID 0 ▼

Duplicate Reset single

Delete Reset all

Save Cancel

▲ Traffic Light Attributes

offset 0.00

parameters

Assign E1 detectors

▲ Expand Phases

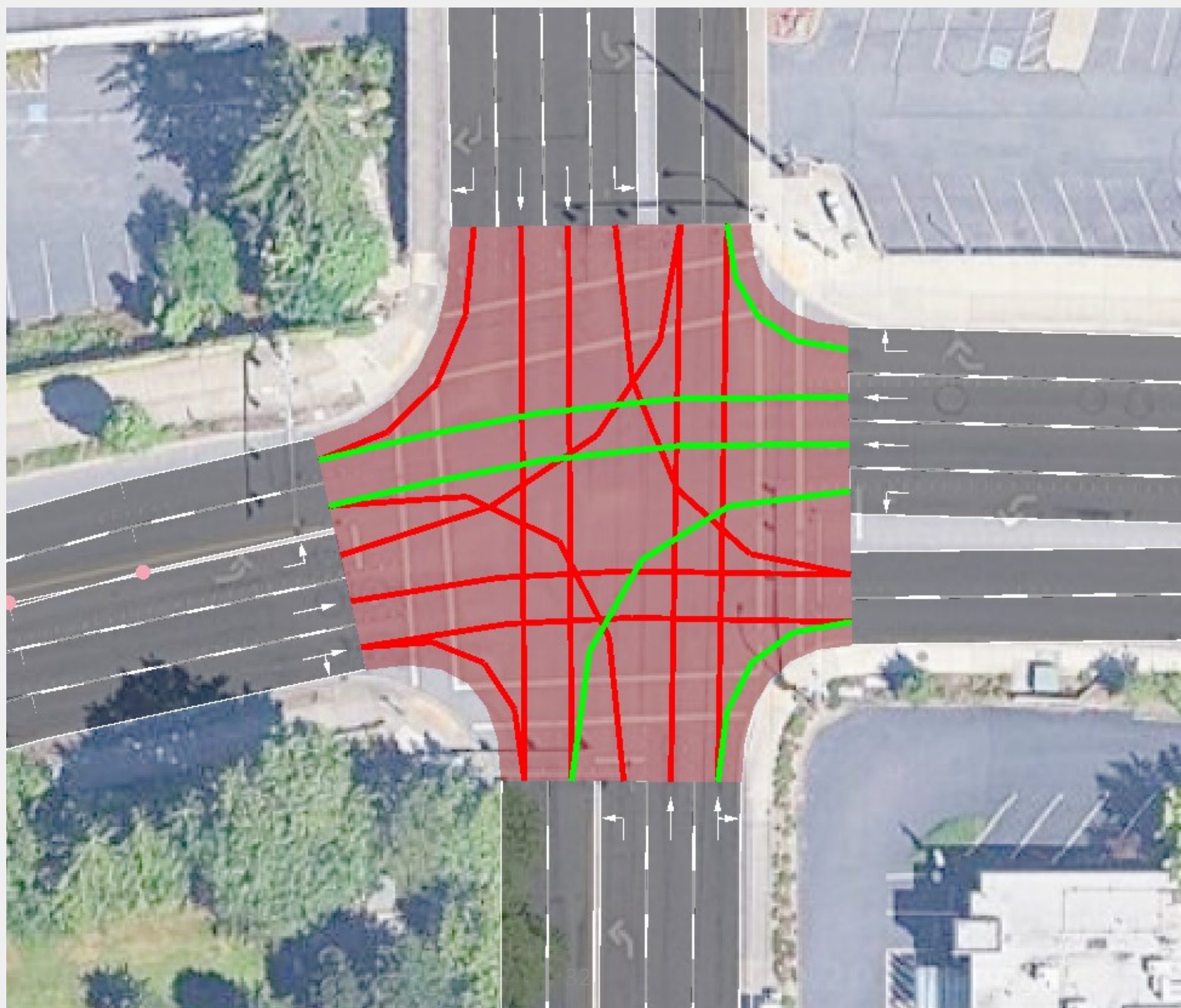
	dur	state
0	17.00	GGGGGrrrrrrr
1	3.00	Gyyyyrrrrrrr
2	20.00	Grrrrrrrrrrr
3	3.00	yrrrrrrrrrrr
4	20.00	rrrrrrrrGGG
5	3.00	rrrrrrrrGyy
6	20.00	rrrrGGGGGrr
7	3.00	rrrryyyyyrr
8	1.00	rrrrrrrrrrrr
Σ	90.00	Links: 16

Clean States Add States

Group Sig. Ungroup Sig.

▲ TLS Program File

Load Save



Traffic Signal

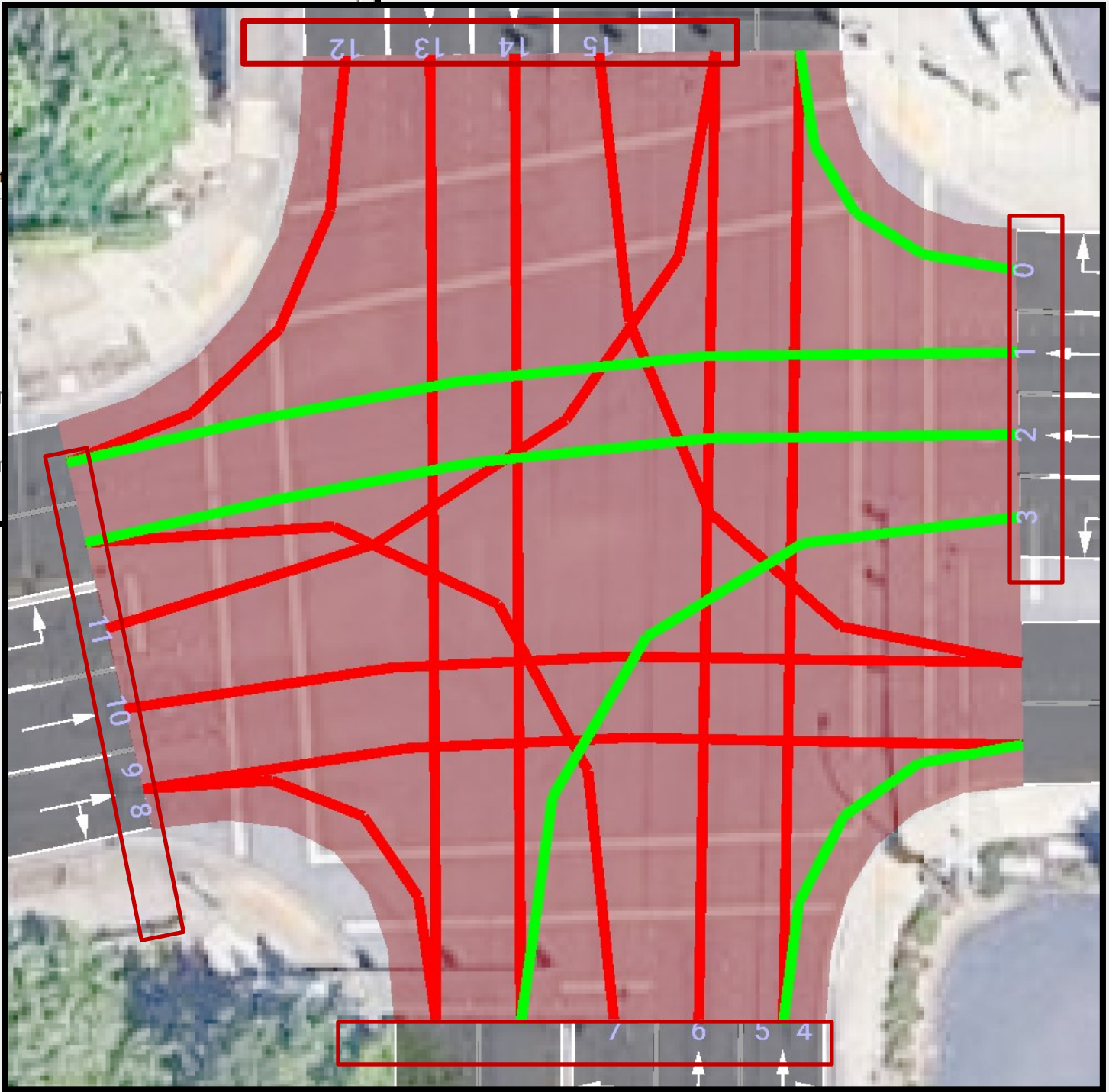
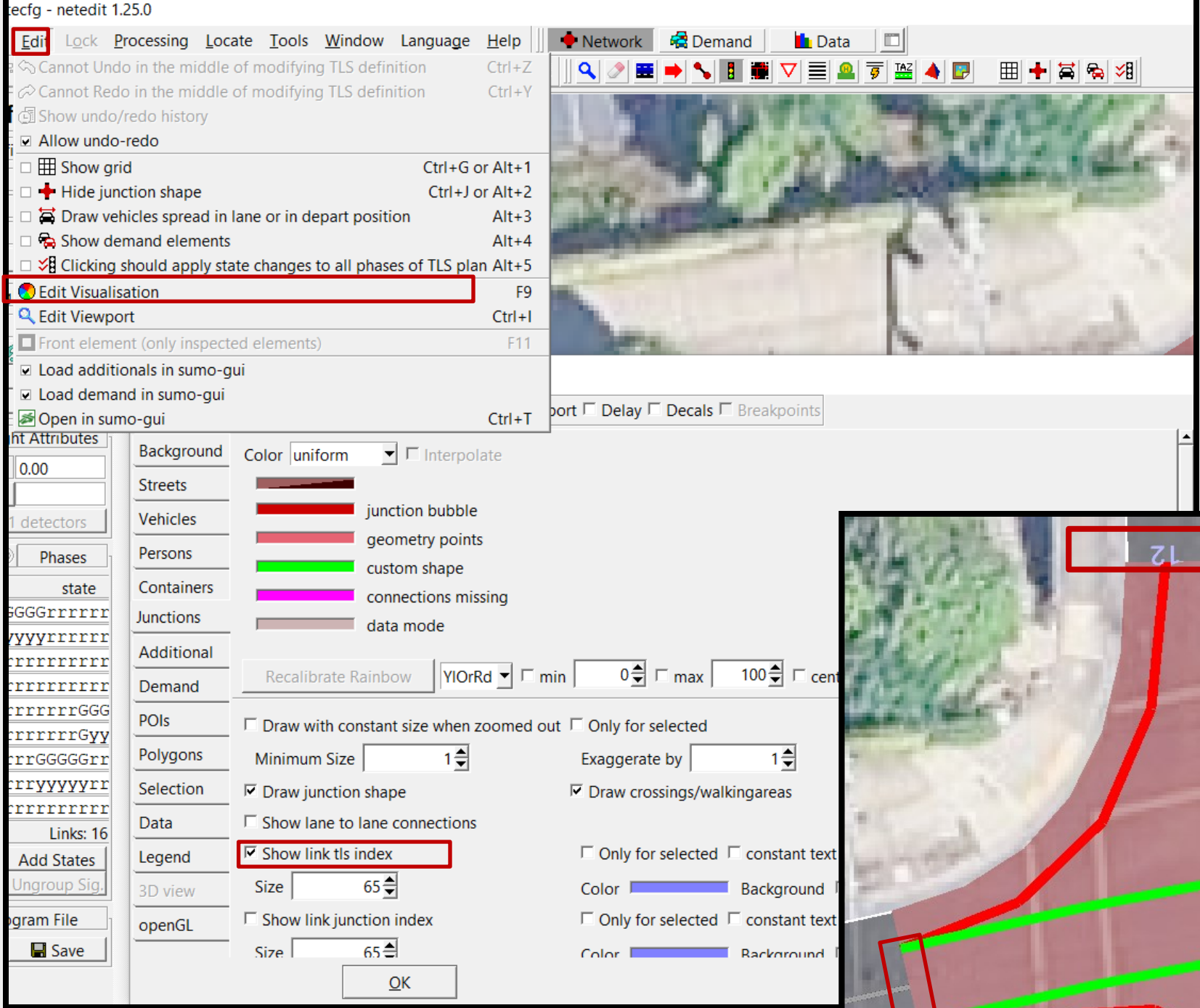
5. Netedit

6. Edit → Edit Visualization

7. Select Junction

8. Show link tls index

9. Do you see each traffic signal head number?



Traffic Signal

10. Review the phase table (left panel) in the “Edit Traffic Light” window:

- Phase 0: 17 s
- Phase 1: 3 s
- Phase 2: 20 s
- Phase 3: 3 s
- Phase 4: 20 s
- Phase 5: 3 s
- Phase 6: 20 s
- Phase 7: 3 s
- Phase 8: 1 s

11. State:

G = Green

y = means Yellow

r = means Red

Edit Traffic Light

Traffic Light

Junction ID

sterJ14_J15_J25_J33_#4more

TLS ID

sterJ14_J15_J25_J33_#4more

type

static

Join

Disjoin

Traffic Light Programs

programID

0

Duplicate

Reset single

Delete

Reset all

Save

Cancel

Traffic Light Attributes

offset

0.00

parameters

Assign E1 detectors

Expand

Phases

	dur	state	next
0	17.00	GGGGGrrrrrrrrrrrr	
1	3.00	Gyyyyrrrrrrrrrrrr	
2	20.00	GrrrrrrrrrrrrGGGG	
3	3.00	yrrrrrrrrrrrrGyyy	
4	20.00	rrrrrrrrGGGGGrrr	
5	3.00	rrrrrrrrGyyyyrrr	
6	20.00	rrrrGGGGGrrrrrrr	
7	3.00	rrrryyyyyrrrrrrr	
8	1.00	rrrrrrrrrrrrrrrr	
Σ	90.00	Links: 16	

Clean States

Add States

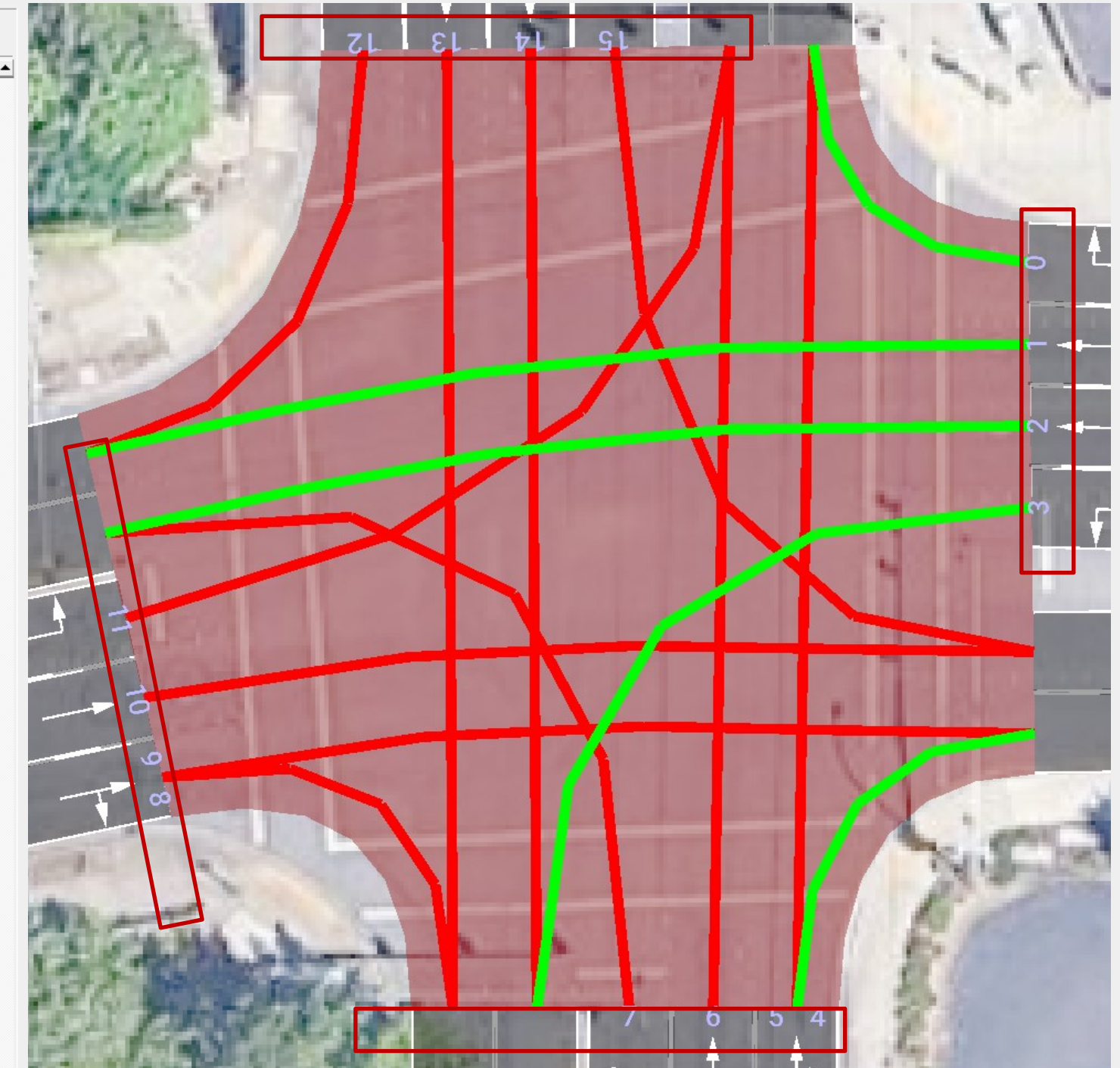
Group Sig.

Ungroup Sig.

TLS Program File

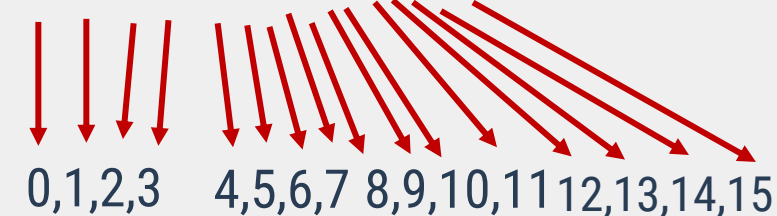
Load

Save



In phase 0:

GGGGGrrrrrrrrrrrr



In phase 1:

Gyyyyrrrrrrrrrrrr

0,1,2,3, 4,5,6,7,8,9,10,11,12,13,14,15

☐ Your phases and durations might be same or different, but the logic is the same.

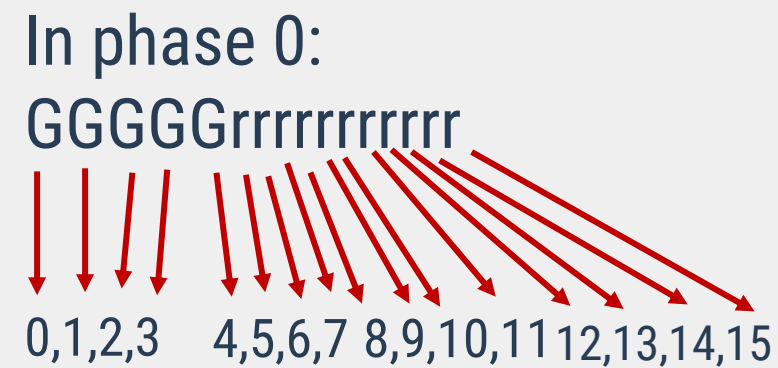
Deliverable 1

Deliverable 1 Folder:

- 1. Open HTML**
- 2. Respond to the Question**
- 3. Generate Final Submission and Submit in Avenue**

Deliverable 1

- Q1. What does it mean by each phase and?
duration?
- Q2: What it mean by each letter “G,r,y”?
- Q3: In phase 0, what does it mean by bellow letter
and number?



Edit Traffic Light

Traffic Light

Junction ID

sterJ14_J15_J25_J33_#4more

TLS ID

sterJ14_J15_J25_J33_#4more

type

static

Join

Disjoin

Traffic Light Programs

programID

0

Duplicate

Reset single

Delete

Reset all

Save

Cancel

Traffic Light Attributes

offset

0.00

parameters

Assign E1 detectors

Expand

Phases

	dur	state	next
0	17.00	GGGGGrrrrrrrrrrrr	
1	3.00	Gyyyyrrrrrrrrrrrr	
2	20.00	GrrrrrrrrrrrrGGGG	
3	3.00	yrrrrrrrrrrrrGyyy	
4	20.00	rrrrrrrrGGGGGrrrr	
5	3.00	rrrrrrrrGyyyyrrrr	
6	20.00	rrrrGGGGGrrrrrrrr	
7	3.00	rrrryyyyyrrrrrrrr	
8	1.00	rrrrrrrrrrrrrrrrrr	
Σ	90.00	Links: 16	

Clean States

Add States

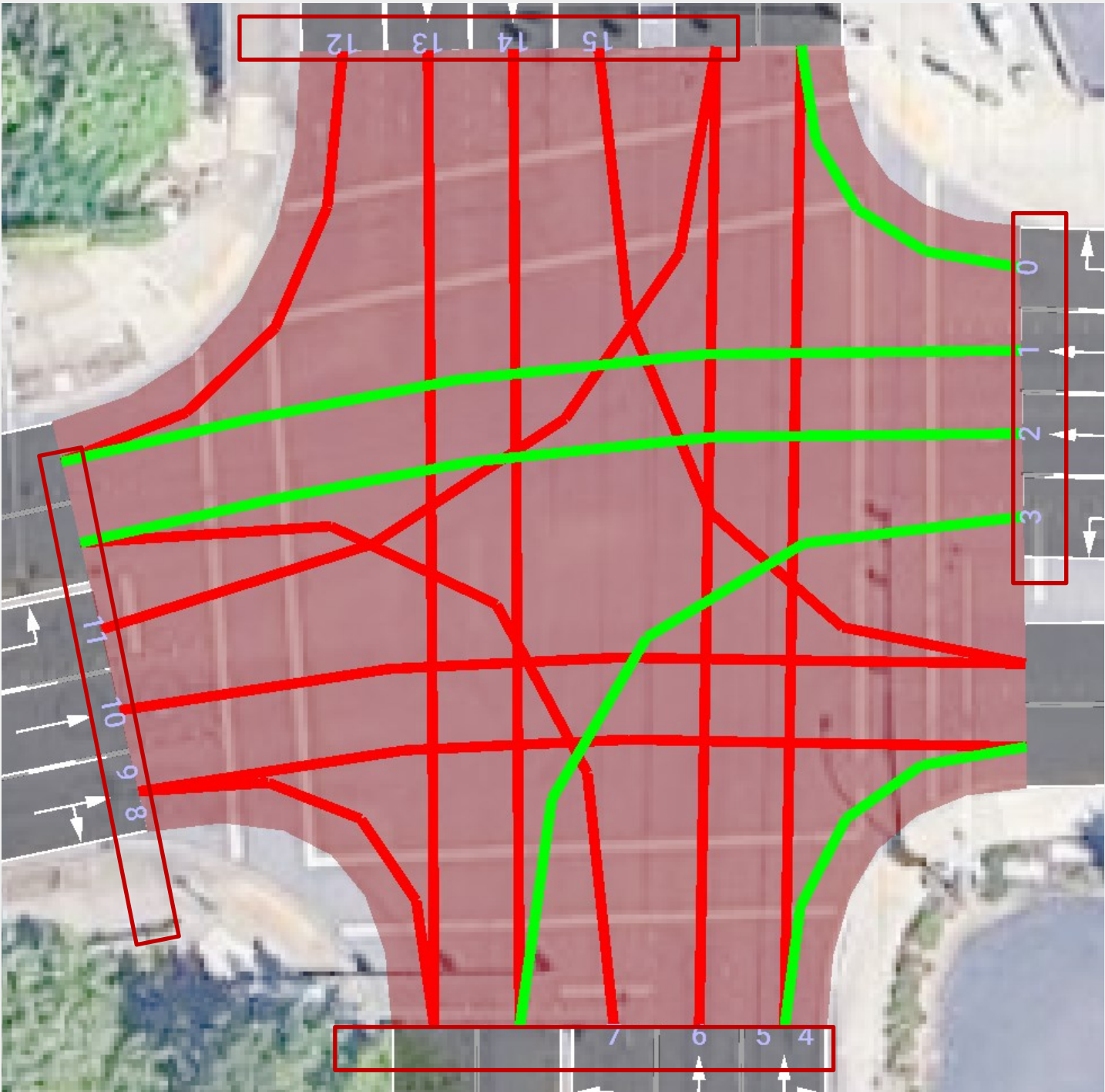
Group Sig.

Ungroup Sig.

TLS Program File

Load

Save



Milestone 2 (Continuing)

Traffic Signal

❑ Keep only four phases (Phases 0–3) →

Edit Traffic Light

Traffic Light

Junction ID: clusterJ14_J15_J25_J33_#4more

TLS ID: clusterJ14_J15_J25_J33_#4more

type: static

Join Disjoin

Traffic Light Programs

programID: 0

Duplicate Reset single

Delete Reset all

Save Cancel

Traffic Light Attributes

offset: 0.00

parameters

Assign E1 detectors

Expand Phases

	dur	state	next	name				
0	17.00	GGGGGrrrrrrrrrrrr			+	x	↑	↓
1	3.00	Gyyyyrrrrrrrrrrrr			+	x	↑	↓
2	20.00	GrrrrrrrrrrrrGGGG			+	x	↑	↓
3	3.00	yrrrrrrrrrrrrGyyy			+	x	↑	↓
4	20.00	rrrrrrrrGGGGGrrrr			+	x	↑	↓
5	3.00	rrrrrrrrGyyyyrrrr			+	x	↑	↓
6	20.00	rrrrGGGGGrrrrrrrr			+	x	↑	↓
7	3.00	rrrryyyyyyrrrrrrrr			+	x	↑	↓
8	1.00	rrrrrrrrrrrrrrrrrr			+	x	↑	↓
Σ	90.00	Links: 16						



❑ Delete Phases 4–8 using the red X

❑ Set the signal phases and values as the next slide

Edit Traffic Light

Traffic Light

Junction ID: clusterJ14_J15_J25_J33_#4more

TLS ID:

type: static

Join Disjoin

Traffic Light Programs

programID: 0

Duplicate Reset single

Delete Reset all

Save Cancel

Traffic Light Attributes

offset: 0.00

parameters

Assign E1 detectors

Expand Phases

	dur	state	next	name				
0	17.00	GGGGGrrrrrrrrrrrr			+	x	↑	↓
1	3.00	Gyyyyrrrrrrrrrrrr			+	x	↑	↓
2	20.00	GrrrrrrrrrrrrGGGG			+	x	↑	↓
3	3.00	yrrrrrrrrrrrrGyyy			+	x	↑	↓
Σ	43.00	Links: 16						

Clean States Add States

Group Sig. Ungroup Sig.

TLS Program File

Load Save

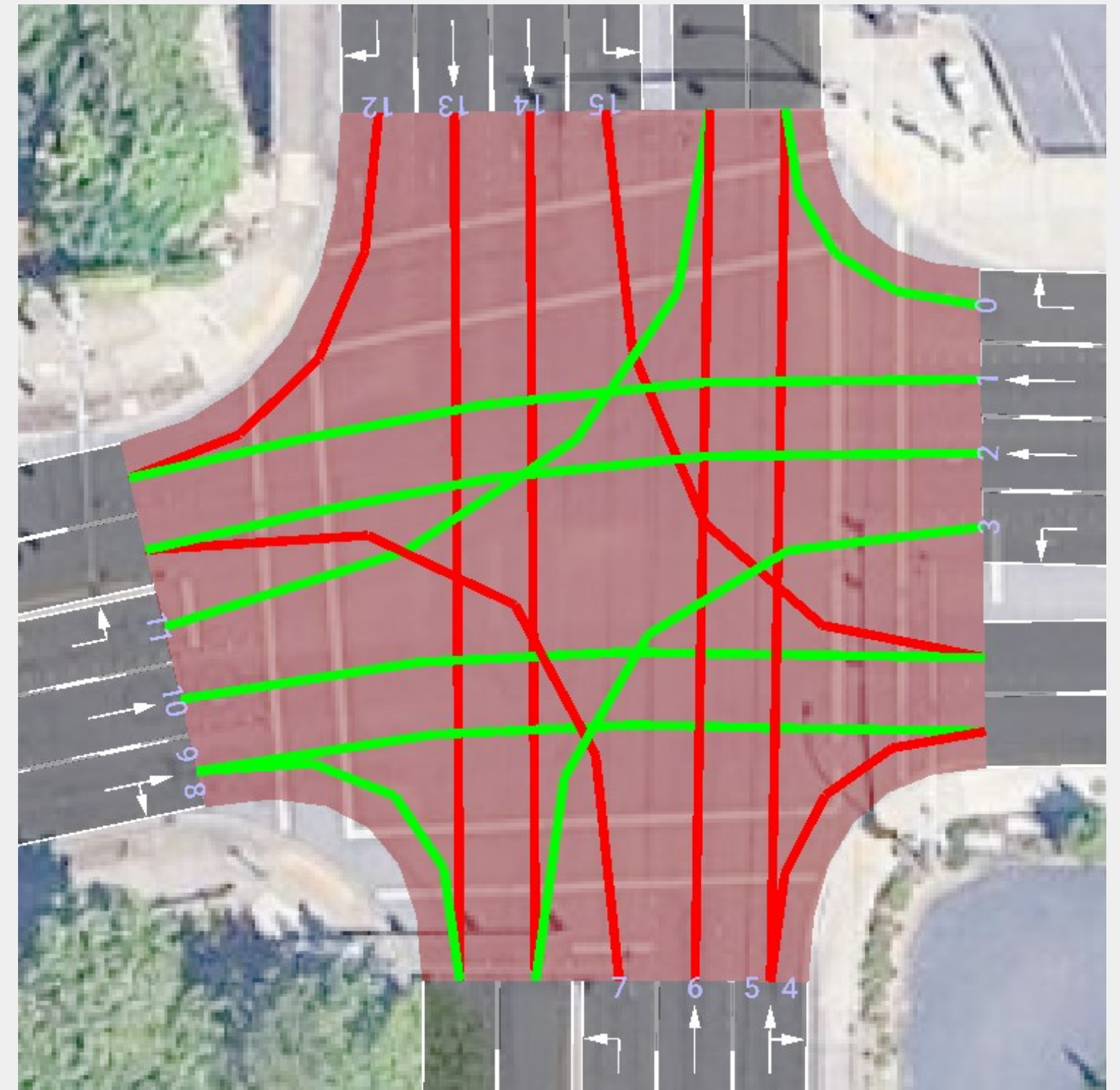
Traffic Signal

☐ Set the signal phases as follows:

Phase 0: Eastbound + Westbound movements - Green

☐ Duration and state should be same as below

Phases					
	dur	state	next	name	
0	27.00	GGGGrrrrGGGGrrrr			+ x ↑ ↓
1					+ x ↑ ↓
2					+ x ↑ ↓
3					+ x ↑ ↓
Σ	60.00	Links: 16			

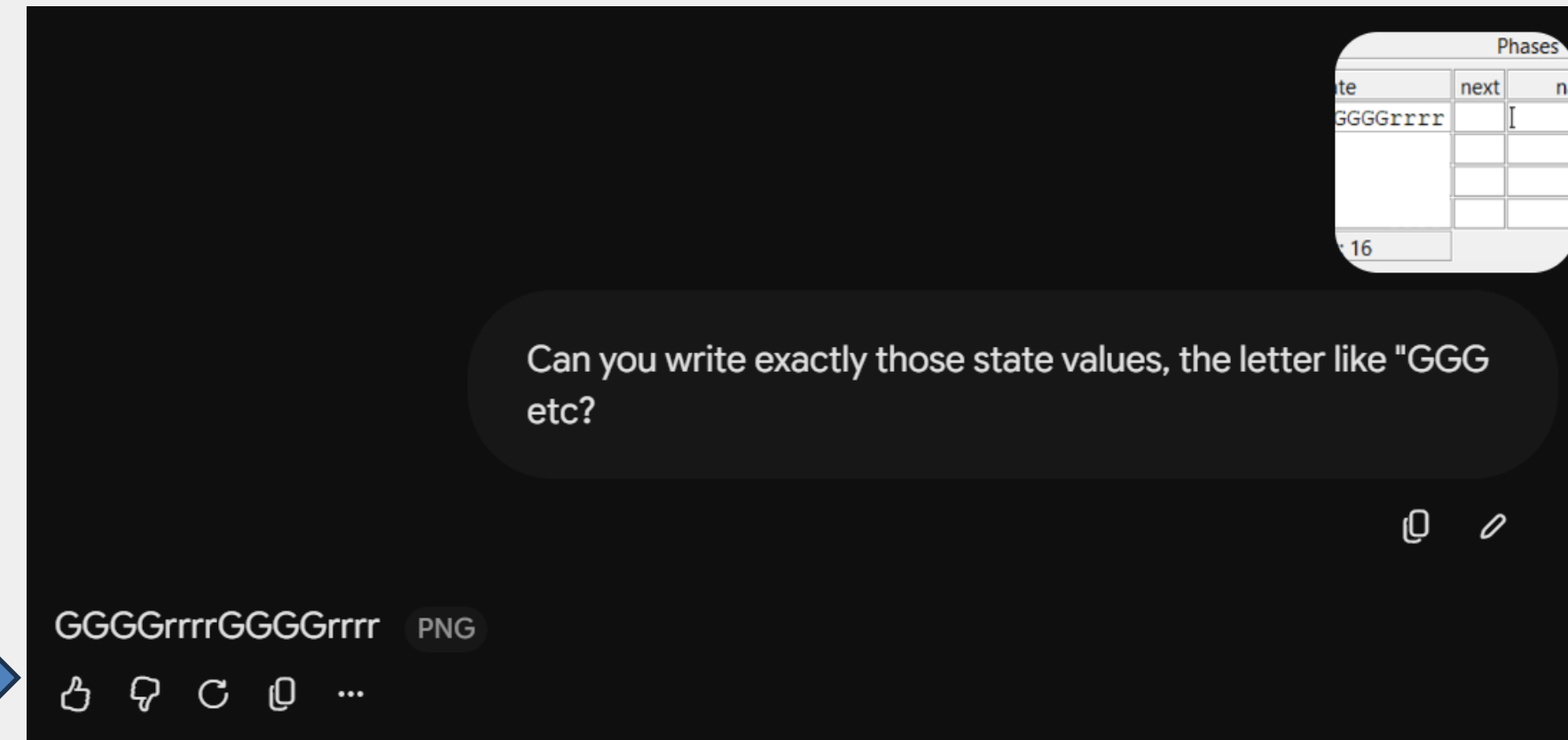


☐ After setting up, your signal colors should look like the image

Use of Generative AI

- ❑ When you have an image with so many texts that you need to rewrite it manually → Just open ChatGPT or any AI (Sign UP) → Give the image → Ask to write exactly the texts

Expand ↻ Phases					
	dur	state	next	name	
0	27.00	GGGGrrrrGGGGrrrr		I	⬆ ⬇ ⬆ ⬇
1					⬆ ⬇ ⬆ ⬇
2					⬆ ⬇ ⬆ ⬇
3					⬆ ⬇ ⬆ ⬇
Σ	60.00	Links: 16			



- ❑ Don't Trust AI, always visually check
- ❑ Then, copy and paste those 😊
- ❑ "GGGGrrrrGGGGrrrr"

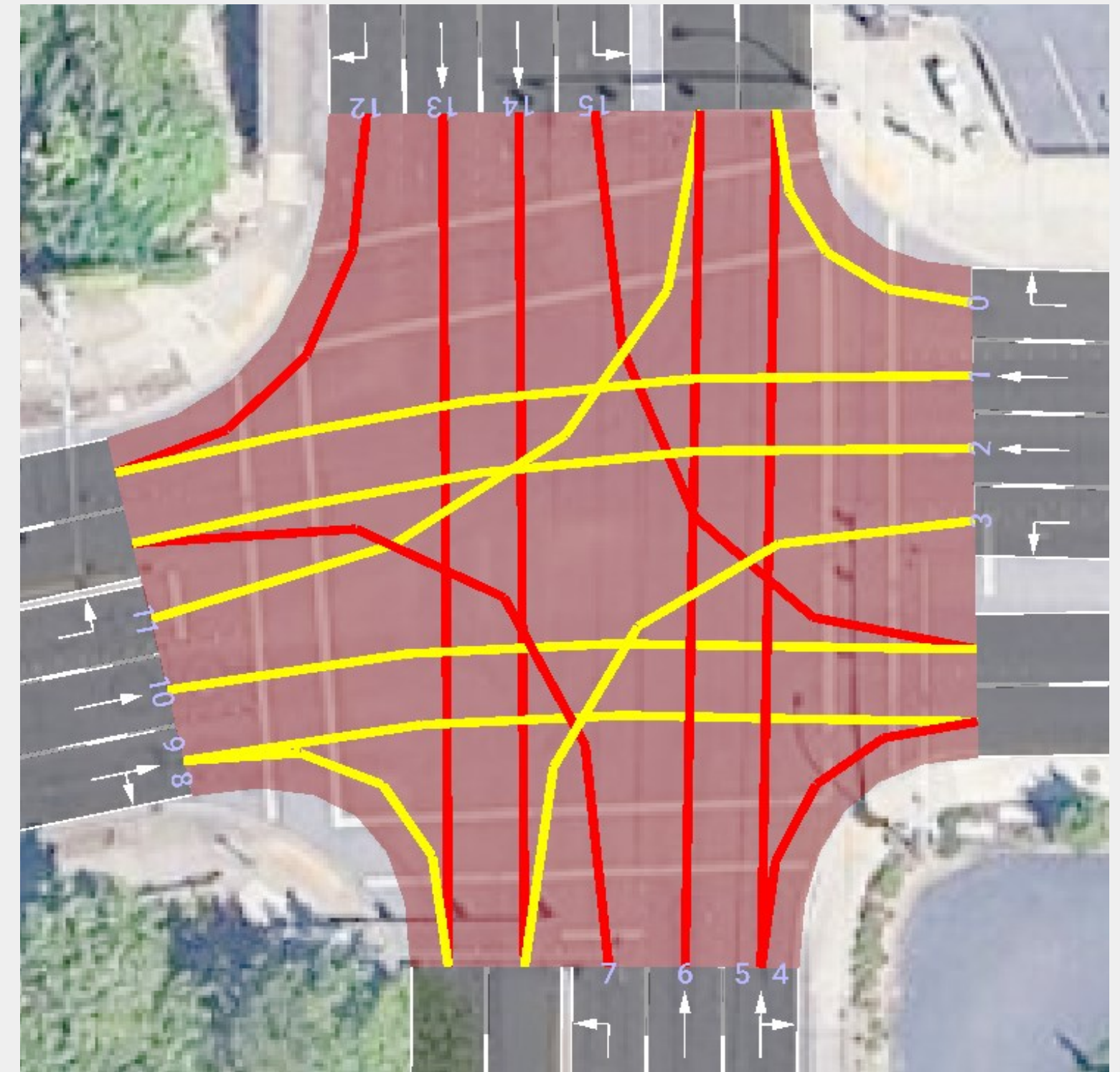
Traffic Signal

❑ Set the signal phases as follows:

Phase 0: Eastbound + Westbound movements - Green

Phase 1: Eastbound + Westbound movements - Yellow (clearance)

Expand		Phases							
	dur	state	next	name					
0	27	GGGGrrrrrGGGGrrrrr		I	+	x	↑	↓	
1	3.00	yyyyrrrrryyyyyrrrr			+	x	↑	↓	
2					+	x	↑	↓	
3					+	x	↑	↓	
Σ	60.00	Links: 16							



Traffic Signal

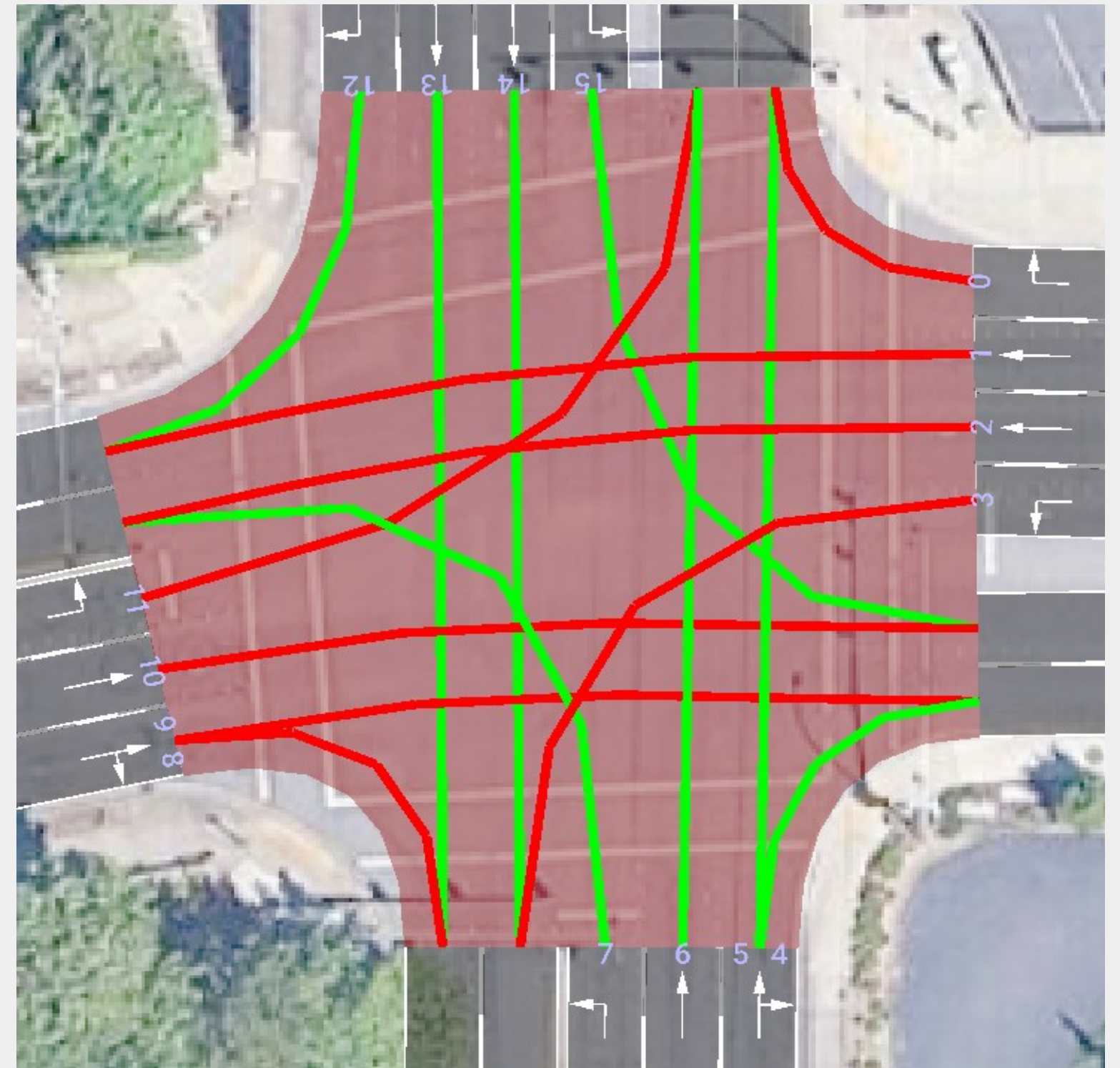
❑ Set the signal phases as follows:

Phase 0: Eastbound + Westbound movements - Green

Phase 1: Eastbound + Westbound movements - Yellow (clearance)

Phase 2: Northbound + Southbound movements - Green

Expand		Phases					
	dur	state	next	name			
0	27.00	GGGGrrrrGGGGrrrr			+	×	↑
1	3.00	yyyyrrrryyyyrrrr			+	×	↑
2	27	rrrrGGGGrrrrGGGG	I		+	×	↑
3					+	×	↑
Σ	60.00	Links: 16					



Adjusting Traffic Signal to Four Phases

- ❑ Set the signal phases as follows:

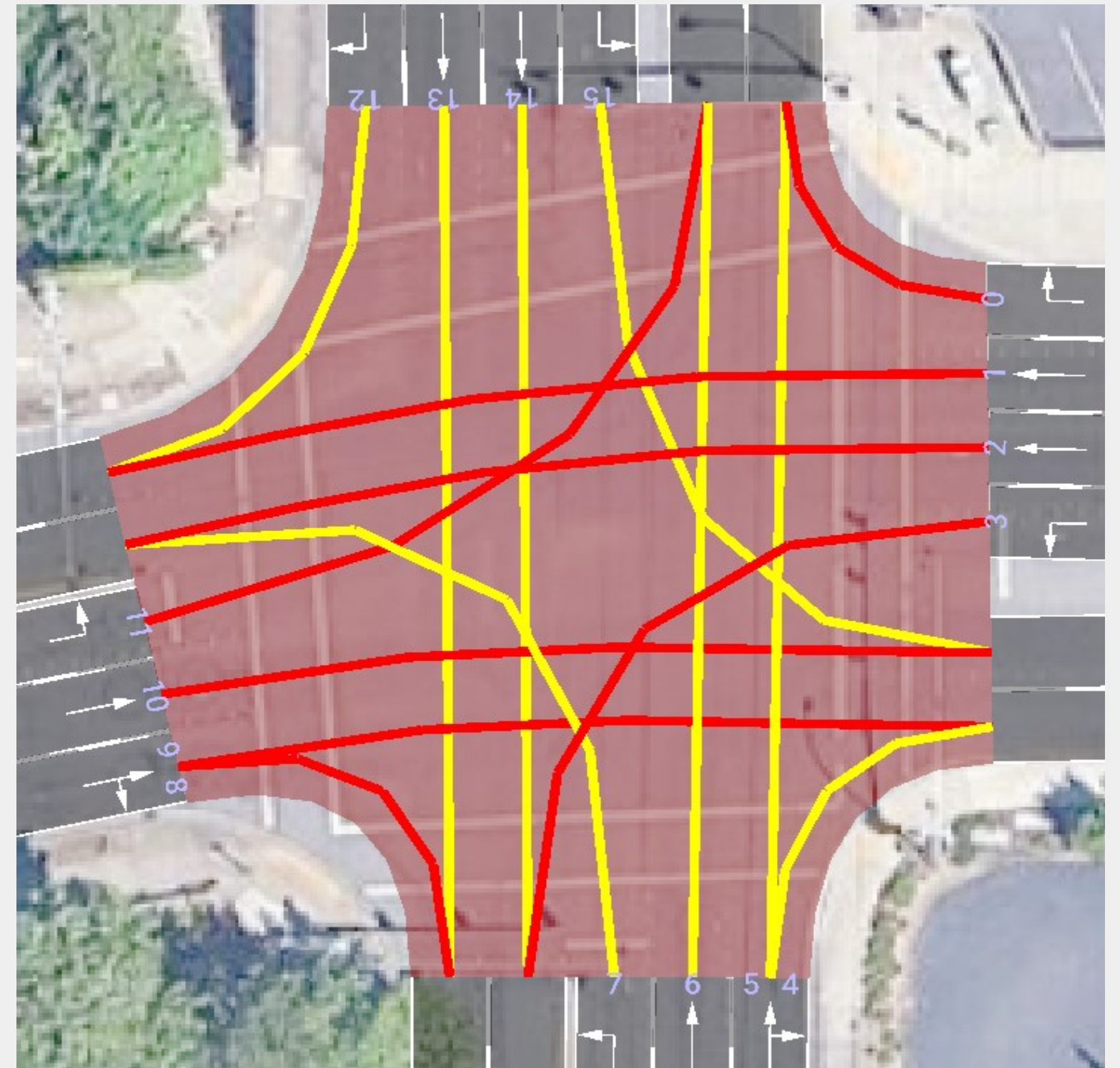
Phase 0: Eastbound + Westbound movements - Green

Phase 1: Eastbound + Westbound movements - Yellow (clearance)

Phase 2: Northbound + Southbound movements - Green

Phase 3: Northbound + Southbound movements - Yellow (clearance)

Expand				Phases				
	dur	state	next	name				
0	27	GGGGrrrrGGGGrrrr			+	x	↑	↓
1	3.00	yyyyrrrryyyyrrrr			+	x	↑	↓
2	27	rrrrGGGGrrrrGGGG			+	x	↑	↓
3	3.00	rrrryyyyrrrryyyy		I	+	x	↑	↓
Σ	60.00	Links: 16						



Adjusting Traffic Signal to Four Phases

❑ Save Files

Edit Traffic Light

Traffic Light

Junction ID

clusterJ14_J15_J25_J33_#4more

TLS ID

type

static

Join

Disjoin

Traffic Light Programs

programID

0

Duplicate

Reset single

Delete

Reset all

Save

Cancel

Traffic Light Attributes

offset

0.00

parameters

Assign E1 detectors

Expand

Phases

	dur	state	next	name				
0	27.00	GGGGrrrrGGGGrrrr			+	x	↑	↓
1	3.00	yyyyrrrryyyyrrrr			+	x	↑	↓
2	27	rrrrGGGGrrrrGGGG			+	x	↑	↓
3	3.00	rrrryyyyrrrryyyy			+	x	↑	↓
Σ	60.00	Links: 16						

Clean States

Add States

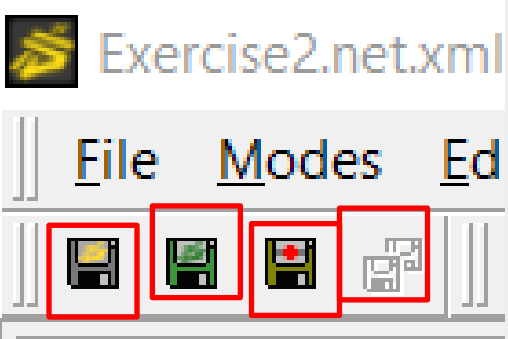
Group Sig.

Ungroup Sig.

TLS Program File

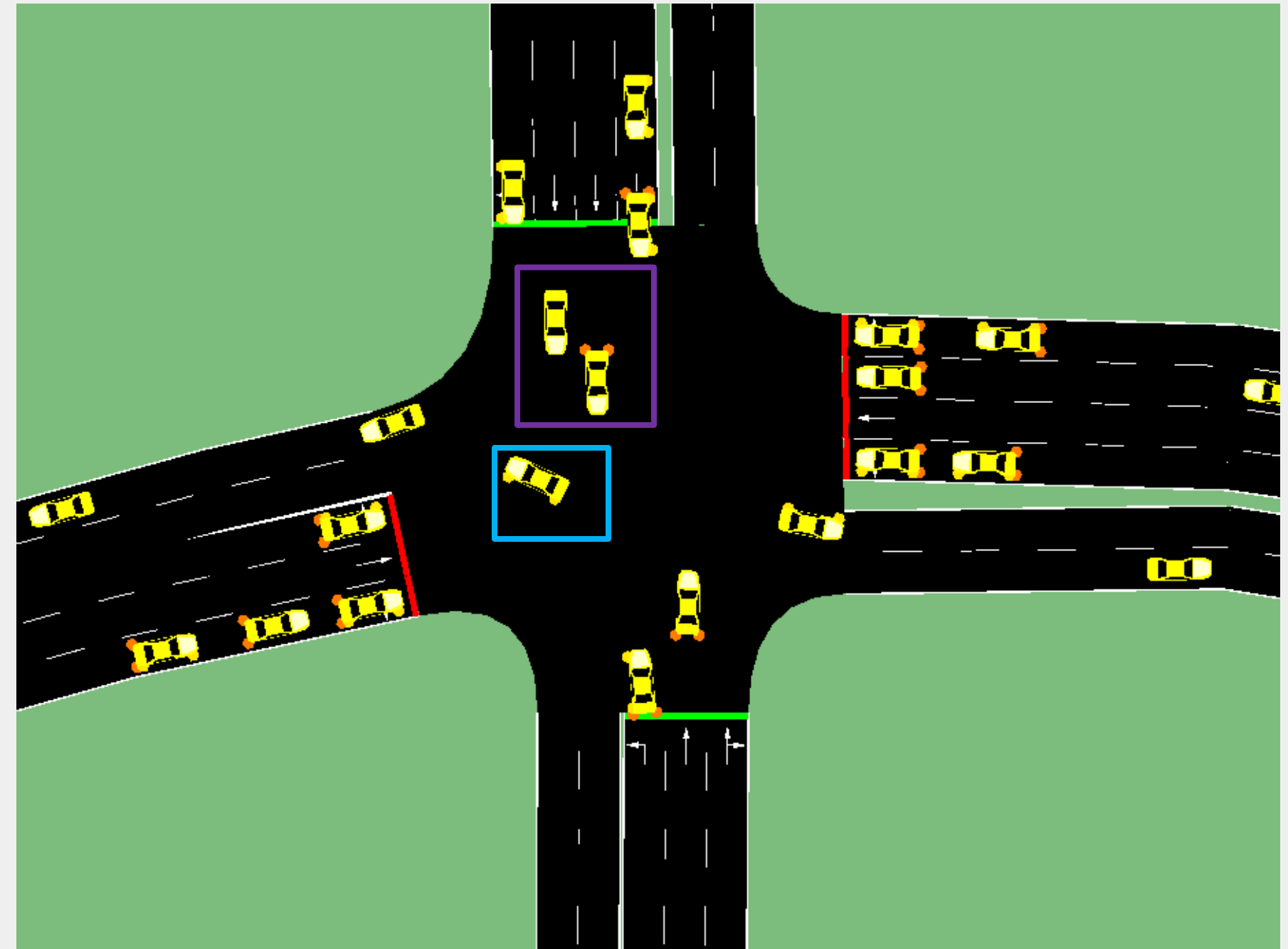
Load

Save



End of Milestone 2: Run Traffic Signal

- ☐ Run the SUMO Gui
- ☐ You will notice that left-turn movement is incorrectly given right of way over the through movement.
- ☐ Don't worry, we will fix it in The Next Milestone
- ☐ Raise Your Hand for Instructor Confirmation



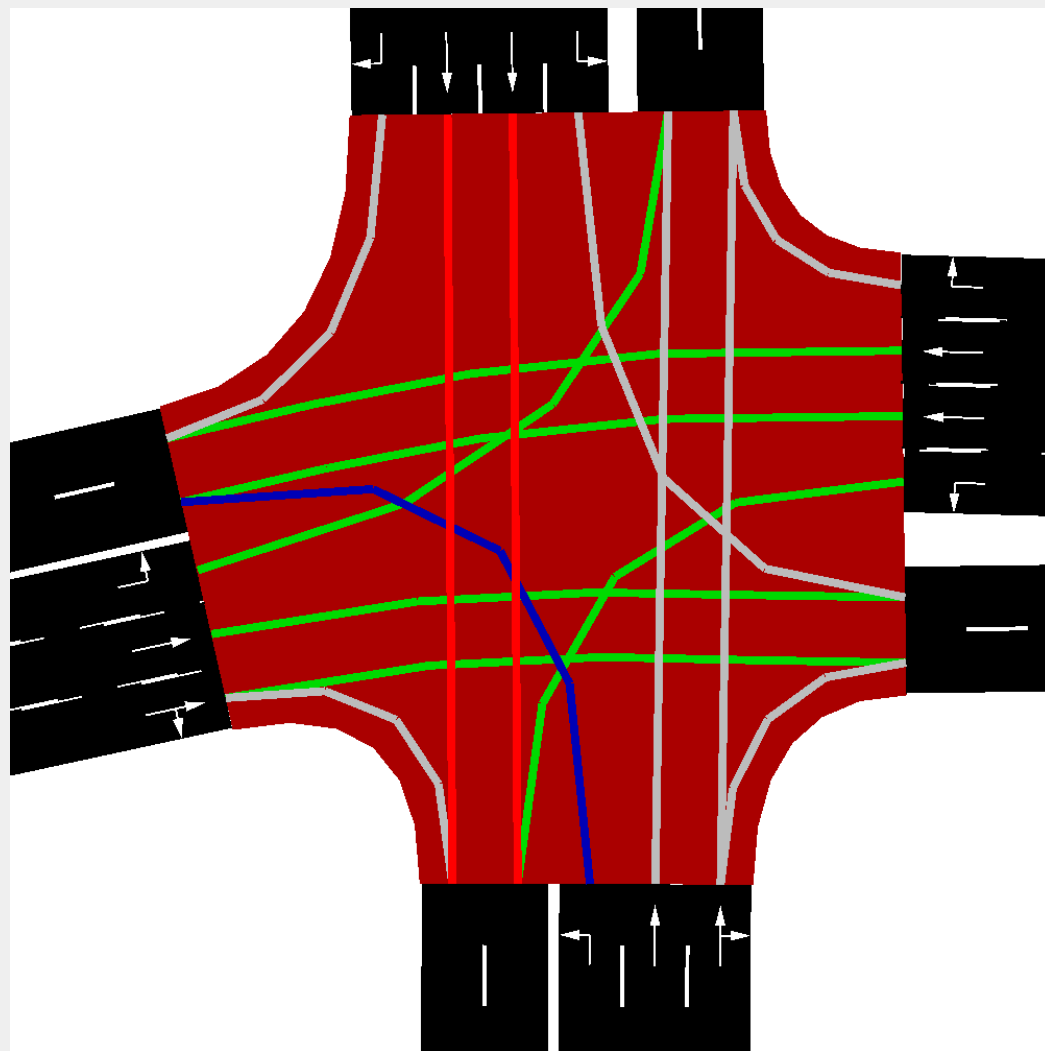
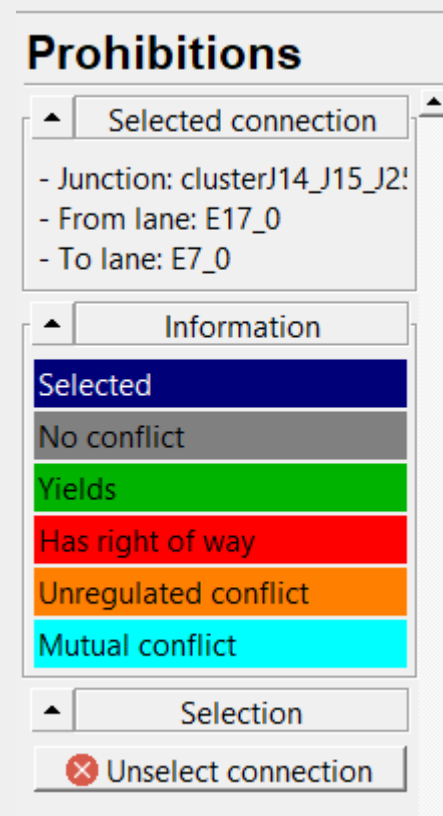
Milestone 3: Adjust Right of Way

Adjust Right of Way

☐ Open Netedit



☐ Select the connection



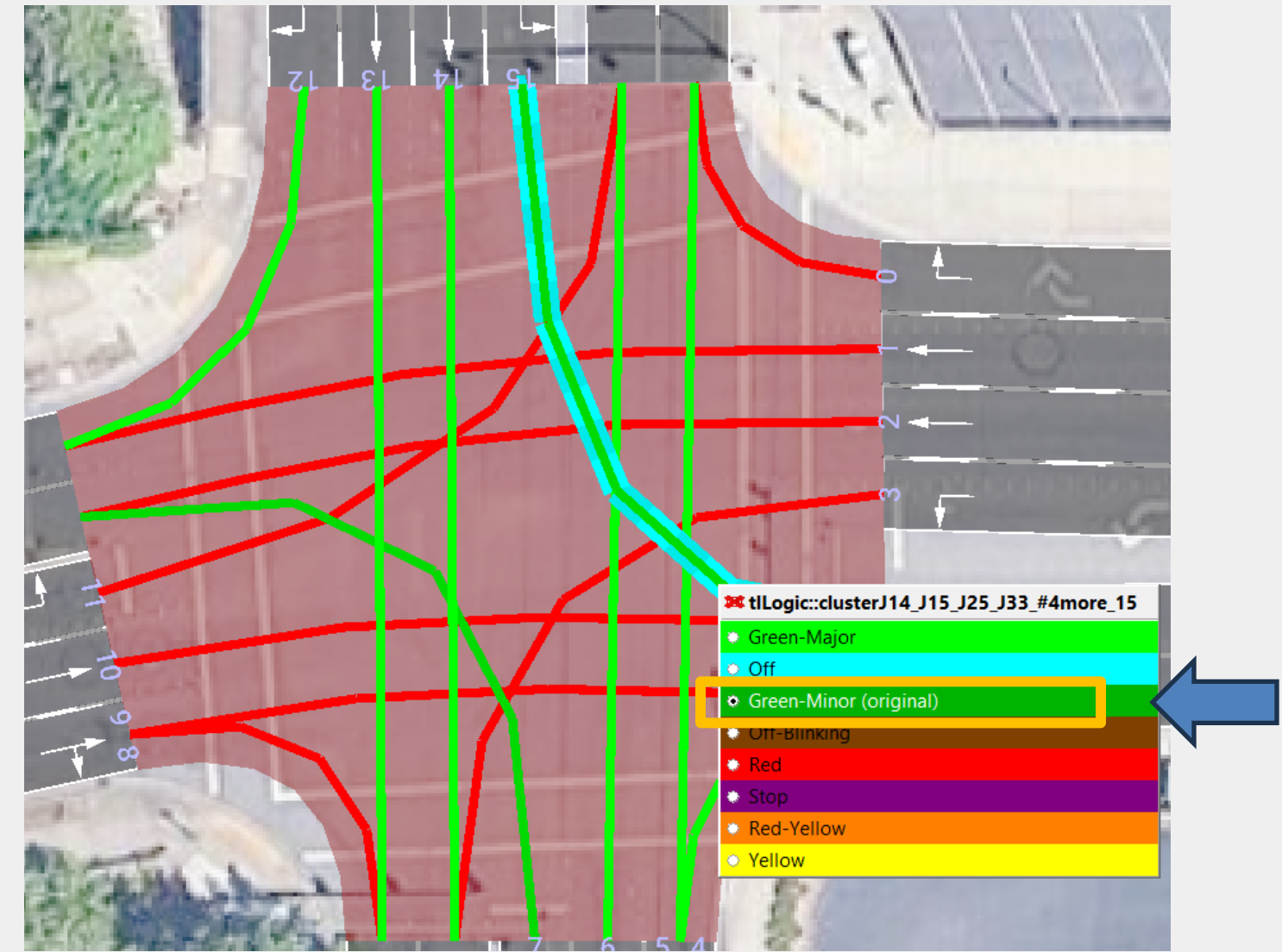
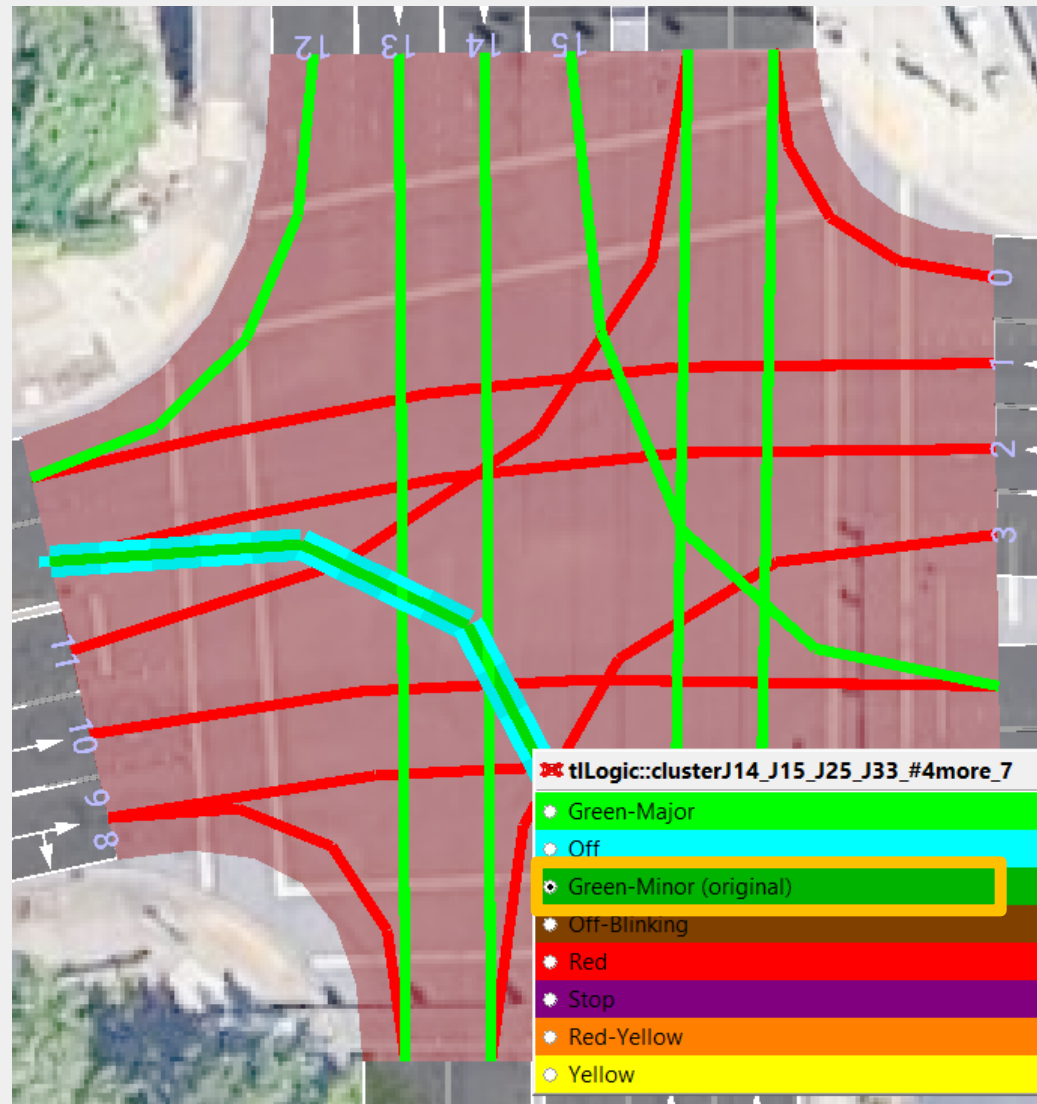
- ☐ Traffic on the blue connection yields to the green connection
- ☐ Traffic on the blue connection has right of way over the red connection

Adjust Right of Way

❑ Left Panel “Edit Traffic Signal” → Select Phase 2

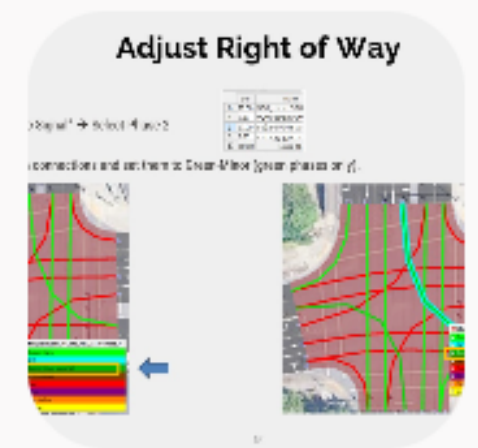
	dur	state
0	27.00	GGGgrrrrGGG
1	3.00	yyyyrrrryyy
2	27.00	rrrrGGGGrrr
3	3.00	rrrryyyyrrr
Σ	60.00	Links: 16

❑ Right-click the left-turn connections and set them to Green-Minor (green phases only).



Use of Generative AI

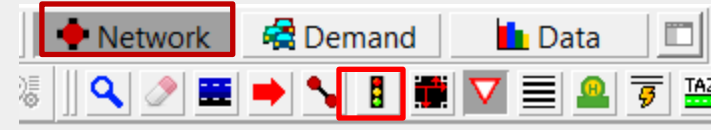
- ❑ Open Chatgpt or any AI (gemini etc)
- ❑ Copy and paste the slide 45 (take whole slide image) and ask
“What does it mean by this slide in SUMO netedit”
- ❑ Do you know understand?
- ❑ Ask about “the difference of minor green and major green in traffic signal in Netedit”



What does it mean by this slide in Sumo Netedit

Adjust Right of Way: Continuing

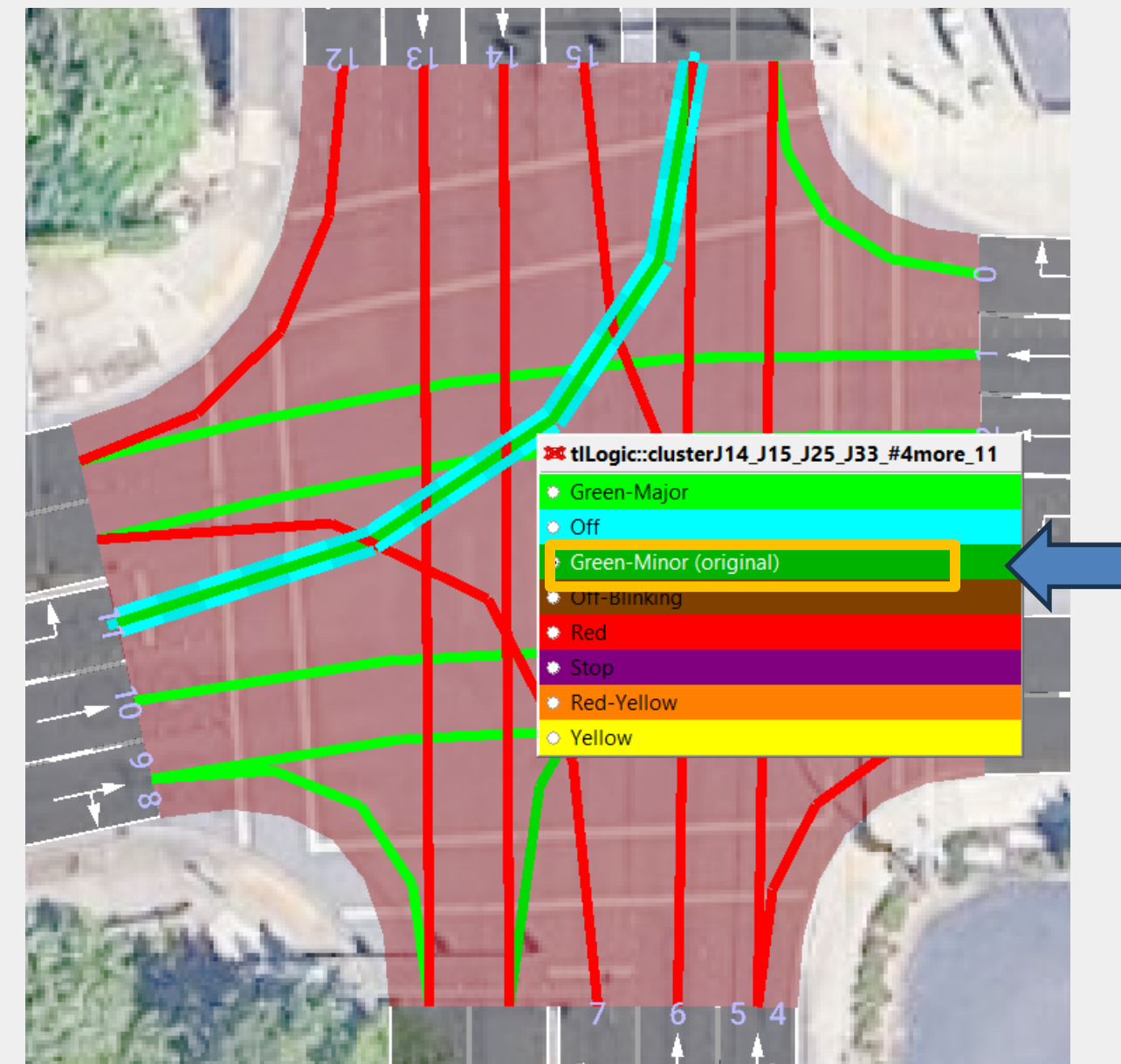
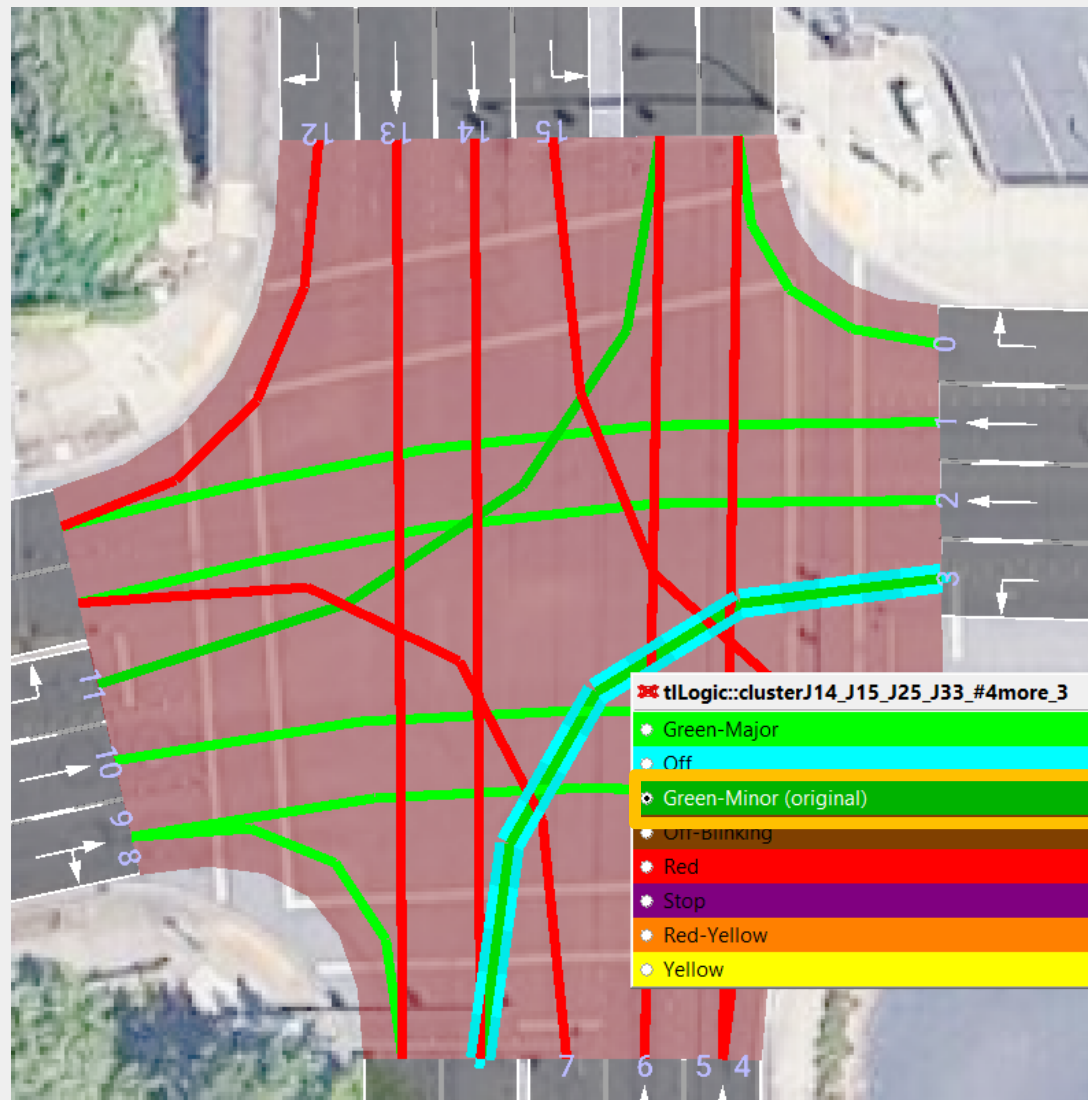
❑ Select Traffic Light



	dur	state
0	27.00	GGGrrrrrGGG
1	3.00	yyyyrrrryyy
2	27.00	rrrrGGGGrrr
3	3.00	rrrryyyyrrr
Σ	60.00	Links: 16

❑ Left Panel “Edit Traffic Signal” → Select Phase 0

❑ Right-click the left-turn connections and set them to Green-Minor during their green phases.



Adjust Right of Way

Save Files

Edit Traffic Light

▲ Traffic Light ▲

Junction ID clusterJ14_J15_J25_J33_#4more

TLS ID

type static ▼

⚡ Join ⚡ Disjoin

▲ Traffic Light Programs ▲

programID 0 ▼

⌘ Duplicate ⌚ Reset single

⌘ Delete ⌚ Reset all

✔ Save ✖ Cancel

▲ Traffic Light Attributes ▲

offset 0.00

parameters

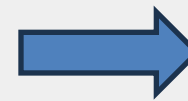
Assign E1 detectors

▲ Expand ↻ Phases ▲

	dur	state	next	name	
0	27.00	GGGgrrrrGGGgrrrr			+
1	3.00	yyyyrrrryyyyrrrr			+
2	27.00	rrrrGGGgrrrrGGGg			+
3	3.00	rrrryyyyrrrryyyy			+
Σ	60.00	Links: 16			
Clean States		Add States			
Group Sig.		Ungroup Sig.			

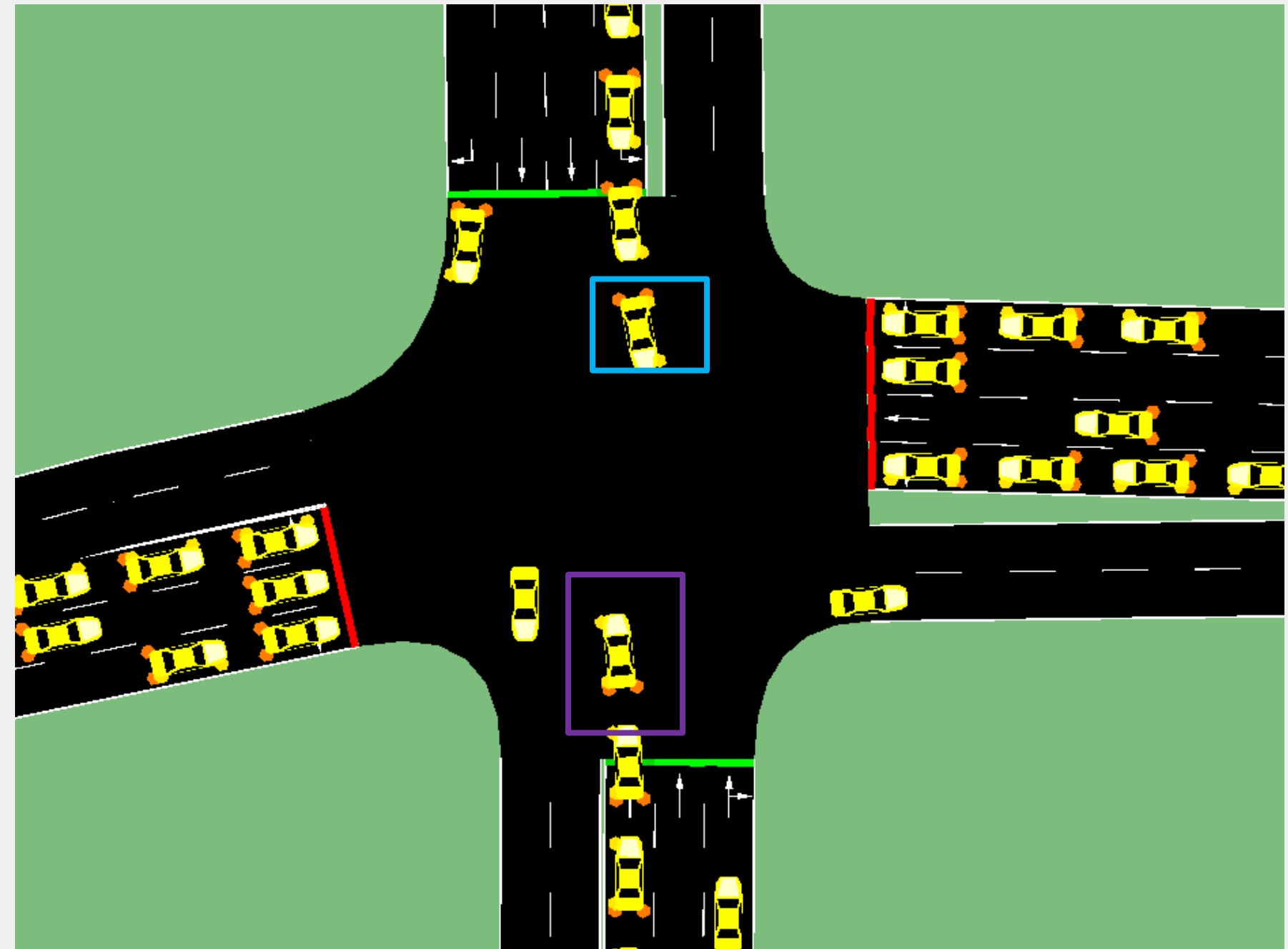
▲ TLS Program File ▲

📂 Load 💾 Save



End of Milestone 3: Run Traffic Simulation

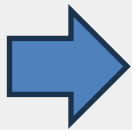
- ☐ Run the SUMO Gui
- ☐ Verify that left-turn movements yield to through movements
- ☐ Raise Your Hand for Instructor Confirmation



Milestone 4:
Simulate
Human Driven Vehicles (HDVs) &
Connected Autonomous Vehicles (CAVs)

Simulate HDVs

❑ Car Following Model for HDV



Types

Type Editor

Create Type

Reset Type

Copy Type

Current Type

DEFAULT VEHTYPE

Internal attributes

id	HDV
vClass	passenc
color	
length	5.00
minGap	2.50
maxSpeed	55.56
siredMaxSpe	2777.78
arkingBadge	
probability	1.00
accel	2.60
decel	4.50
sigma	0.50
tau	1.00

Extended attributes

Edit extended attributes

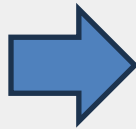
Parameters

Edit parameters

Netedit attributes

File e2.rou.xml

Help



❑ Find the values from the slide “**Vehicle Characteristics & Dynamics Parameter**” and change it in below

Set up Car Following and Lane Changing Model Parameters for HDVs

Accept Cancel Reset

Simulate HDVs

❑ Lane Changing Model for Normal Car:

 Edit vType 'DEFAULT_VEHTYPE' ✕

Vehicle Type attributes				Lane Change Model attributes		Car Following Model attributes	
vClass	 passenger	quiShape	 passenger	strategic	1.00	Algorithm	Krauss
id	DEFAULT_VEHTYPE	probability	1.00	cooperative	0.5	accel	2.60
color		personCapacity	4	speedGain	1.00	decel	4.50
length	5.00	containerCapacity	0	keepRight	1.00	apparentDecel	4.50
minGap	2.50	boardingDuration	0.50	sublane	1.00	emergencyDecel	9.00
maxSpeed	55.56	loadingDuration	90.00	opposite	1.00	sigma	0.50
desiredMaxSpeed	2777.78	latAlignment	center	pushy	0.00	tau	1.00
speedFactor	nc(1.00,0.10,0.20,2.00)	minGapLat	0.12	pushyGap	0.00		
emissionClass	A4/PC petrol Euro-4	maxSpeedLat	1.00	assertive	1.00		
width	1.80	actionStepLength	0.00	impatience	0.00		
height	1.50	Edit parameters	driverState.device=true	timeToImpatience	infinity		
imgFile				accelLat	1.00		
osgFile	car-normal-citrus.obj			lookaheadLeft	2.00		
laneChangeModel	LC2013			speedGainRight	0.10		
Junction Model attributes				axSpeedLatStandir	0.00		
crossingGap	10.00	ignoreFoeProb	0.00	maxSpeedLatFacto	1.00		
ignoreKeepClearTim		ignoreFoeSpeed	0.00	turnAlignDistance	0.00		
driveAfterYellowTim		sigmaMinor	0.00	overtakeRight	0.00		
driveAfterRedTime		timegapMinor	1.00	oRightAcceptanceT			
driveRedSpeed	0.00	impatience	0.00	oRightAcceptanceT	0.00		

✓ Accept ✕ Cancel ↺ Reset

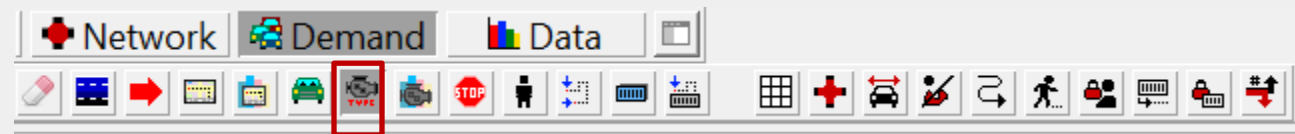
Always Save Files

Save Files



Simulate CAVs

❑ Car Following Model for CAV



Types

Type Editor

Create Type

Delete Type

Copy Type

Current Type

CAV

Internal attributes

id	CAV
vClass	passenc
color	
length	5.00
minGap	2.50
maxSpeed	55.56
siredMaxSpe	2777.78
arkingBadqe	
probability	1.00
accel	2.60
decel	4.50
sigma	0.50
tau	1.00

Extended attributes

Edit extended attributes

Parameters

Edit parameters

Netedit attributes

File 5-e2.rou.xml

Help



Set up Car Following and Lane Changing Model Parameters for CAVs

Simulate CAVs: Car Following Model

❑ Car Following Model for CAV

TABLE II
CAR-FOLLOWING MODEL PARAMETERS FOR SIMULATED CAVS

Parameter	Value	
Speed deviation	0.05	⑥
Time headway	0.6 s	④
Minimum gap	1.5 m	⑤
Max acceleration	2.9 m s ⁻²	①
Deceleration	7.5 m s ⁻²	②
Emergency deceleration	9 m s ⁻²	③

Change Color to Grey

Second value

Edit vType 'CAV'

Vehicle Type attributes

vClass	passenger	guiShape	passenger
id	CAV	probability	1.00
color	112,112,112,0	personCapacity	4
length	5.00	containerCapacity	0
minGap	1.50	guardingDuration	0.50
maxSpeed	112,112,112	loadingDuration	90.00
desiredMaxSpeed	2777.78	latAlignment	center
speedFactor	normc(1.00,0.05)0	minGapLat	0.12
emissionClass	PC petrol Euro-4	maxSpeedLat	1.00
width	1.80	actionStepLength	0.00
height	1.50		
imgFile			
osgFile	-normal-citrus.obj		
laneChangeMod	default		

Junction Model attributes

crossingGap	10.00	ignoreFoeProb	0.00
oreKeepClearTi		gnoreFoeSpeed	0.00
veAfterYellowTi		sigmaMinor	0.00
iveAfterRedTin		timegapMinor	1.00
driveRedSpeed	0.00	impatience	0.00

Lane Change Model attributes

strategic	1.00
cooperative	1.00
speedGain	1.00
keepRight	1.00
sublane	1.00
opposite	1.00
pushy	0.00
pushyGap	0.00
assertive	1.00
impatience	0.00
meToImpatience	infinity
accelLat	1.00
lookaheadLeft	2.00
speedGainRight	0.10
SpeedLatStand	0.00
axSpeedLatFact	1.00
urnAlignDistanc	0.00
overtakeRight	0.00
RightAcceptance	
akeDeltaSpeedl	0.00

Car Following Model attributes

Algorithm	CACC
accel	2.9
decel	7.5
emergencyDecel	9.00
tau	0.6
isionMinGapFac	

Some attributes wasn't implemented yet

58

AcceptCancelReset

Simulate CAVs: Lane Changing Model

- Lane Changing Model for CAVs: LC 2013 → cooperative 1

Edit vType

Vehicle Type attributes

vClass	 passenger	guiShape	default
id	t_0	probability	1.00
color		personCapacity	4
length	5.00	containerCapacity	0
minGap	2.50	boardingDuration	0.50
maxSpeed	55.56	loadingDuration	90.00
desiredMaxSpeed	2777.78	latAlignment	center
speedFactor	normc(1.00,0.00)	minGapLat	0.12
emissionClass	HBFA3/PC_G_EU4	maxSpeedLat	1.00
width	1.80	actionStepLength	0.00
height	1.50	carriageLength	-1.00
imgFile		locomotiveLength	-1.00
osgFile	car-normal-citrus.obj	carriageGap	1
laneChangeModel	LC2013	Edit parameters	ent=2 has.driverState.device=true

Junction Model attributes

crossingGap	10	ignoreFoeProb	0.0
ignoreKeepClearTime	-1	ignoreFoeSpeed	0.0
driveAfterYellowTime	-1	sigmaMinor	0.0
driveAfterRedTime	-1	timegapMinor	1
driveRedSpeed	0.0	impatience	0.0

Lane Change Model attributes

strategic	1.0
cooperative	1
speedGain	1.0
keepRight	1.0
sublane	1.0
opposite	1.0
pushy	0.00
pushyGap	0.00
assertive	1.0
impatience	0.00
timeToImpatience	infinity
accelLat	1.0
lookaheadLeft	2.0
speedGainRight	0.1
maxSpeedLatStanding	0.00
maxSpeedLatFactor	1.00
turnAlignDistance	0.00
overtakeRight	0.00
keepRightAcceptanceTime	-1
overtakeDeltaSpeedFactor	0.00

Car Following Model attributes

Algorithm	CACC
accel	2.60
decel	4.50
emergencyDecel	9.00
tau	1.00
collisionMinGapFactor	

Some attributes wasn't implemented yet

Accept

Cancel

Reset

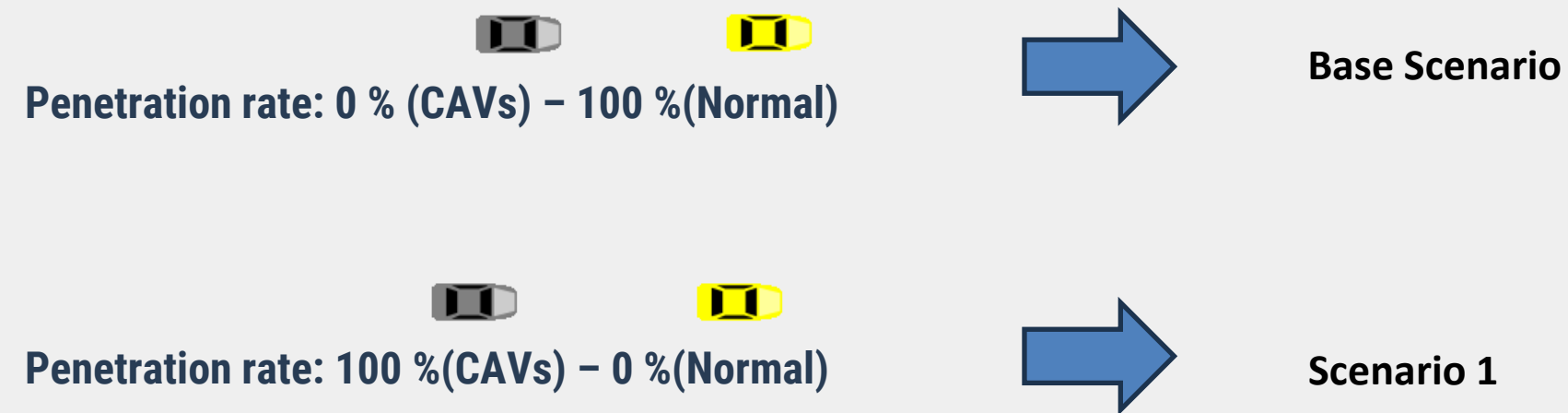
Always Save Files

Save Files



Develop Mixed Traffic for CAVs and HDV

❑ In Your Network, create the following scenarios



❑ Hint 1: Duplicate the whole project folder two more times, name them Base Scenario, Scenario 1

❑ Hint 2: Change the Vehicle type for each traffic demand (See next slide for hint)

BaseScenario: All HDV

❑ Do this for all vehicles

6

Exercise2.netecfg - netedit 1.26.0

File Modes Edit Lock Processing Locate Tools Window Language Help Network Demand Data

3

standard

3

5

Vehicle: flow

Internal attributes

id

f 0

type

DEFAULT VEHTYPE

from

Type to search...

to

CAV

via

DEFAULT_BIKETYPE

color

DEFAULT_CONTAINERTYPE

departLane

DEFAULT_PEDTYPE

departPos

DEFAULT_RAILTYPE

departSpeed

DEFAULT_TAXITYPE

arrivalLane

DEFAULT_VEHTYPE

arrivalPos

DEFAULT_VEHTYPE

arrivalSpeed

HDV

line

personNum

0

containerNum

0

departPosL

center

arrivalPosL

center

accelerationCheck

all

begin

0.00

Flow attributes

terminate

end

spacing

vehsPerHour

end

3600.00

vehsPerHour

500

Parameters

Edit parameters

Netedit attributes

1

2

4

Scenario 1: All CAV

Do this for all vehicles

6

Exercise2.netecfg - netedit 1.26.0

File Modes Edit Lock Processing Locate Tools Window Language Help Network Demand Data

standard

3

Vehicle: flow

Internal attributes

5

id

f 0

type

DEFAULT VEHTYPE

from

Type to search...

to

CAV

via

DEFAULT_BIKETYPE

color

DEFAULT_CONTAINERTYPE

departLane

DEFAULT_PEDTYPE

departPos

DEFAULT_RAILTYPE

departSpeed

DEFAULT_TAXITYPE

arrivalLane

DEFAULT_VEHTYPE

arrivalPos

HDV

arrivalSpeed

line

personNum

0

containerNum

0

departPosL

center

arrivalPosL

center

terminationCheck

all

begin

0.00

Flow attributes

terminate

end

spacing

vehsPerHour

end

3600.00

vehsPerHour

500

Parameters

Edit parameters

Netedit attributes

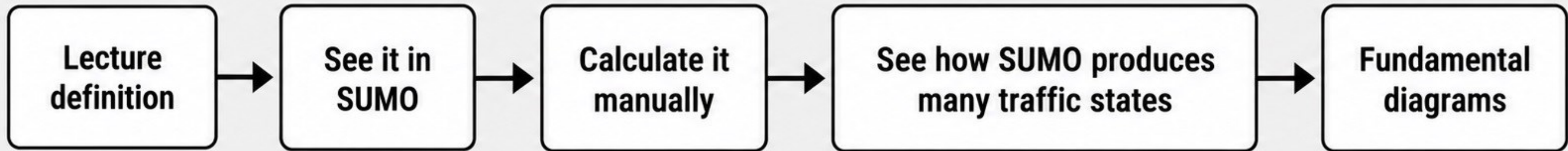
1

2

4

How the Lab Connects to the Course

- ❑ The learning progression in this lab follows a simple sequence:



Deliverable 2

Deliverable 2 Folder:

- 1. Open HTML**
- 2. Respond to the Question Using the SUMO Files**
- 3. Submit only the Generated html in Avenue**

Recap of Week 3 Course Slides

☐ Download Course Slide for Week 3 → Skim Over

Slides 3 – 17 → Answer next questions Q1-5

Question 1: Traffic Density

☐ Run base Scenario Simulation and Stop Simulation at Second 383,

A) How many vehicles do you see on the road?

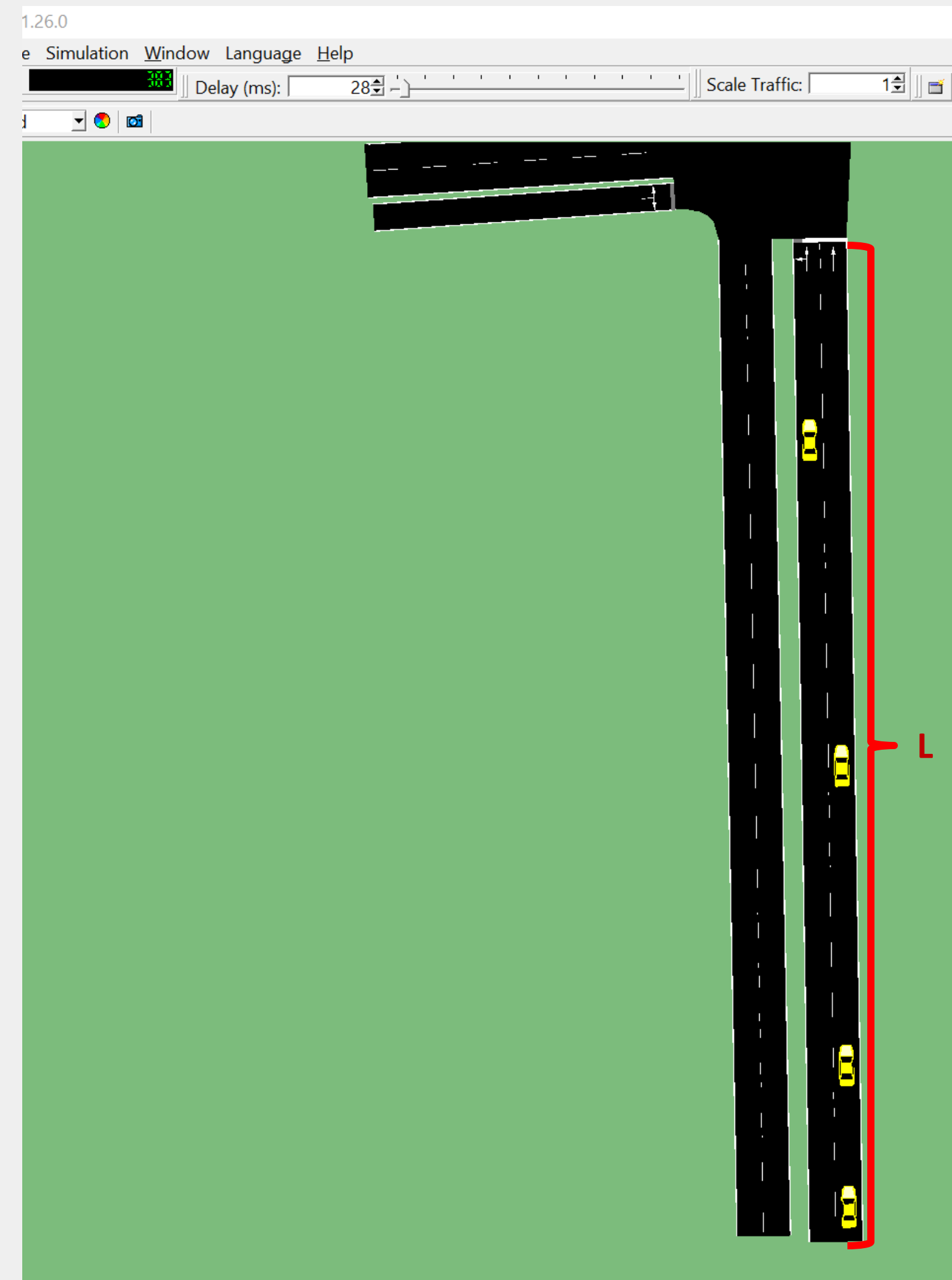
B) Calculate the Traffic Density at this second?

C) Whats traffic density per lane (Traffic Lane Density)?

☐ **Traffic Density (k)**

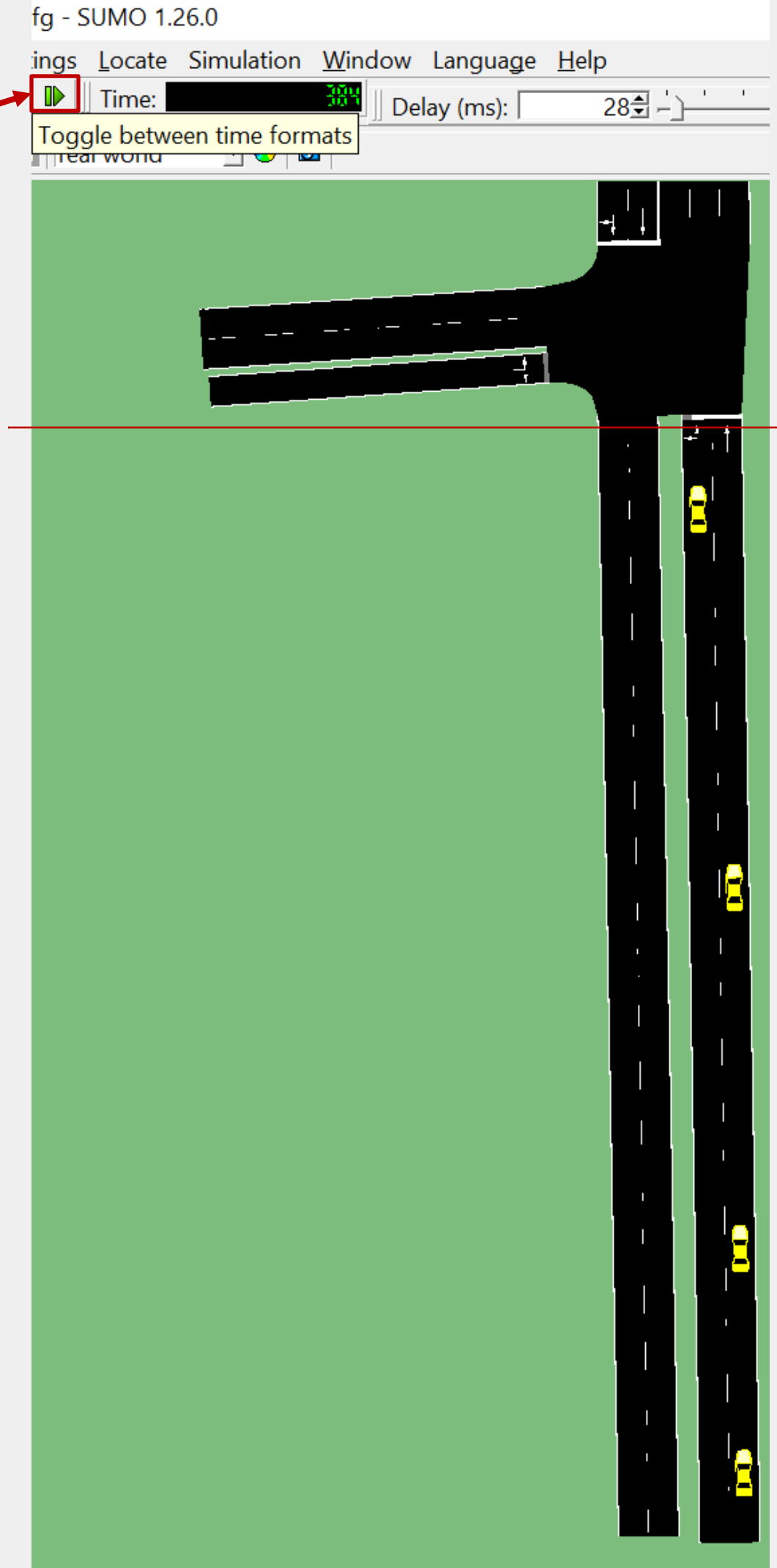
Density k = the **number of vehicles present on a roadway segment per unit length**, usually expressed as **veh/km**.

$$k = \frac{N}{L}$$



Question 3: Traffic Volume & Flow Rate

- ☐ Click on the next step, go to second 384, see vehicles are moving. Similar to when you see a street which vehicle moves in each second. Gradually, increase the simulation second until 443 (1 min), and record how many vehicles completely crossed the intersection (red line)
- D) Calculate traffic volume during this one min?
- E) Calculate traffic flow rate (hourly)?



Question 3: Speed Measurements – Space Mean Speed

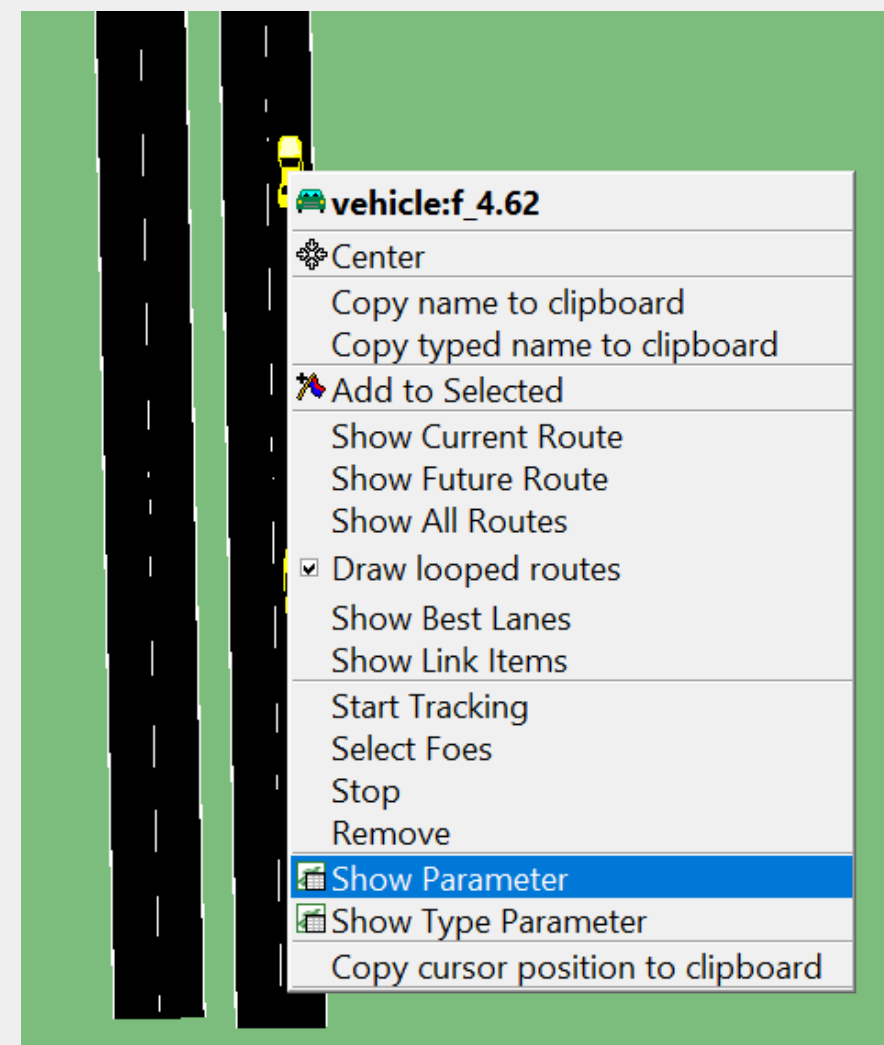
❑ Restart simulation and stop at second 383 → Right Click on Each vehicle → Show parameter → Record their Speed

❑ Considering Week 3 - course Slide 16,

F) Calculate Space Mean Speed values?

G) Run simulation for one more second and calculate Space Mean Speed aggregated over 383 and 384 seconds?

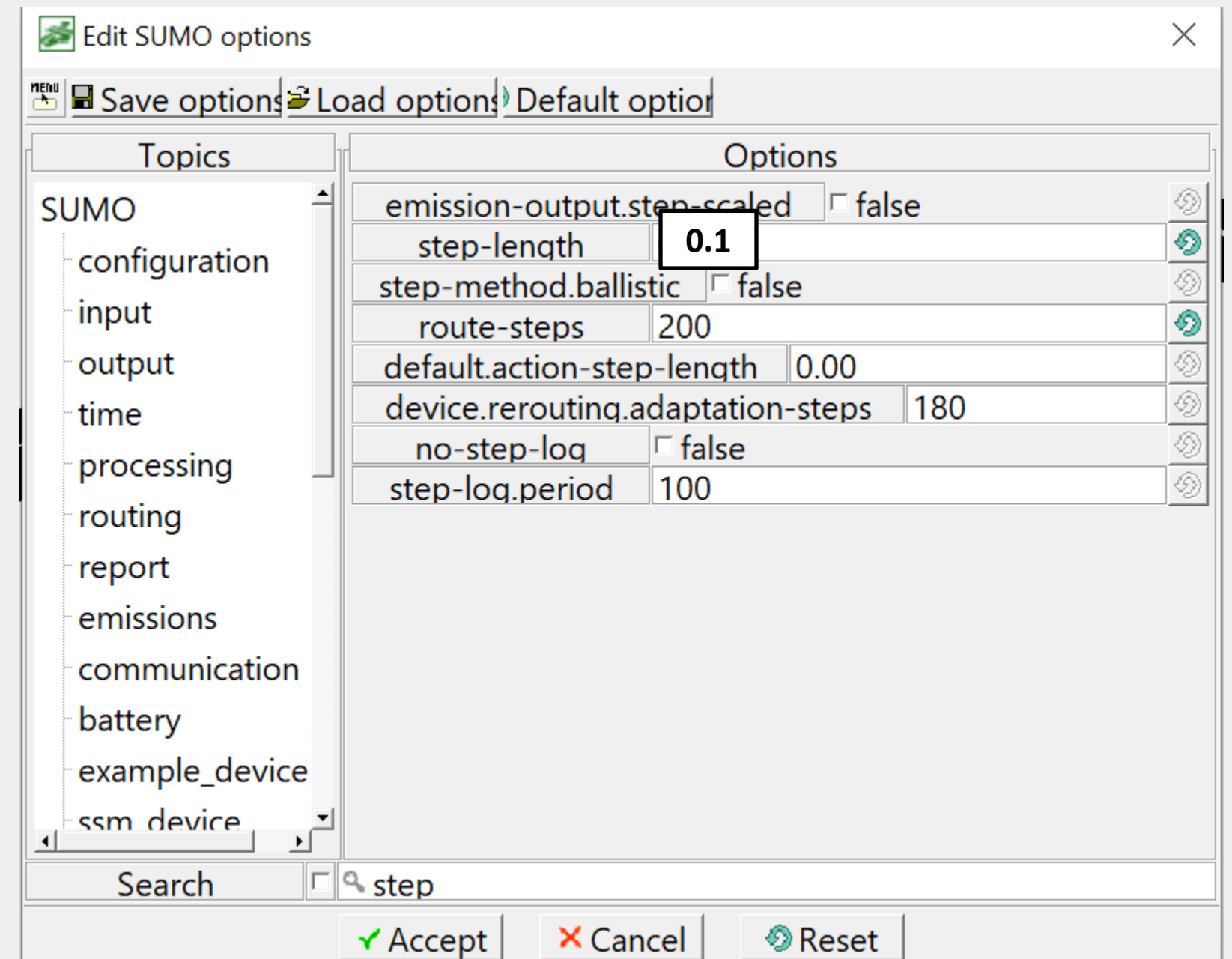
Note: Space Mean Speed can be defined for a road section at a single instant, or aggregated over a time interval such as 1 minute, 5 minutes, or 1 hour.



vehicle:f_4.62 para...			
Name	Value	Dynamic	
lane [id]	E16_0	✓	
position [m]	59.69	📐	
lateral offset [m]	0.00	📐	
speed [m/s]	12.83	📐	
lateral speed [m/s]	0.00	📐	

Question 4: Preparation

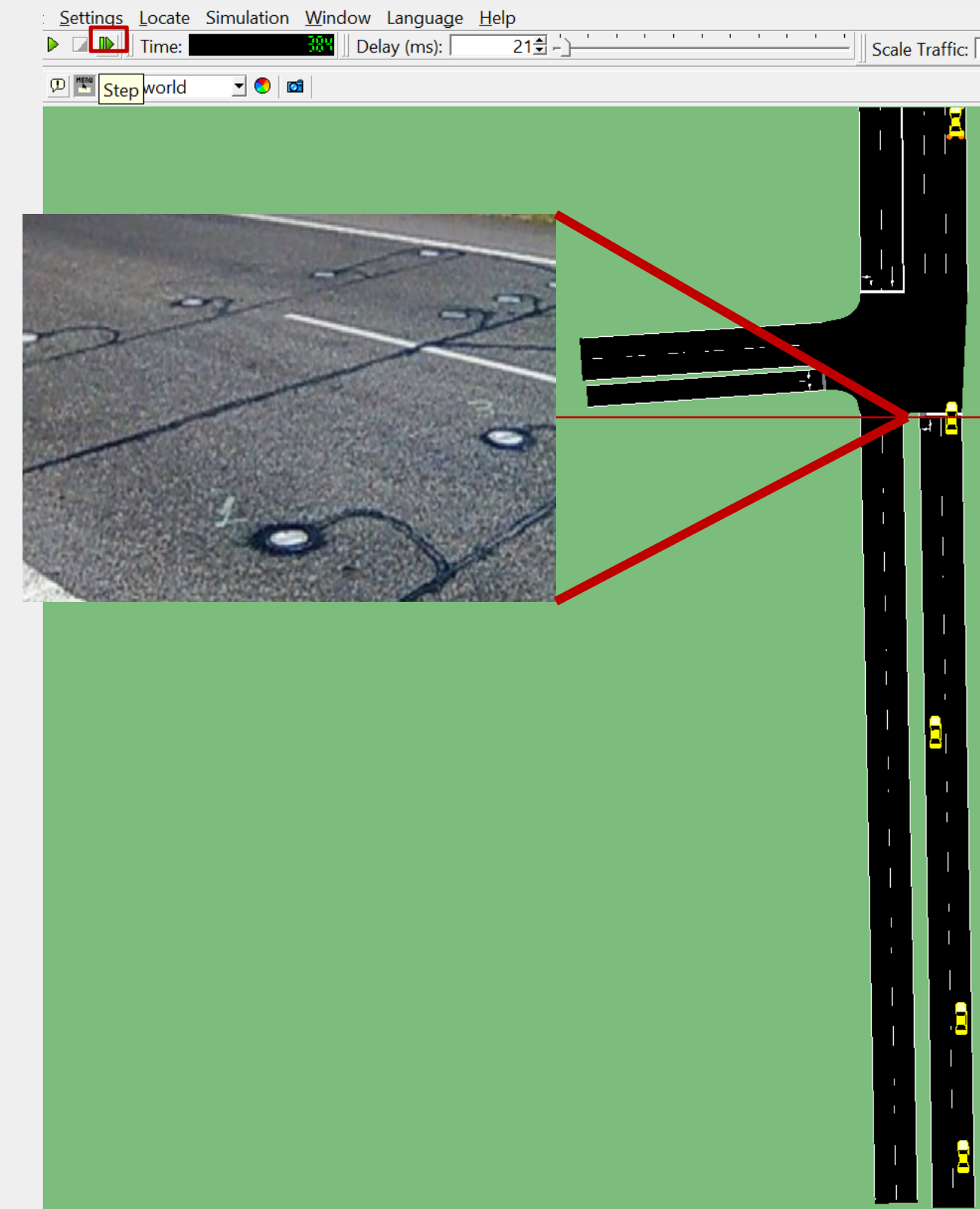
- ☐ Open netedit → Processing → SUMO option →
- ☐ In search bar → Step-length → change the value to 0.1
- ☐ Step-length tells SUMO how often it updates the simulation.
- ☐ 0.1 means 0.1 second
- ☐ Save



Question 4: Speed Measurements – Spot Speed and Time Mean Speed

- ☐ Restart the simulation and pause at $t=384$ s
- ☐ Refer to Slide 16 from the Week 3 lecture. Assume that an **inductive-loop vehicle detector** is installed at the location shown by the red line. The detector measures the **spot speed of each vehicle at the instant it crosses the detector**.
- ☐ Run the simulation from **384 s to 443 s (1 minute)**. Each time a vehicle crosses the red line, pause the simulation, right-click the vehicle, and record its speed. **Do this for the first 10 seconds.**
- ☐ You can create an excel and write something like below

Vehicle	Spot Speed (m/s)
1	
2	
3	
4	



Question 4: Table

- ☐ Compare your excel with this Table
- ☐ Now we have the data for all the seconds see (the next slide), respond the following question
- H)** Record the **spot speed** of each vehicle that crosses the detector during the 1-minute observation period.
- I)** How many vehicles crossed the detector during the 1-minute observation period?
- J)** Calculate the **Time Mean Speed (TMS)** of the vehicles during this 1-minute period:

Vehicle	Spot Speed (m/s)
1	12.87
2	14.36
3	13.61
4	13.76
5	11.67
6	15.68
7	13.15
8	13.67
9	11.85
10	13.47
11	14.08
12	13.31
13	14.83
14	14.68
15	14.58
16	16.83
17	15.31
18	12.76
19	13.61
20	12.94
21	13.30
22	16.35

Question #5 (Density)

Now, you learnt, how to calculate Traffic Density and Space Mean Speed

F) Calculate Flow Rate for this second (384) using slide 17 formula

$$\square \quad q = v_s \times k$$

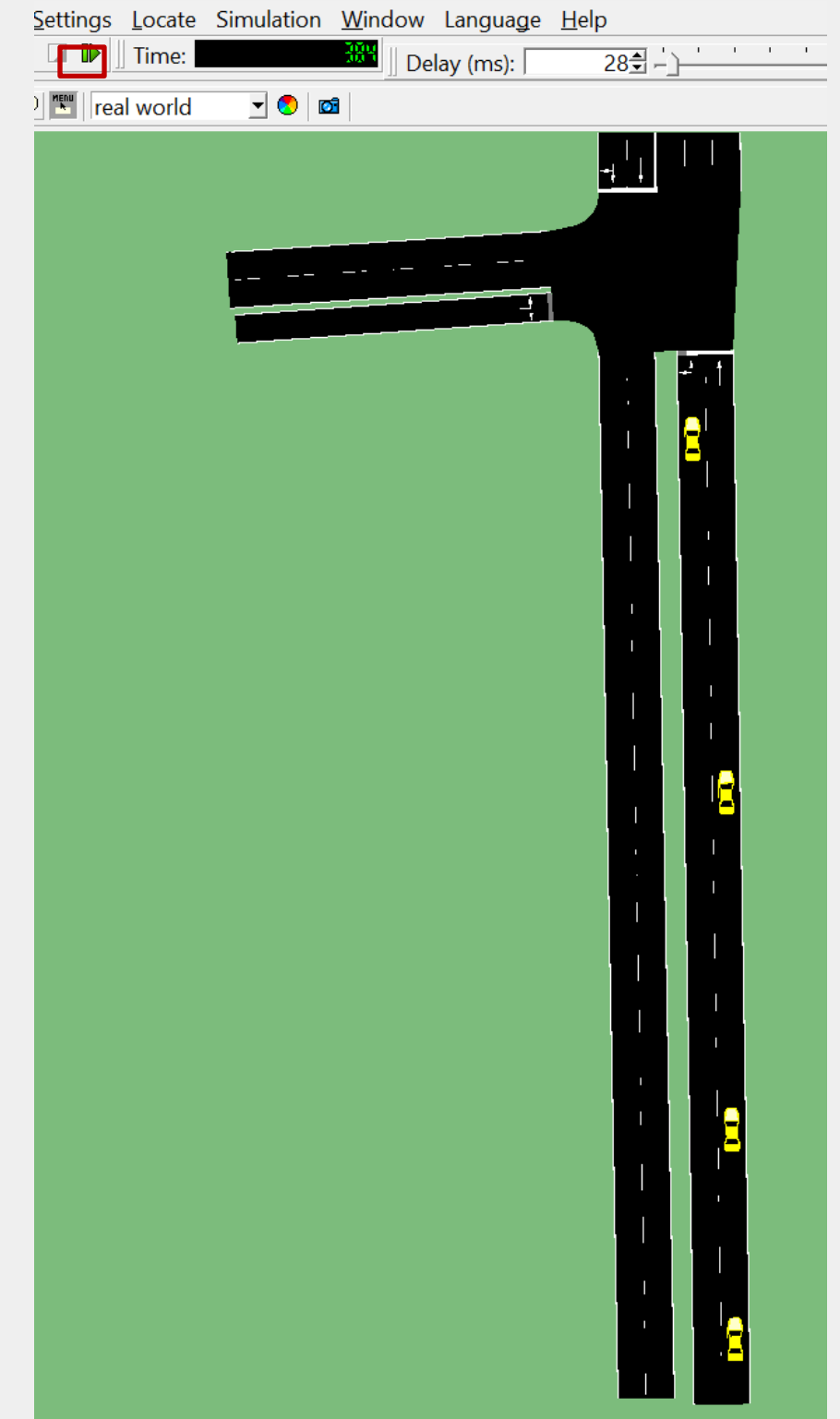
Where:

v_s = Space Mean Speed (km/hr)

q = Traffic Flow Rate (veh/hr)

k = Traffic Density (veh/km)

Note: You need to retrieve Space Mean Speed for second 384 from Q3



**Milestone 4 (Continuing :
Simulate
Human Driven Vehicles (HDVs) &
Connected Autonomous Vehicles (CAVs)**

From One Traffic State to a Fundamental Diagram

- ❑ So far, we manually measured traffic conditions at one instant or during one short observation period.

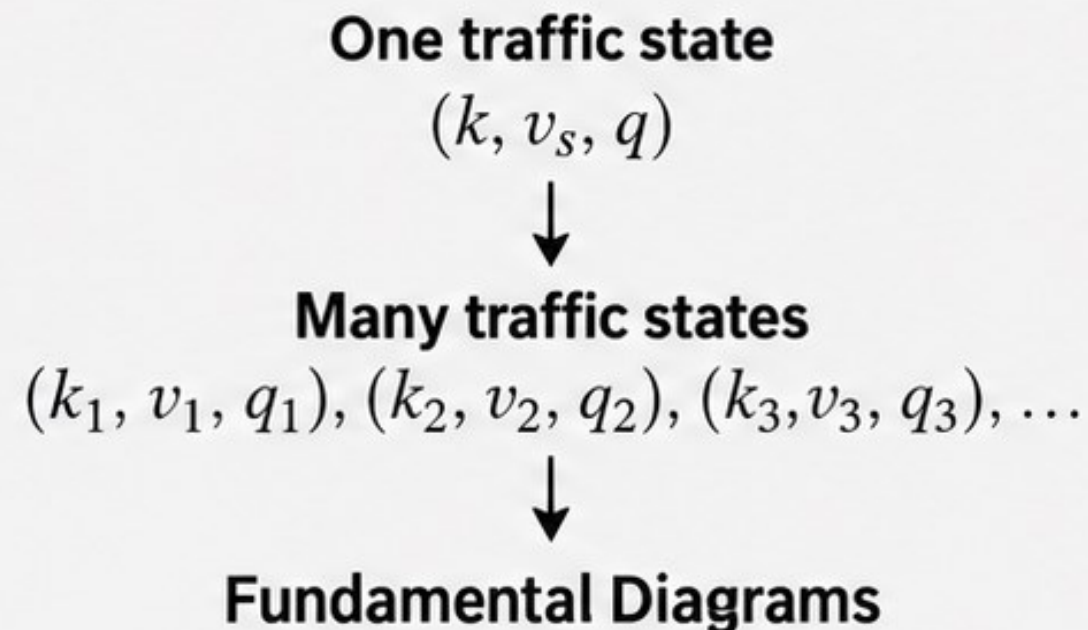
During the full simulation, SUMO continuously records the same traffic variables for Edge E16:

$$\boxed{k \quad v_s \quad q}$$

Each observation represents one traffic state.

By plotting many traffic states, we obtain the three fundamental diagrams introduced in Week 3:

$$\boxed{v_s - k} \quad \boxed{q - k} \quad \boxed{v_s - q}$$

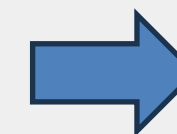
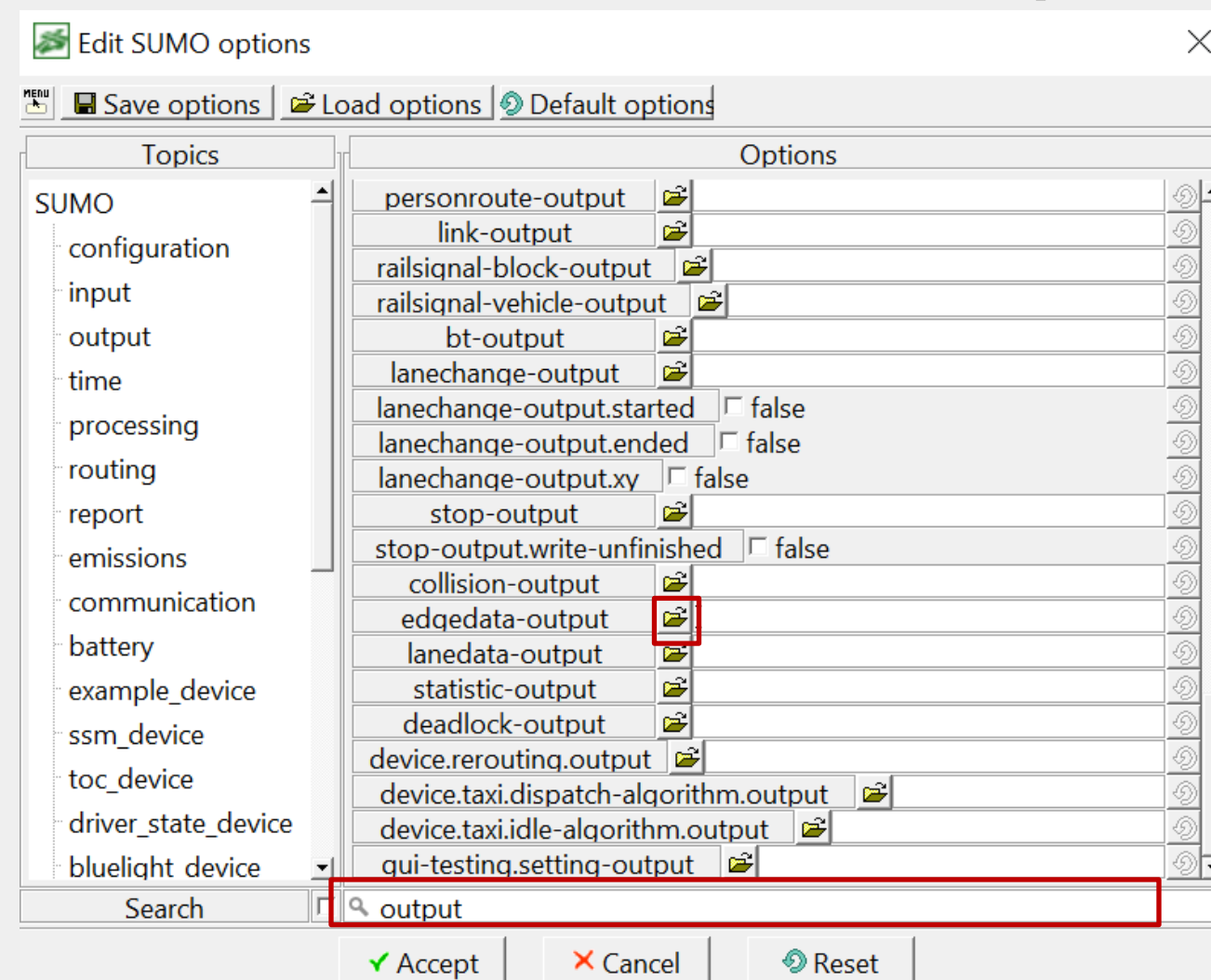
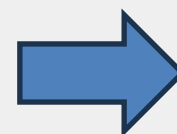
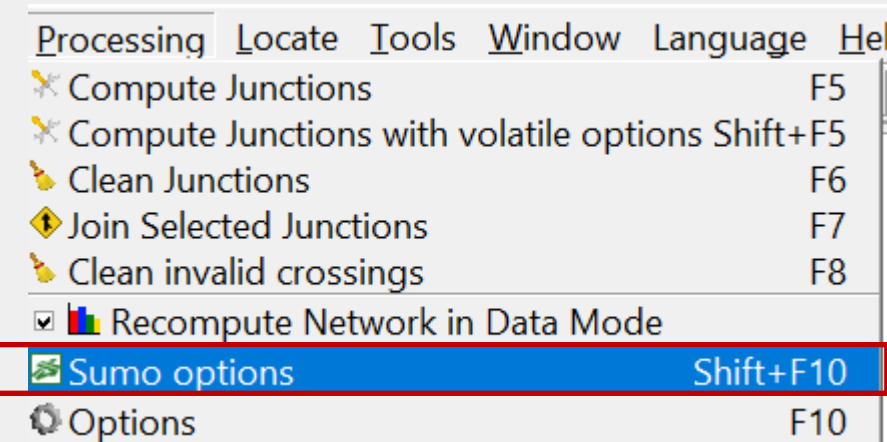


How to get output (Speed, volume etc) from Simulation?

❑ **Goal:** How to get output (Speed, volume etc) from Simulation?

❑ Open BaseScenario netedit and (**This is not SUMO files in deliverable 2, open SUMO in the Base Scenario folder**)

❑ Extract output from SUMO



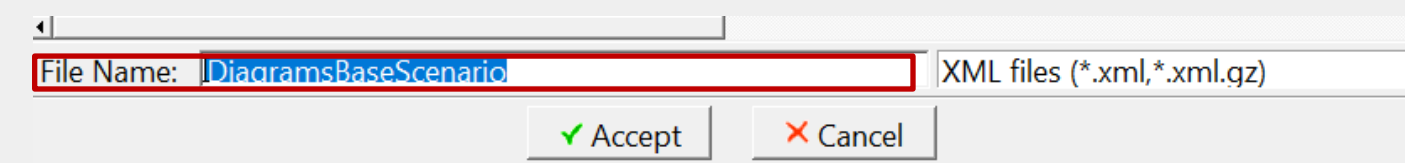
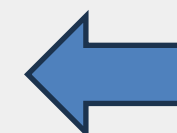
Search Output → Select edgedata-output → Select the icon  for saving



Find Base Scenario Folder → Set the name "DiagramsBaseScenario"



Save Files



How to get output (Speed, volume etc) from Simulation?

- ❑ Create an empty txt file

- ❑ Open it and copy and paste this

```
<additional>
```

```
  <edgeData id="edge_data" file="DiagramsBaseScenario.xml" period="100" excludeEmpty="true"/>
```

```
</additional>
```

- ❑ Change the name of file to "Exercise2_add.xml"

- ❑ Open the Exercise2.sumocfg with notepad and add the below following line:

- ❑ <additional-files value="Exercise2_add.xml"/>

- ❑ Save

```
<sumoConfiguration xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:noNamespaceSchemaLocation="http://www.sumo-project.org/sumo.xsd">
  <input>
    <net-file value="Exercise2.net.xml"/>
    <route-files value="Exercise2.rou.xml"/>
    <additional-files value="Exercise2_add.xml"/>
  </input>
  <output>
    <edgedata-output value="C:/Users/ahmad/Downloads/DiagramsBaseScenario.xml"/>
  </output>
</sumoConfiguration>
```

How to get output (Speed, volume etc) from Simulation?

- ❑ Run SUMO
- ❑ This creates the DiagramsBaseScenario in the output folder

images	2026-05-12 11:48 AM	Microsoft Edge HTML...	319 KB
 DiagramsBaseScenario.xml	2026-05-12 11:48 AM	Microsoft Edge HTML...	319 KB

- ❑ Open the file and observe the output (Find Density, Flow and Speed)

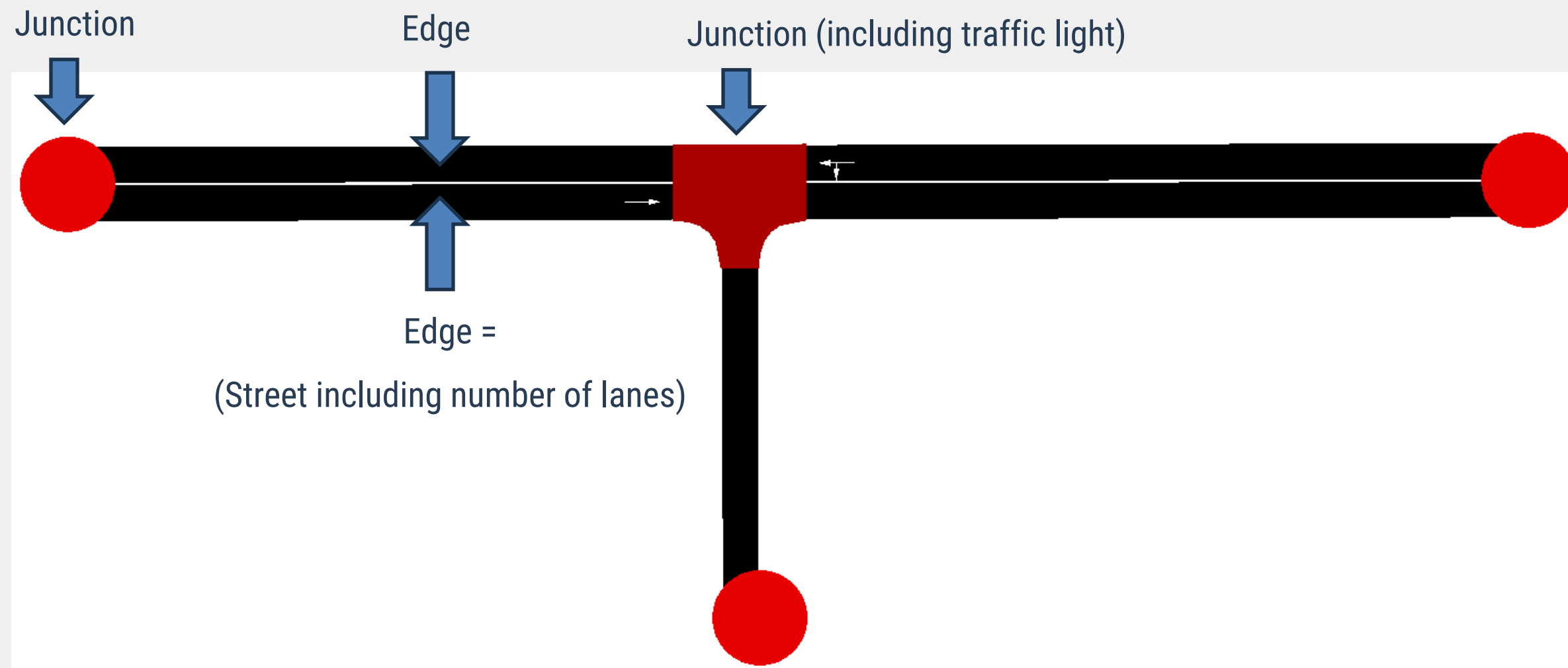
```
<meandata xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:noNamespaceSchemaLocation="http://sumo.dlr.de/xsd/meandata_file.xsd">
  <interval begin="0.00" end="100.00" id="edge_data">
    <edge id="E10" sampledSeconds="31.15" traveltime="1.60" overlapTraveltime="1.99" density="12.49" overlapDensity="15.17" laneDensity="6.24" occupancy="3.09" waitingTime="0.00" timeLoss="2.29" speed="12.85" speedRelative="0.92"
    departed="0" arrived="16" entered="16" left="0" laneChangedFrom="0" laneChangedTo="0" flow="576.00"/>
    <edge id="E13" sampledSeconds="184.05" traveltime="4.95" overlapTraveltime="5.52" density="69.07" overlapDensity="80.40" laneDensity="23.02" occupancy="12.53" waitingTime="0.00" timeLoss="117.19" speed="5.05" speedRelative="0.36"
    departed="42" arrived="0" entered="0" left="41" laneChangedFrom="41" laneChangedTo="41" flow="1149.29"/>
  </interval>
</meandata>
```

- ❑ This file provide different measurements (density, flow, speed) for each edge (street) in random simulation second in each period

Base Scenario: Flow, Density, Speed

- ☐ Open Deliverable 3 folder
- ☐ Run html
- ☐ Select the Speed and Density chart is fit ($R^2 > 0.7$)
- ☐ Zoom in → You can observe the fundamental diagrams (Speed, Density) for each edge (street)
- ☐ What is an edge? (see next slide)

Edge vs Junction

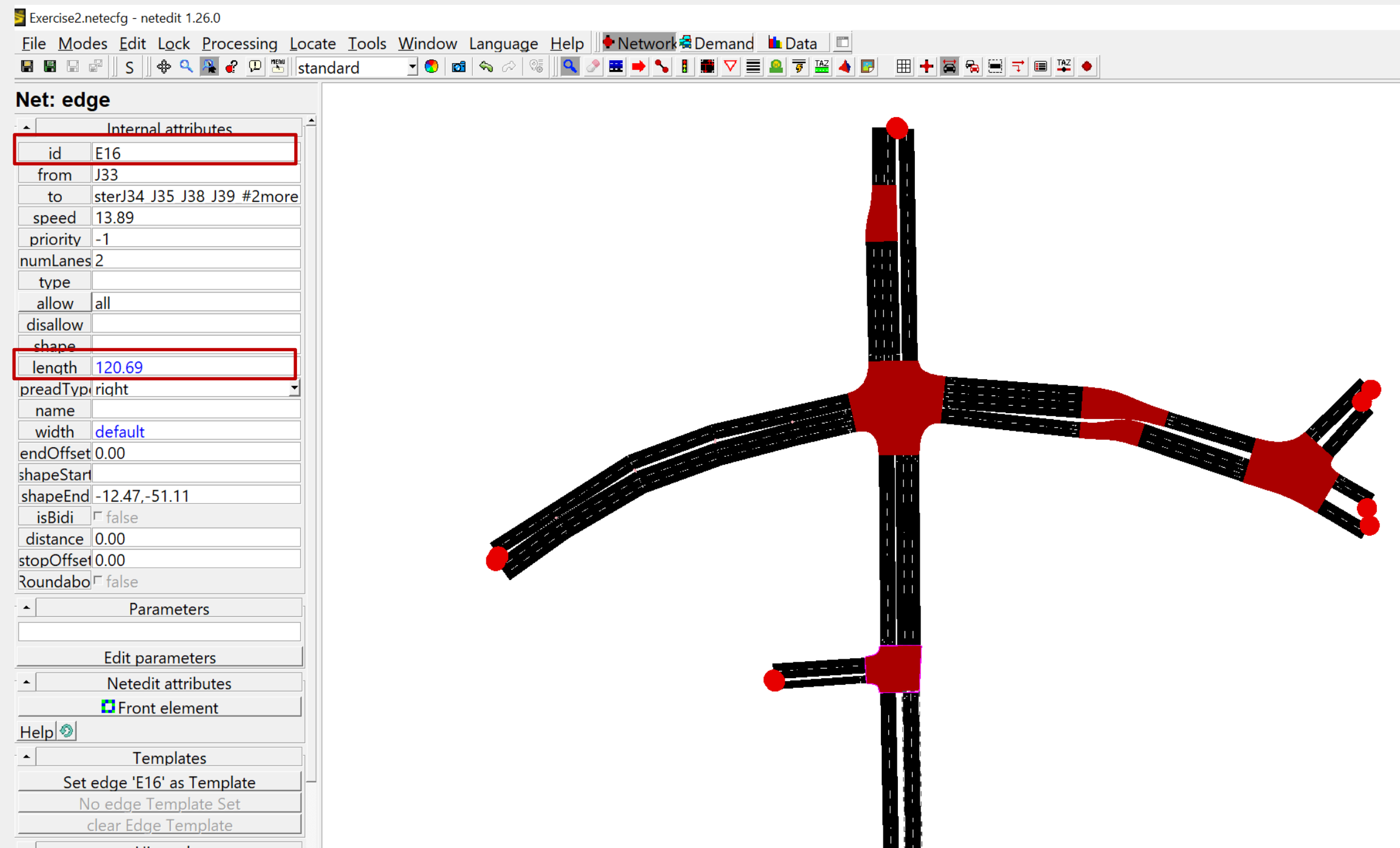


Next Slides are Some Example Calculation

Read Them Carefully before Deliverable 3

Identify Edge 16

- ❑ Open netedit → Network → Magnifier
- ❑ Select each edge (street) to identify Edge 16
- ❑ Record its length

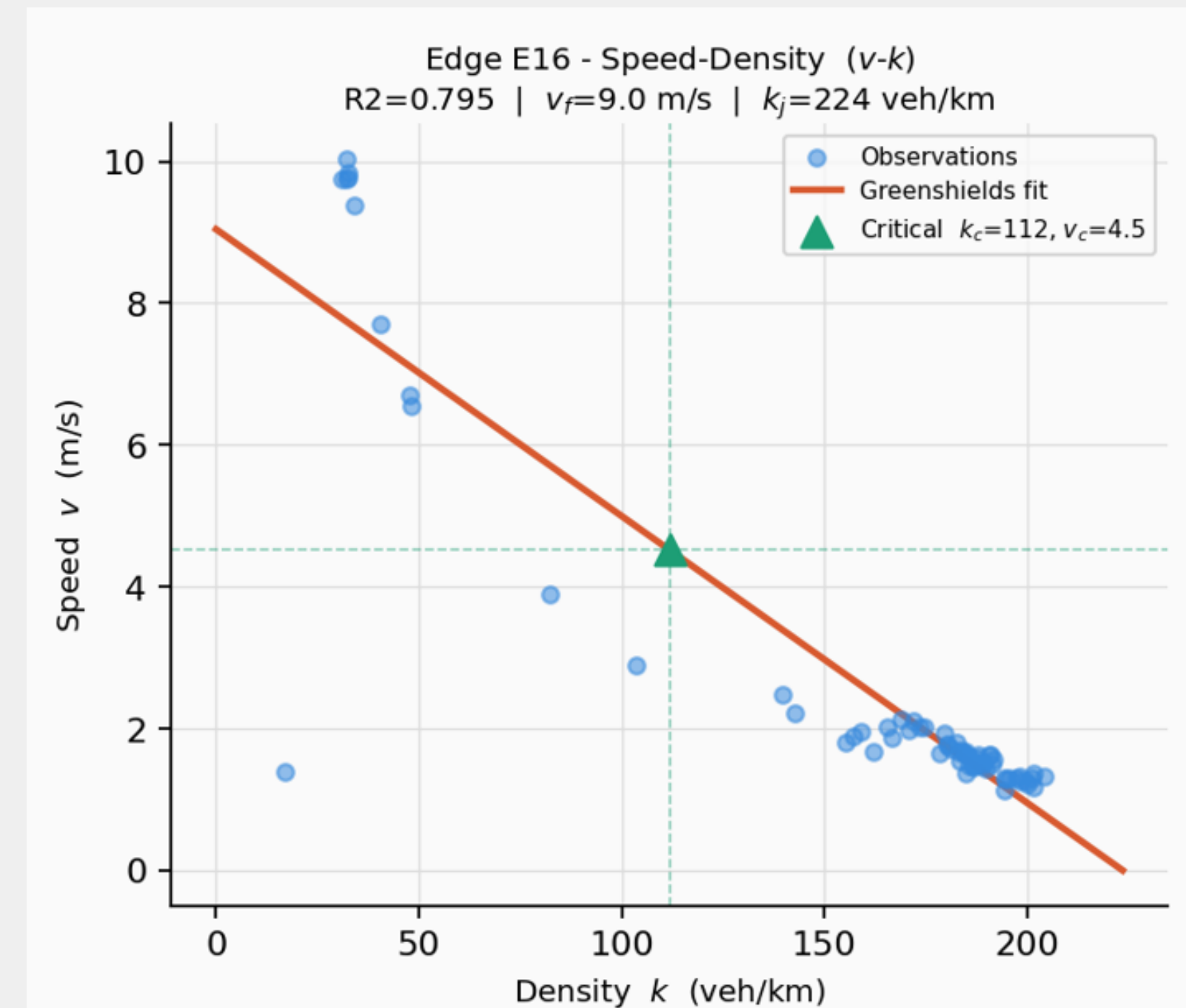


From SUMO Results to Fundamental Diagrams

□ Keep only three ideas:

- Blue points = SUMO traffic observations
- Orange line/curve = Greenshields fitted model
- Green marker = critical traffic condition

□ Each SUMO observation gives a traffic state (k, v_s, q) .
Plotting many traffic states gives the three
fundamental diagrams.



Speed–Density Diagram

- ☐ **Question 6: Speed–Density Relationship:** As density increases, vehicles interact more and speed generally decreases.
- ☐ From the fitted diagram, identify the free-flow speed (v_f) and jam density (k_j).

Use the Greenshields equation:

$$v_s = v_f \times (1 - k / k_j)$$

For Edge E16: $v_f = 9.0$ m/s, $k_j = 224$ veh/km

At $k = 112$ veh/km:

$$v_s = 9.0 \times (1 - 112 / 224)$$

$$v_s = 9.0 \times 0.5$$

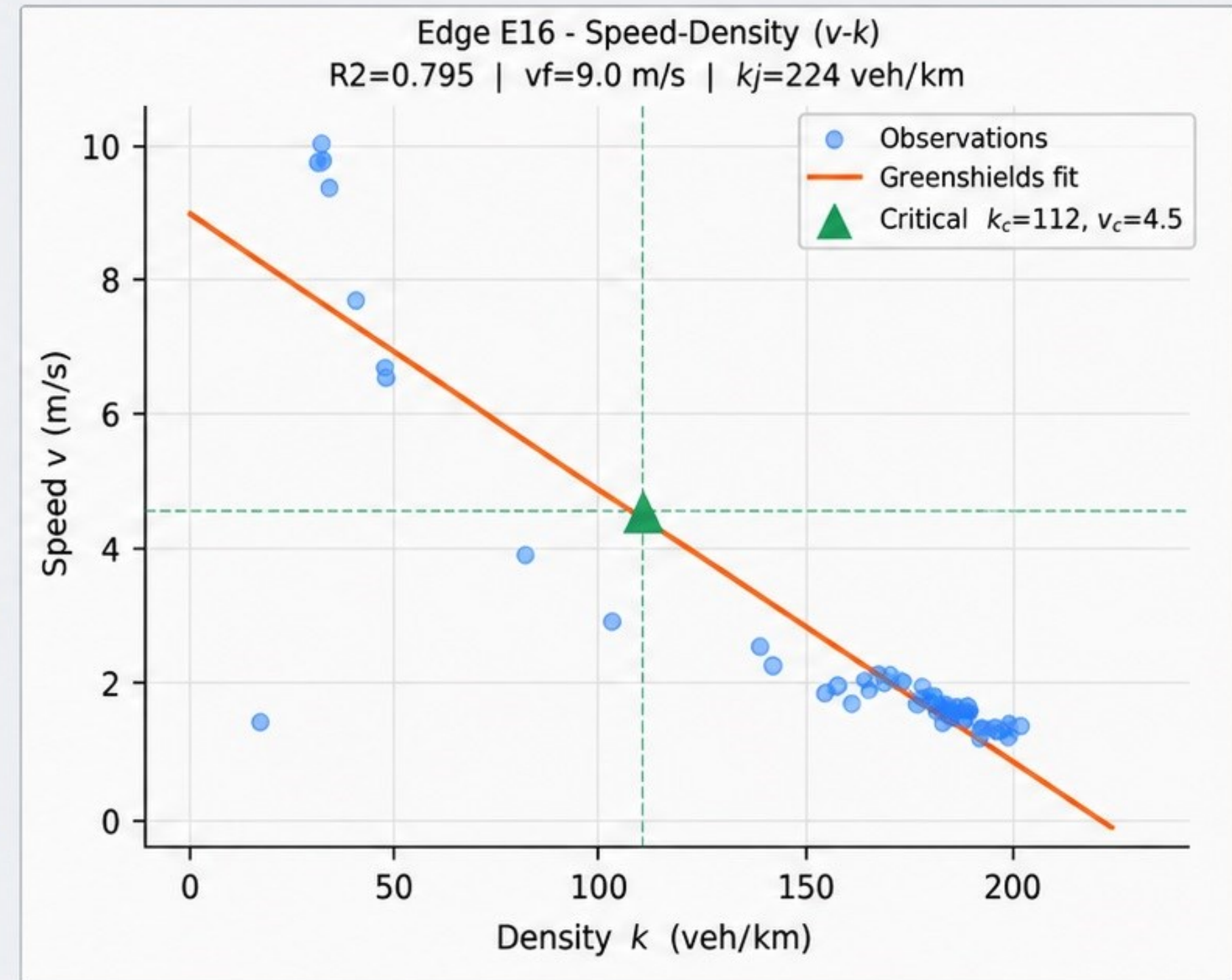
$$v_s = 4.5 \text{ m/s}$$

Does this match the green point on the diagram?

Check the two extreme conditions:

- When $k = 0$, $v_s = 9.0$ m/s
- When $k = 224$, $v_s = 0$ m/s

Interpretation: When density is zero, vehicles travel at free-flow speed. When density reaches jam density, vehicles stop.



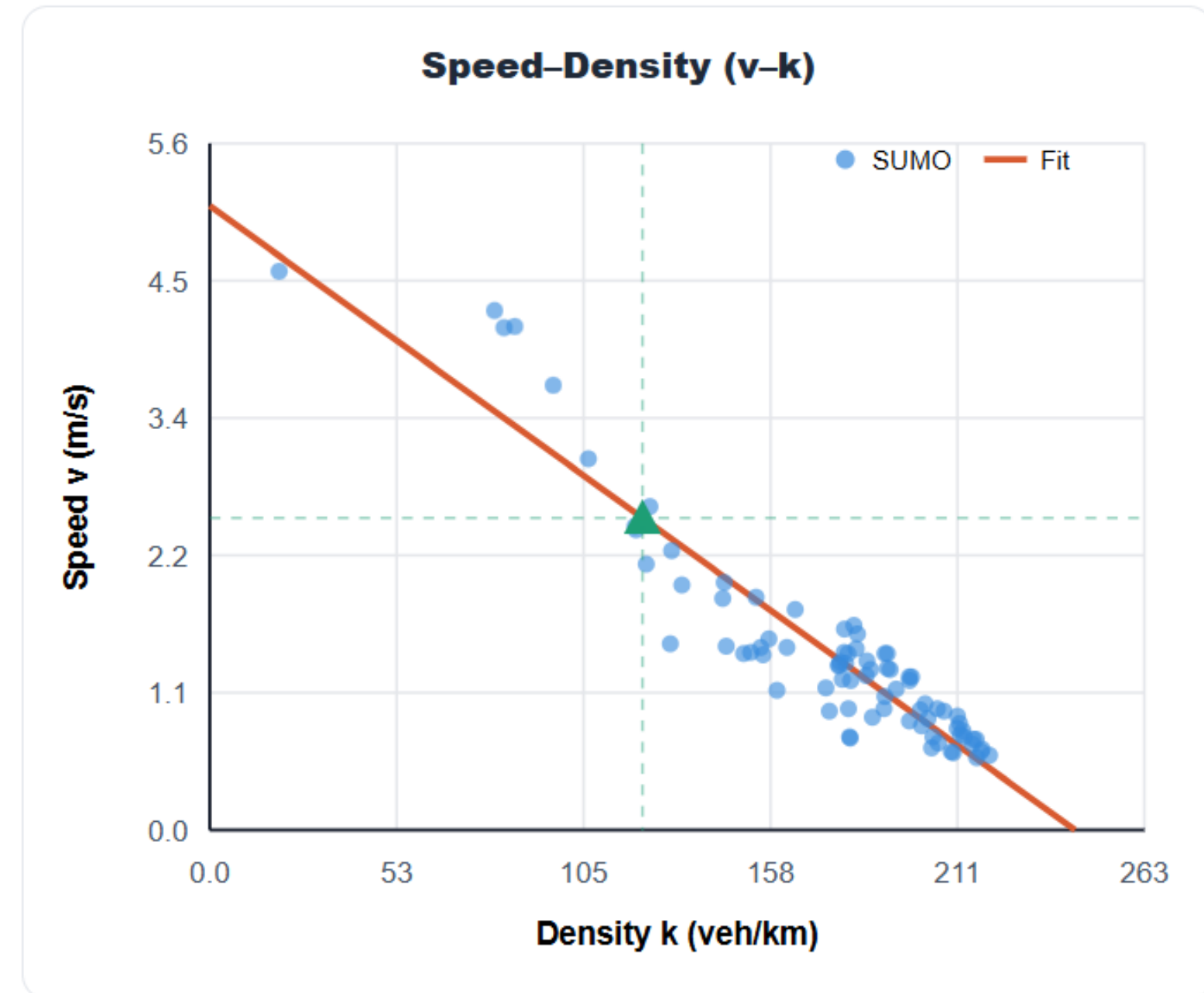
Deliverable 3

Deliverable 3 Folder:

1. Open HTML
2. Respond to the Question Using the SUMO Files
3. Submit only the Generated two htmls in Avenue

Deliverable 3

- ☐ Open Folder Deliverable 3
- ☐ Open html and Drag and Drop your BaseScenario DiagramsBaseScenario.xml into HTML
- ☐ Tab: Select an Edge (Its better to select an Edge with $R^2 > 0.7$) tagged with Green Color)
- ☐ Select the edge with green condition
- ☐ Go to tab 2
- ☐ Answer question (For that, carefully observe the previous slide how we calculate fundamental diagrams parameters)
- ☐ generate HTML and Submit

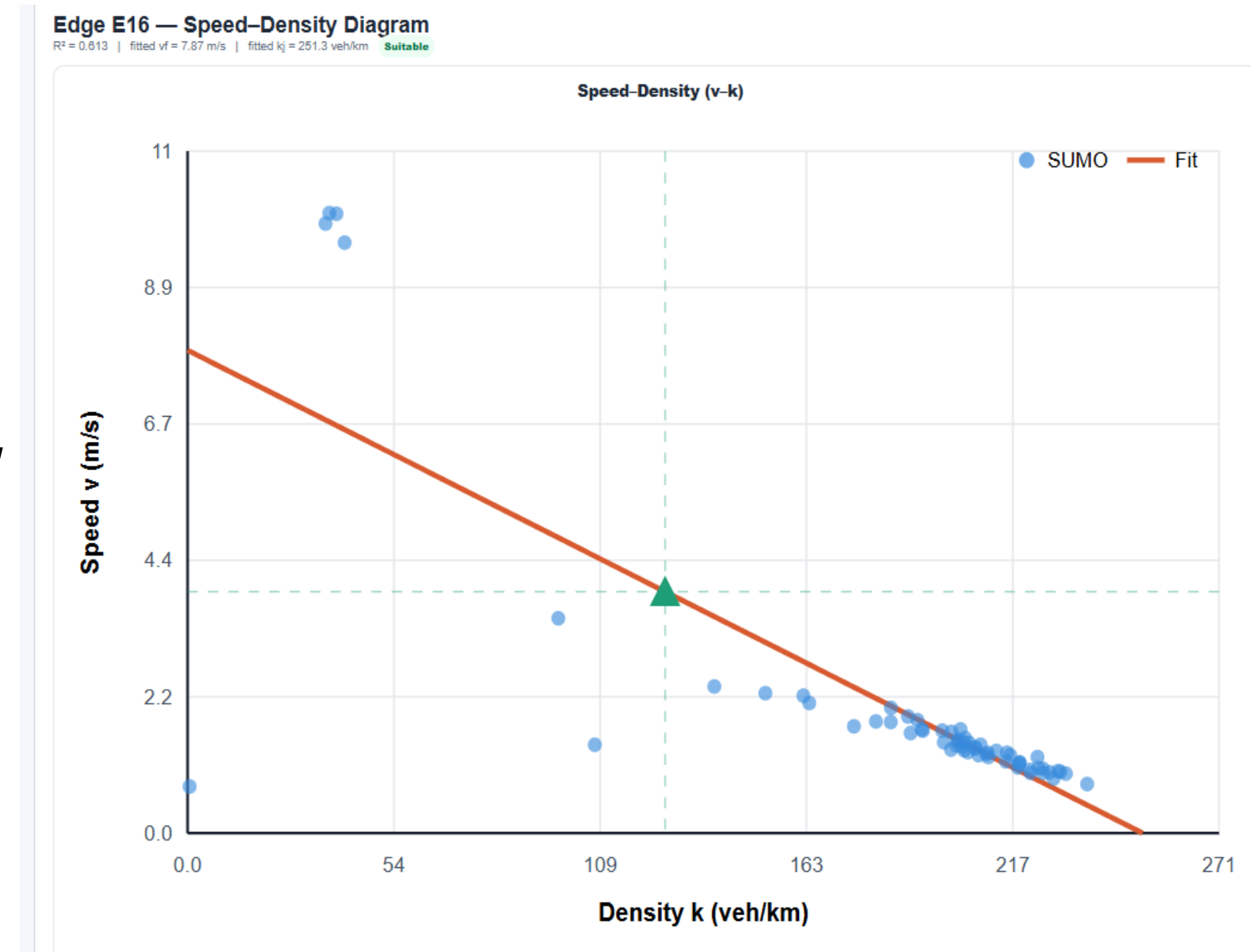


Scenario 1 (CAVs) : Flow, Density, Speed

- ❑ Conduct the same practice for Scenario 1 to export DiagramsBaseScenario.xml and export fundamental diagrams and answer the next questions

Deliverable 3

- ☐ Open Folder Deliverable 3
- ☐ Open html and Drag and Drop your BaseScenario DiagramsBaseScenario.xml into HTML
- ☐ Tab: Select the same Edge you selected in BaseScenario
- ☐ Make Sure both of Them should be Green
- ☐ Select the edge with green condition
- ☐ Go to tab 2
- ☐ Answer question (For that, carefully observe the previous slide how we calculate fundamental diagrams parameters)
- ☐ generate HTML and Submit



Use of Generative AI

- ❑ **Upload both diagrams to a generative AI tool** (e.g., ChatGPT or Gemini) and ask it to compare the two images.
- ❑ **Provide enough context in your prompt.** Explain what the diagrams represent, which scenarios are being compared, and that both diagrams are for the **same SUMO edge**.

Example prompt:

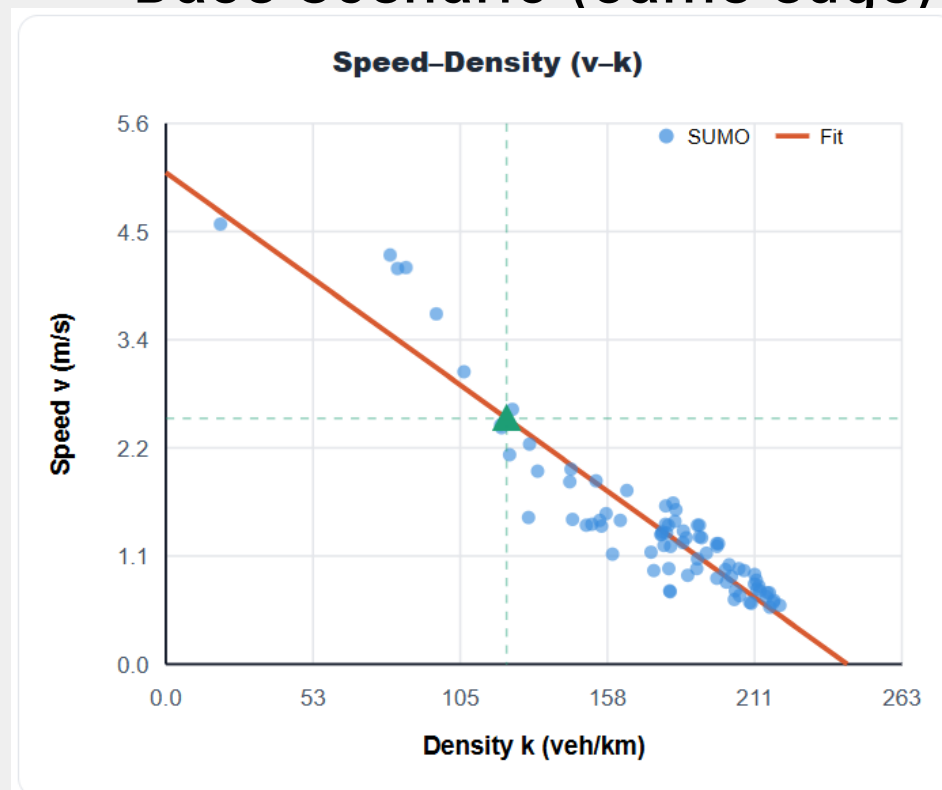
These two Speed–Density fundamental diagrams were generated from SUMO for the same roadway edge.

The **left diagram** is the Base Scenario with **100% human-driven vehicles (HDVs)**.

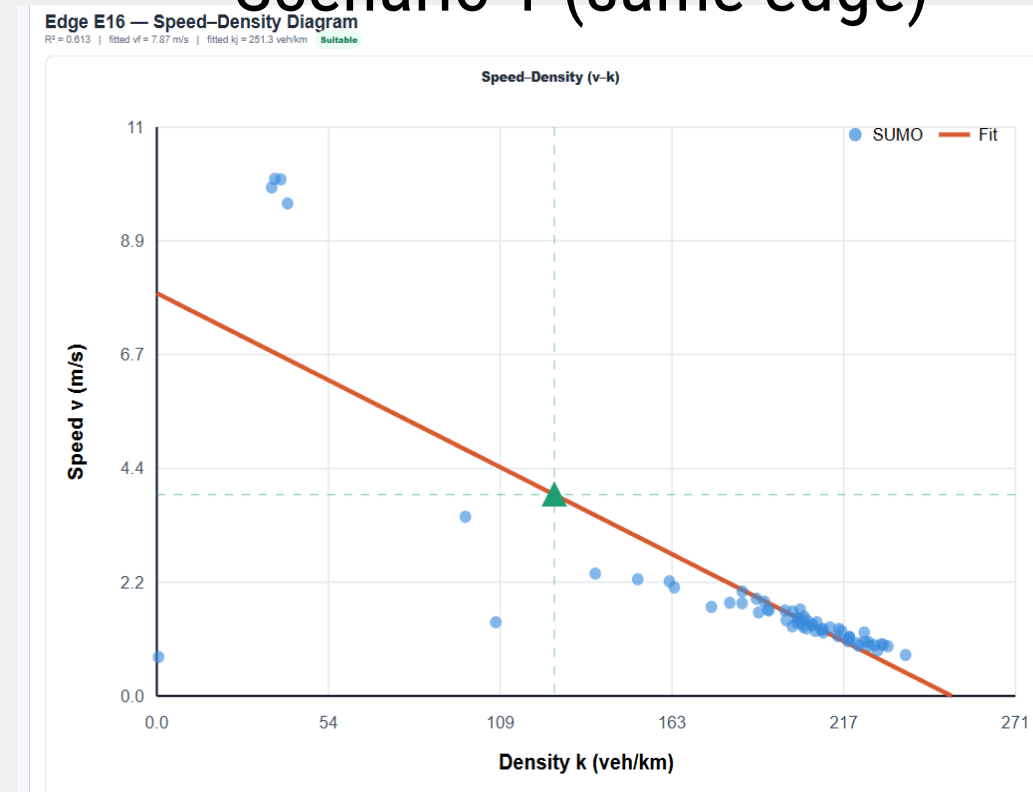
The **right diagram** is Scenario 1 with **100% autonomous vehicles (AVs)**.

Compare the two diagrams and explain the main differences in traffic behavior. Focus on the fitted speed–density relationship, free-flow speed, jam density, critical point, and how speed changes as density increases.

Base Scenario (same edge)



Scenario 1 (same edge)



Use of Generative AI

☐ Read and understand the AI output (Do not simply accept it) and think if the explanation makes sense

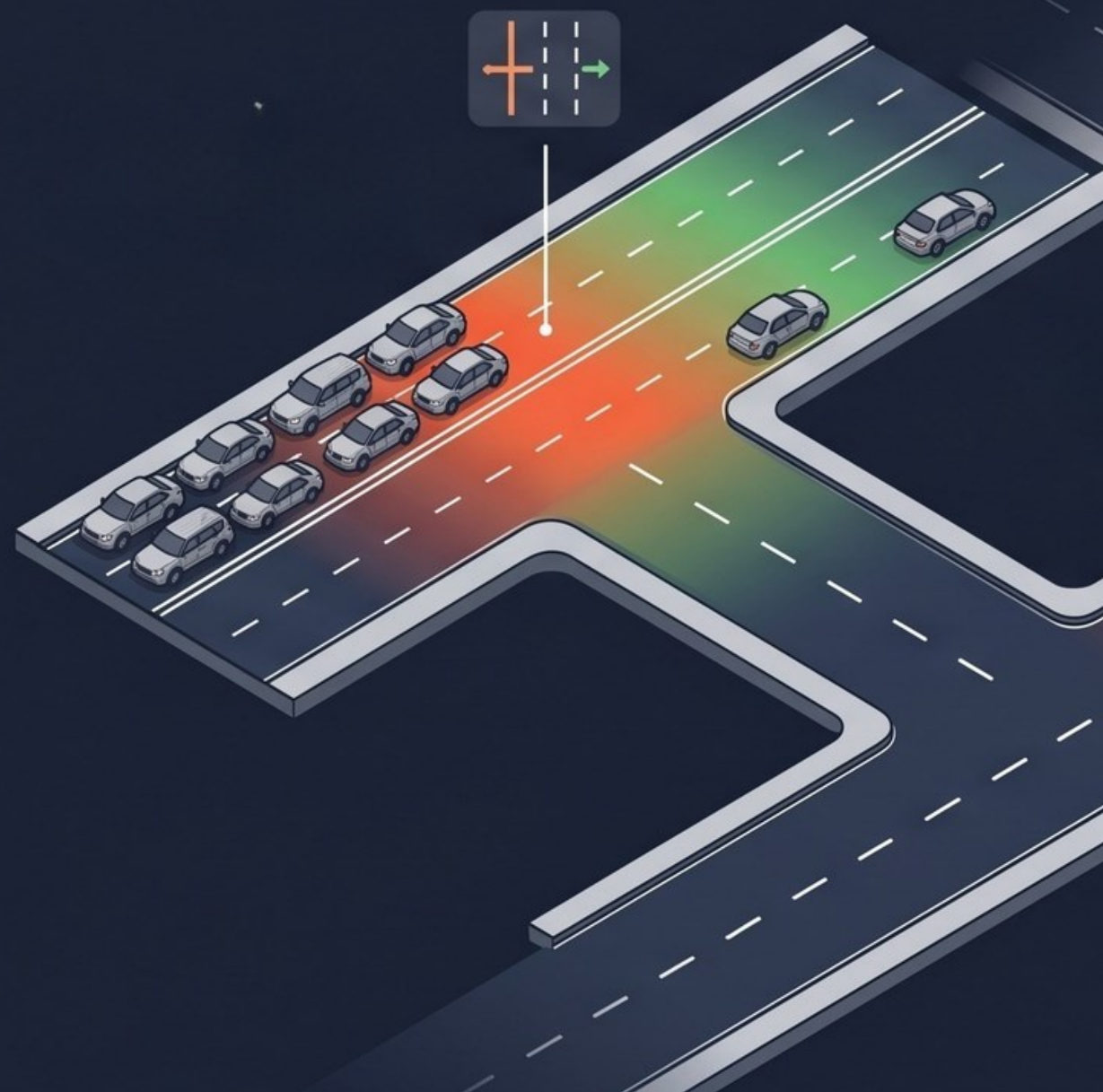
☐ **Write your understand now in a paragraph and raise your hands for instructor**

Mini-Project – Stage A2 & Report A

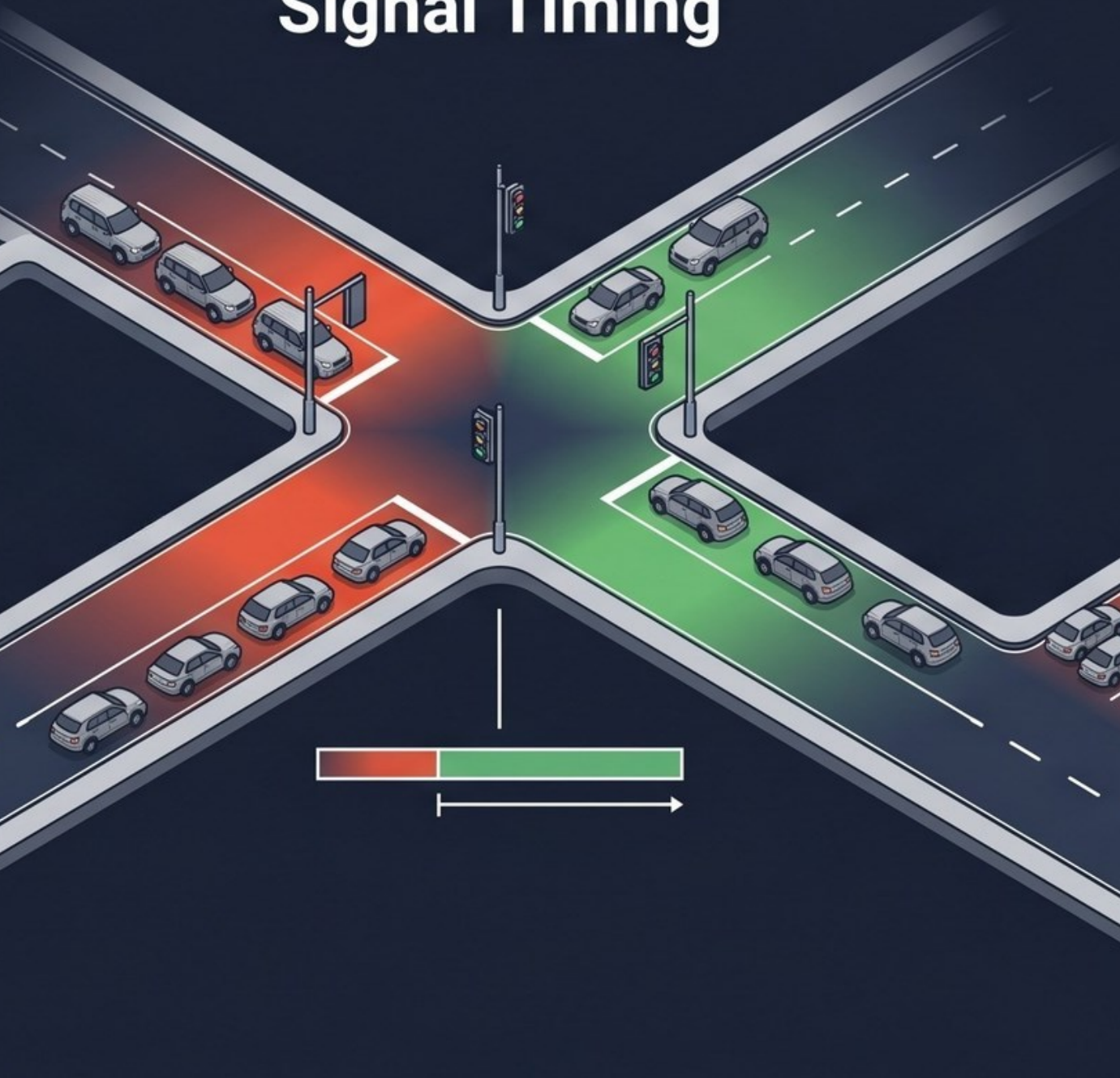
Mini-Project – Stage A2 & Report A

- ❑ Mini Project A includes
- ❑ Stage A1 → You have done this
- ❑ Stage A2 → Needs to be done (**Only for one intersection inside your study area**)
- ❑ Report A → Needs to be done
- ❑ On Friday night (Oct 9), You will download below required materials:
- ❑ Download **SMRTTECH-4SC3-MiniProjectsA_Complete** and read it
- ❑ Download **WritingStructure.docx** and read it
- ❑ Download **Template.docx** and read it
- ❑ In Simple words: You need to implement this lab milestones to your already developed network in Stage A1 (**Only for one intersection**) plus one more scenario (Scenario 2: which is one intervention- see next slide)

Lane Addition



Signal Timing



Demand Adjustment



Milestone 5 (Intervention):

This is to help more about MiniProject A,
read this after reading the previous slide word file

Background

Your task is to **intervene in the simulation** to shift traffic conditions on at least one edge from its current state toward a more balanced traffic condition, then re-calibrate the fundamental diagram to evaluate the impact of your intervention.

Task Description

Choose ONE of the following three intervention strategies (next slides) and apply it to your SUMO scenario. Your intervention must be justified, implemented, documented, and evaluated through fundamental diagram re-calibration.

Submission note: Include all three diagrams as figures in your report with proper axis labels, units, and captions. Your Python script output file should be submitted alongside the report.

Strategy A - Geometric Intervention: Lane Addition or Removal

Modify the number of lanes on one or more edges in NetEdit to redistribute traffic capacity across the network.

- Open your network in **NetEdit** and select a congested edge (select one those red)
- Add one additional lane to increase capacity
- Re-run the simulation and extract a new edgeData XML output
- Re-calculate the fundamental diagram for your modified edge

Example justification: "Edge E14 operates at near-zero speed with density exceeding 300 veh/km. Adding a lane increases capacity and should reduce density and increase speed toward free-flow conditions."

Strategy B - Operational Intervention: Traffic Signal Timing

Modify the signal timing at one or more intersections in your network to redistribute green time and reduce queue buildup on congested approaches.

- Identify which junction controls flow onto your congested edge
- Open the signal timing in SUMO (via NetEdit or directly in the .net.xml) and increase green time for the congested approach
- Re-run the simulation and extract a new edgeData XML output
- Re-calibrate the fundamental diagram for the affected edge

Example justification: "Edge E3 experiences heavy queuing due to insufficient green time on its downstream signal. Increasing the green phase from 30s to 50s should reduce waiting time and allow the edge to discharge more vehicles per cycle."

Strategy C - Demand Intervention: Traffic Volume Adjustment

Modify the traffic demand in your route file to either increase flow on a free-flow edge (pushing it toward congestion) or decrease flow on a congested edge (relieving it toward free-flow).

- Open your rou.xml and locate the vehicle flow definitions
- Increase or decrease the vehsPerHour or number attribute for the relevant route
- Re-run the simulation and extract a new edgeData XML output
- Re-calibrate the fundamental diagram for your target edge

Example justification: "Edge E10 remains in free-flow throughout the simulation because demand on its upstream route is low. Increasing vehicle flow by 50% on that route will push E10 toward capacity and allow a valid fundamental diagram to be calibrated."